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From BBS to Blackboard: The Historical Pedagogy of Hacker Culture in Phrack Magazine (1985-2016)

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Abbreviations

BBS: Bulletin Board System

CDA: Critical Discourse Analysis

DIY: Do-It-Yourself

FLOSS: Free/Libre and Open-Source Software

ICT: Information and Communication Technologies

IRC: Internet Relay Chat

MIT: Massachusetts Institute of Technology

MOOC: Massive Open Online Course

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Abstract

This thesis explores Phrack Magazine as an important historical site of informal learning within the context of hacker culture, and that it is not just a technical magazine but also a discursive space where knowledge, identity, and values are negotiated, taught and analyzed. Based on sociocultural theory of learning and considering theories on digital informal learning, the research investigates how hacker subjectivity is negotiated through linguistic praxis, instructional genres, and cultural narratives. The research methodology integrates Critical Discourse Analysis with computer text analysis by employing Voyant Tools to enable both qualitative and quantitative comprehension of learning undertaken across decades of Phrack publications. The outcome reveals that the magazine sustains peer-initiated and participatory forms of knowledge transfer alongside, simultaneously exerting boundaries of ideology and cultivating a selective perception of openness. Despite its emphasis on decentralization and anti-authoritarianism, Phrack's rhetoric is still largely homogeneous in tone and culturally exclusive. This work adds to the research on digital education by reinterpreting hacker rhetoric as persuasive, but bounded pedagogy practiced outside of institution-based contexts.

Keywords: Hacker Culture, Informal Learning, Phrack Magazine, Digital Pedagogy, Critical Discourse Analysis, Peer Learning, Alternative Education, History of Phrack magazine

Extended Summary

This dissertation is an analysis of the learning practices of hacker culture by a close reading of Phrack Magazine, an online magazine produced by and for hackers since 1985. The general argument is that Phrack is an informal yet highly structured learning environment wherein knowledge, identity, and ideology are constructed and transmitted through discourse. Although traditionally neglected in educational studies, hacker culture presents an interesting example of how self-directed, socially situated learning can occur outside institutional settings. By drawing on sociocultural learning theory, digital informal learning, and critical discourse analysis in an analysis of Phrack, this study demonstrates how hackers learn and teach through collective textual practices.

The thesis begins with a conceptual exploration of informal learning, its main features being non-institutional, interest-led, and integrated in daily or community life. Unlike formal learning, which is often top-down, curriculum-led, and assessed by standardized measures, informal learning is learner-led and context-dependent. Drawing from theorists such as Vygotsky, Rogoff, and Lave and Wenger, the thesis is predicated on learning as a social process accomplished through participation in culturally meaningful activity. Utilization of digital technologies has enhanced the prospect of informal learning and thus the development of what is known as digital informal learning (DIL). The model supports personalized, self-managed, and socially networked learning through a number of platforms and media. It is within this evolving digital learning landscape that Phrack Magazine is situated.

Chapter one presents the theoretical context, explaining how informal learning is different and how it overlaps with formal learning. It explains how DIL, facilitated by digital tools and platforms, allows for decentralized knowledge creation and collective meaning-making. The chapter also introduces the concept of Personal Learning Environments (PLEs) as online environments defined by learners to support learning and the application of knowledge across media and communities. The culture of hackers is then introduced as a model example of DIL in action, where social interaction and technical experimentation are the foundation of learning.

The second chapter historically maps the formation of hacker identity. Starting from the phone phreaking of the 1950s and 1960s, it illustrates how hackers have been shaped by technological change, legal pressure, and popular resistance. The chapter highlights the importance of the early hacking hangouts like the Homebrew Computer Club and MIT's Tech Model Railroad Club for developing the ideological and practical foundations of hacker culture. As computing and the internet at home developed, hackers gathered around Bulletin Board Systems (BBSes), where they exchanged files, advice, and status markers. These systems were early digital affinity spaces where reputations were made, hierarchies were formed, and "hacker" began to imply both technical proficiency and subcultural identification. The term later earned stigma as well from media and law enforcement in mainstream society, which associated hacking with criminality and deviance. Hackers responded by strategically reclaiming the term, using forums like Phrack to redefine public perception and reconstruct their identity around the ideals of creativity, autonomy, and ethical objection.

Phrack Magazine is also introduced in chapter two as a discursive field where hacker identity is constantly constructed and negotiated. More than a newsletter or message board, Phrack became a site of education, boundary-drawing, and ideological development. It offered tutorials, editorials, manifestos, and scene reports that modeled technical knowledge but also created group norms and ethics. By establishing both what to know and how to be, Phrack became an unofficial yet authoritative pedagogical authority within hacker culture.

Chapter three is the main empirical exploration of the thesis, in which critical discourse analysis is combined with computational text mining using Voyant Tools. The text corpus analyzed is 69 Phrack issues, since the debut issue in 1985 through 2016. The extensive textual archive was analyzed for lexical patterning, genre conventions, and rhetorical moves. The chapter identifies the primary pedagogical features in the magazine, such as the use of step-by-step tutorials, in-group jargon, technical modeling, and common themes of resistance and ethical hacking. It indicates that learning within Phrack is not only a question of transmitting technical expertise but also of forming identity, authority, and community. The language of the magazine works to include insiders and exclude outsiders, marking out levels of expertise, ideological dedication, and subcultural affiliation.

Through this multi-method examination, the thesis illustrates that Phrack facilitates informal learning through the entanglement of identity construction and knowledge sharing. The magazine discourse places readers in the role of more than recipients of passive knowledge; instead, they are framed as participants in a collective cultural project. Readers are situated as learners, as mentors, and as moral agents within a community at large. In teaching readers to think, write, and behave like hackers, Phrack continues the hacker ethos from generation to generation.

The fourth and final chapter discusses the broader implications of these findings. It highlights that informal learning environments like Phrack challenge traditional boundaries of formal education. Rather than peripheral or ancillary, informal learning in subcultures can be structured, recursive, and transformative. The thesis contributes to a number of bodies of scholarly literature. In informal learning theory, it extends the concept of learning beyond institutional environments and highlights its cultural and ideological dimensions. In hacker historical studies, it provides a close reading of the constructions of hacker identity in pedagogical discourse. In educational technology, it demonstrates the values of peer-led, autonomous learning environments outside of mainstream systems. Finally, in discourse analysis and digital humanities, it showcases a mixed-methods approach to the analysis of large corpora of text with both computational precision and interpretive depth.

The study also addresses limitations. It acknowledges that the priority given to textual analysis makes it impossible to observe reader response or community dynamics directly. The ethical dilemma of studying a legally gray subculture also required tactful care, especially in how pseudonyms and authorship were handled. Nonetheless, the thesis remains founded upon publicly accessible data using an open and rigorous methodological approach.

In conclusion, the thesis affirms that Phrack Magazine is not just a technical archive but a pedagogical space in which informal learning is enacted repeatedly. It is a fluid and dynamic space in which technical knowledge, cultural identity, and moral values are produced and reproduced through language. This result challenges educators and researchers to reconsider what constitutes a learning environment and to think about the ways in which informal, subcultural, and digitally mediated environments can be important to an understanding of education in the twenty-first century.

Chapter 0: Introduction

0.1 Rationale and Context

The expansion of digital technologies has revolutionized the way people engage with learning, often beyond traditional schools and curriculum. This shift has developed new forms of learning, interest-driven, participatory, and socially embedded, but ones that are not always observable in dominant institutional or theoretical models of education (Ito et al., 2013). Informal learning, which historically de-emphasized in conventional education research, is now particularly salient in online spaces, where students form communities of shared practice and interest (Jenkins et al., 2009). These online affinity spaces, like fan fiction sites, video game forums, and maker cultures, enable knowledge construction through shared action, shared language, and shared knowledge (Gee, 2005). These environments challenge traditional teacher-student, formality-informality boundaries and underscore the need for new theoretical frameworks that can account for these hybrid learning environments (Reinders & Benson, 2017). One of the most striking and under-theorized examples of this is the hacker community, as it is depicted and remade in its own communicative texts.

Phrack Magazine was a BBS-based [1], e-zine produced by and for hackers from 1985 before the existence of world wide web. It offered a rich, lengthy corpus to help hackers explore learning as a culturally positioned and ideologically invested practice. Phrack was more than a newsletter or archive of exploits; it was a reflexive community forum, manifesto space, tutorial resource, and cultural toolkit for hackers participating in hacker subculture. Its integration of technical tutorial, countercultural commentary, and identity formation took it to the status of an exemplary site for examination of informal learning as a discourse-mediated and socially negotiated process.

This thesis argues that Phrack is conducting a form of "hacker pedagogy", a peer-to-peer, non-institutional, informal learning process founded on cultural performance, ideological critique, and communication. It examines how Phrack teaches, positions, and excludes through language, and how its texts mediate a continuing process of learning and shaping a hacker's identity.

0.2 Problem Statement

Even as academic attention to hacker culture has grown, the place of informal learning within it has been under-theorized. Research on hacker subcultures tends to emphasize legal, security, or political aspects (Jordan & Taylor, 2004; Coleman, 2013), yet fewer look at how hackers learn, share information, or promulgate ethical ideals through ongoing discursive activity.

At the same time, studies of informal learning also tend to concentrate on participatory learning (Gee, 2004), digital literacies, or open educational practices (Dron & Anderson, 2014) and less so on subcultures that are at the edge of legality and social acceptability. Phrack itself is widely referenced in the literature on hacker history and identity (Sterling, 1992; Lunceford, 2009) but less so analyzed as a structured, informal, learning community.

This thesis addresses this gap by bringing together sociocultural learning theory and discourse analysis in an analytical framework for examining Phrack as a historical pedagogical discursive space wherein instruction, ideology, and identity are intertwined.

0.3 Research Aims and Questions

The general aim of this thesis is to examine how informal learning is realized discursively in hacker subcultures through and in the textual practices of Phrack Magazine from the early days of it works to its last publication. More specifically, it seeks to determine how technical knowledge, ethical position, and subcultural membership are built, shared, and solidified through discourse.

The research is guided by the following questions:

- How are informal, peer-based learning practices constructed through discourse in Phrack magazine?
- What rhetorical and discursive strategies does Phrack use to position its readers as learners, mentors, or members of a hacker learning community?
- How do Phrack articles enable hacker cultural values of experimentation, curiosity, and ethical knowledge sharing?
- How does Phrack discourse build boundaries of inclusion and exclusion within hacker learning communities?

0.4 Scope and Structure

The thesis is composed of four main chapters, each building a cumulative case for viewing Phrack as a learning environment and cultural artifact.

Chapter 1 theorizes informal learning research, distinguishing it from formal and informal education. It situates digital informal learning within sociocultural theory, emphasizing learner

agency, peer support, and context-sensitive participation. It introduces Digital Informal Learning (DIL) as a relevant framework.

Chapter 2 traces the history and evolution of hacker identity and culture, highlighting how the hacker as a social actor has evolved through resistance, technical innovation, and countercultural values. It situates Phrack within the general discursive formation of hacker subcultures, illustrating how the magazine documents and creates hacker life.

Chapter 3 presents the primary empirical examination. By synthesizing Critical Discourse Analysis (CDA) and Voyant Tools, the chapter examines how learning is written into Phrack's lexical patterns, rhetorical moves, and genre conventions. The chapter teases themes such as peer-authorship, in-group terminology, technical modeling, and ideological framing.

Chapter 4 concludes by summarizing the findings and declaring their significance for informal learning theory, hacker studies, and digital education. Recommendations for further research are also made.

0.5 Theoretical Framework

The study is grounded theoretically in sociocultural learning theory, which views learning as a social process affected by participation in culturally meaningful activity (Vygotsky, 1934; Lave & Wenger, 1991; Rogoff, 1990). Rather than the transmission of knowledge, this view emphasizes learning as becoming a socially negotiated, incremental alignment with a community's language, practices, and norms.

This is complemented by a body of literature on Digital Informal Learning that showcases how autonomy, self-direction, and peer scaffolding come to the forefront in digital environments (Reinders & Benson, 2017; Doleck et al., 2019). Learning in this framework is typically horizontal in nature instead of being hierarchical, embedded in media practice, and sustained through knowledge-making with others.

Finally, Critical Discourse Analysis (Fairclough, 1995; Gee, 2011) provides a methodological emphasis whereby examining how power, identity, and ideology are negotiated through words. CDA enables Phrack to be examined not just as a medium of information, but as a discursive arena where hacker literacy is constructed, struggled over, and controlled. These frameworks may be blended to enable a two-level analysis: sociocultural theory informs what learning is as a social practice, Digital Informal Learning sharpens the emphasis on peer-to-peer[1] learning online, and Critical Discourse Analysis enables consideration of how these learning processes are acted out and reproduced through language.

[1] **Peer pedagogy** is a term describing learning approaches where meaning is co-developed in the context of collaborative efforts among students as opposed to top-down delivery by the instructor to the student. It relies on the suggestion that the student teaches and supports fellow students through mutual participation in actual tasks (Boud, 2001; Corneli & Danoff, 2011).

0.6 Methodology Overview

This thesis employs a mixed method of discourse analysis, blending computational text analysis with qualitative interpretation. The corpus comprises 69 Phrack Magazine issues between the years 1985 and 2016, chosen to reflect the evolution of hacker pedagogy under shifting sociopolitical and technological conditions.

Texts were preprocessed for analysis in Voyant Tools, allowing for the determination of lexical frequencies, keyword collocations, and diachronic trends. These computational findings offered points of entry for more interpretive analysis via CDA. Phrack's major genres were selected for analysis, including tutorials, manifestos, editorials, and scene reports; each of which was treated as a pedagogical instrument within the magazine's learning environment.

Throughout the analysis, interpretive sensitivity was aimed at rhetorical devices, indexicality, and how authority, authorship, and expertise are discursively constructed. This enabled macro and micro-level observations on how hacker pedagogy is said and learned.

0.7 Original Contribution

This thesis contributes to four key areas of scholarship by bridging theoretical insight with empirical analysis.

- Within informal learning theory, it contradicts narrow views of education as institutional and neutral by proving that informal learning can be cyclical, organized, and ideologically loaded. This gives education researchers and theorists of learning a more nuanced view of how learning is lived in non-institutional and subcultures.
- Within hacker and digital subculture studies, it offers a rich analysis of how subcultural identity is negotiated through pedagogic discourse, illustrating the interdependence of learning affiliation, resistance, and boundary work on culture. Its value lies at the level of researchers of digital culture, hacker research, and subcultural sociology who would be interested to know about how discourse mediates learning and group formation.
- In e-pedagogy and educational technology, it underscores the potential of decentralized peer-to-peer learning models to enable identity construction, critical thinking, and ethical self-reflection outside of institutional settings. This is relevant for researchers, instructional designers, and learning technologists interested in non-traditional learning environments and learner autonomy.
- In discourse analysis, it is a methodology that integrates computational methods with interpretive critique for analysis of corpora of texts of culturally diverse origin on large scales. It provides discourse analysts and digital humanities scholars with a model for mixed-method text analysis balancing scale against contextual sensitivity.

- Historically, it forms a part of a revisionist history of the ways in which learning and expertise have shifted away from formal institutions in the late 20th and early 21st centuries. Through an analysis of Phrack Magazine as a pedagogical mode, it charts how hacker cultures developed and disseminated knowledge across shifting technological environments, from BBS networks to early web sites, while subverting dominant educational models. This is an orientation that benefits education historians, media historians, and scholars of digital knowledge culture who are concerned about how informal learning infrastructures evolve.

0.8 Limitations and Ethical Considerations

While the analysis offers a comprehensive view of Phrack's discursive learning structures, it is limited by its textual focus. It does not incorporate ethnographic or interactional data nor directly study reader response or communal engagement. In addition, the ethical complexity of hacker subculture, like its contested legality and use of anonymity, requires delicacy in corpus selection and interpretive framing.

All the texts analyzed were in the public domain and were chosen according to explicit selection criteria for learning discourse. No usernames or pseudonyms were tracked beyond their textual role, and no inference was made regarding the offline identity of a particular author.

0.9 Conclusion

Phrack Magazine illustrates how learning takes place in spaces created through countercultural values, peer-to-peer talk, and prolonged textual interaction. Beyond the function of a static repository, Phrack is an active pedagogical infrastructure in which technical knowledge, group membership, and ideological stance are actively cultivated. Placing hacker discourse as a site of informal education invites broader questions about the boundaries of acceptable learning, including who sets them and whose knowledge counts as authentic.

This breakdown aligns with the existing understanding of Digital Informal Learning, which stresses learner independence and social construction of knowledge by engaging participants within online communities. This accredits that informal learning can take place outside institutional controls based on personal interest, social access, and collaborative interaction.

The use of Critical Discourse Analysis allowed close attention to the way language mediates inclusion, authority, and identity. Phrack is, in this sense, more than an information source. It is a space that one reads and traverses, and one that is pedagogically organized through its discursive habitus.

Chapter 1: Informal Learning: Theoretical Foundations and Digital Contexts

1.1 What is Informal Learning?

Scholars prefer to make a distinction between formal and informal learning, although the terms are typically interchanged in descriptive accounts. Colley, Hodkinson, and Malcolm (2003) assert that informal learning describes organized learning activity occurring outside the formal education system, often involving a systematic but voluntary engagement, such as workshops in the community or programs aimed at adult education. Informal learning, then, is usually unstructured,

unplanned, and embedded within the activity of daily life, e.g., learning by experience, social contact, or engaging in work. Both occur outside institutional environments, but informal learning still encompasses some degree of purposeful planning and curriculum, whereas informal learning occurs on more of a routine basis through participation in social and cultural practice. To provide some foundation for an understanding of these, there must be some foundation in general theories of learning.

Behaviorist, cognitive, and sociocultural theorists like Skinner, Piaget, Bruner, and Vygotsky have developed various models to explain the process of learning. Cognitive theorists, like Piaget (1970) and Bruner (1960), concentrated on internal mental processes like reasoning, memory, and problem-solving. Building from and moving away from these positions, sociocultural theorists, Vygotsky (1934/1986) and Rogoff (1990), focused on the fundamentally social nature of learning. From this perspective, knowledge is constructed through collaboration by way of interaction and firmly embedded in cultural tools, practices, and social relations. The theory tradition, and especially the sociocultural tradition, plays a prominent role in the study of informal and formal learning. Sociocultural theories are thus an important lens to examine the workings of these various modes of learning in varying settings and population groups. These theoretical shifts, especially those advocated by sociocultural theorists like Vygotsky (1934/1986) and Rogoff (1990), and by researchers like Colley, Hodkinson, and Malcolm (2003) and Livingstone (2001) have prompted scholars to challenge not only mechanisms of learning but the conditions and environments in which learning occurs.

To further understand the organization of informal learning in relation to other extra-institutional forms, it is helpful to make sharp distinctions about how it differs from formal education.

Though the terms 'informal' and 'formal' learning are often used interchangeably, they refer to distinct learning modalities. Informal learning is typically unstructured, unintentional, and embedded in everyday life, whereas formal learning refers to organized educational activities that occur outside the formal school system but still involve planned instruction, objectives, or certification. Examples include adult literacy programs, community workshops, or vocational training (McGivney, 1999; Ward, 1998). At the same time, scholars have also differentiated learning more broadly into planned and unplanned, or conscious and unconscious forms (Reber, 1967; Krashen, 1981), further complicating attempts to draw neat boundaries. Still, recognizing these distinctions is essential for understanding the flexible and dynamic landscape of informal education.

This particular difference that is a subject of debate is the one between formal and informal learning. While useful, others contend that this dichotomous model underestimates the dynamic, context-dependent, and multifaceted character of real learning experiences (Colley, Hodkinson, & Malcolm, 2003; Livingstone, 2001). Informal learning is an umbrella term for a broad variety of learning processes that occur outside the institutionalized structure of schooling. Typically, both spontaneous, voluntariness and embedded in social contexts, informal learning is strikingly

different from the curriculum-driven, formal character of schooling. Nevertheless, the difference between these learning modes is not always clear.

Callanan et al. (2011) argue that informal learning is better explained through internal characteristics, such as its focus on the learner, its embedding in daily life, and its disconnection from formal evaluation, rather than by the physical environments in which it occurs. Learning often occurs in ways that transcend traditional descriptions, especially if one reflects on the path of human development. Young children, for example, acquire language, socialization, and fundamental thinking well before they ever attend school. Through discussion, observation, problem-solving, and engagement in worthwhile activities, adults continue to learn throughout their lives, in their families, workplaces, and communities. These examples challenge the perception that learning takes place only in institutions. Just as formal learning can take place outside traditional classroom settings, informal learning can take place outside formal learning settings.

Though the use of the words "informal" and "formal" learning is usually interchangeable, these words imply two different kinds of learning. Informal learning is largely unstructured, unintentional, and incorporated into the way of living, e.g., socialization, solution-finding, or observation. Informal learning is subsequently formally structured education activities operated outside of the formal school system but still entail some organized elements, e.g., objectives, teacher or facilitators, or certification. Examples include community workshops, vocational training, and adult education. As McGivney (1999) and Ward (1998) note, informal learning is more likely to address specific community or learner needs, prioritizing flexibility, accessibility, and relevance. Colley, Hodkinson, and Malcolm (2003) caution that too great a differentiation between forms of learning can obscure the dynamic, intersecting nature of lived learning experience; but for analytical purposes, it remains useful to make this distinction.

Informal learning, according to Callanan et al. (2011), is typically characterized by five features: (1) it is non-didactic, (2) socially interactive, (3) embedded in activity that is meaningful, (4) triggered by the learner's interests or needs, and (5) typically free of formal evaluation. This view is consistent with Vygotsky's (1934) crucial distinction between everyday concepts and scientific concepts. Everyday concepts arise spontaneously from participation in everyday activities, including play, social communication, and problem-solving activities. Everyday concepts are directly linked to personal experiences and culturally mediated. Scientific concepts, in contrast, are likely to be learned through formal educational processes; they are explicitly taught, abstract in nature, and arranged in systematic structures. Most importantly, Vygotsky did not see these kinds of knowledge as distinct or hierarchically arranged, but as interdependent.

He maintained that although common experience furnishes the experiential foundation of abstract thinking, scientific ideas are accountable for organizing, enriching, and expanding daily knowledge. Thus, the learning process is optimized when both these settings meet and mutually influence one another. This synthesis confronts strict differences between formal and informal

learning and highlights their dynamic interrelatedness. At the heart of this vision is the belief that collaboration and mutual support are not only sound teaching methods but are also historically and culturally embedded in the learning process. The enduring power of collaborative learning is demonstrated in ancient traditions, such as the Jewish Talmudic method of paired study, which emphasizes the value of deep, dialogical engagement. Likewise, classical theorists like Quintilian and Comenius promoted peer learning among classmates, arguing that teaching others improves one's own understanding. The Bible also supports this principle; for instance: "Two are better than one… For if they fall, one will lift up the other… A threefold cord is not quickly broken. (Ecclesiastes 4:9–12, New International Version)." These time-tested customs illustrate that cooperative learning is not a recent practice in education but a fundamental aspect of human education throughout various cultures and time (Johnson, Johnson, & Smith, 1991). This mutual and collaborative learning process is characteristically expressed in contemporary formal and informal learning settings. Community-based learning initiatives which is typically classified as informal education, emphasize values such as contextual relevance, accessibility, and learner autonomy, as highlighted by McGivney (1999) and Ward (1998) in their work on adult and community education.

These activities are typically negotiated with students and tailored to meet specific community needs, offering flexible and socially embedded learning experiences. Voluntary organizations frequently embrace hybrid models that merge formal education components, such as certification and workshops, with informal components like mentoring, peer learning, and on-the-job trainings. Such frameworks have proven particularly valuable for marginalized groups, like women returners to education and ethnic minorities learning in the context of culturally appropriate projects (Ward, 1998). In each of these instances, the learning experience is not isolated but a collective endeavor that enhances knowledge as well as social capital.

The Common School Movement focused on the implementation of peer instruction and cooperative learning practices. These programs stress the importance of mutual instruction and social interaction as critical elements not only in informal educational programs but also in formal educational systems (Johnson et al., 1991). Together, these examples gesture toward a broad agreement: learning of any kind—formal or informal—is more than mere individual cognitive exertion and is instead a social, contextual, collaborative phenomenon—a perspective that is strongly coupled with sociocultural theory of learning. The significance of informal learning is particularly pronounced in early childhood since basic skills and knowledge arrive a long time ahead of when formal schooling even starts. From a sociocultural viewpoint, learning is socially constructed and mediated by culture through the tools, the collective practices, and the interpersonal relations. Dewey (1916) stressed the importance of learning through experience and warned against alienating education from daily life. Similarly, as noted above, Vygotsky's (1934) differentiation between scientific concepts and everyday concepts clarifies how children's every day, lived experiences provide the intellectual and cultural basis for subsequent formal education.

Children's language learning is a compelling example: through immersed, interactive, and socially situated interaction with caregivers and peers, children learn to express complex meanings without didactic lessons or examinations. Meanwhile, social-emotional competencies arise through implicit exchanges in the home. Several studies (e.g., Callanan et al., 2011; Del Prette & Del Prette, 2021) show that parents' use of emotionally expressive language is positively associated with children's emotional competence. These modes of learning emerge from shared narrative construction, observational practices, and reciprocal social engagement, elements that map onto the defining characteristics of informal learning. Early children begin to learn about scientific concepts prior to the onset of formal education. Informal settings, like family discussions, exposure to various media, and trips to science museums, promote curiosity, causal reasoning, and hands-on learning. Children's incessant "why" questions demonstrate an innate inclination to question and show how informal learning settings provide an early foundation for subsequent scientific knowledge (Callanan et al., 2011).

The importance of informal learning is not limited to childhood but goes on throughout adulthood and is attained in everyday routines, hobbies, community involvement, and occupational activities. Adult learners, too, learn through observing, engaging, and taking part in significant activities, as do children. In the workplace, this is often in tandem with Rogoff's (1990) "intent participation," where learning is supported through structured practice and social interaction rather than typical methods of education.

Adult informal learning facilitates not just personal development but also more general forms of advancement. It can aid in enhanced self-esteem, independence, and literacy (personal development); reinforce social connections and civic engagement (social development); improve employability and job mobility and therefore, economic independency; and, for some, prompt re-entry into formal learning (FEDA, 1997). As FEDA (1997) and Maxwell (2023) point out, such dimensions (personal, social, economic, and academic) are not distinct but interrelated, often overlapping each other in the learner's quotidian experience shaped by the learner's environment and socio-institutional variables. Informal learning across a person's lifespan facilitates the development of self-efficacy and lifelong agency which are the central notions in education and civic affairs.

In brief, informal learning defies categorical fixation, encompassing a wide range of socially situated, context-dependent, and learner-directed practices. The distinction between informal and formal learning lies less in setting, but in process, motivation, and relational dynamics. Informal learning typically occurs outside institutional boundaries; but no less significant in building knowledge, identity, and participation across the life span. At the core of these processes is a conception of learning agency, or the ability of the individual to initiate, direct, and maintain their own learning in manners that are relevant to their needs, contexts, and aspirations. Agency is far from a strictly individual characteristic; rather, it is a product of the interplay among internal motivations, cultural resources, and social support networks. As digital technologies pervade more aspects of daily life, they not only increase but also alter the terms upon which this agency is able

to be realized. The next section discusses how these technological developments bring about Digital Informal Learning (DIL), a model of learning that is characterized by its emphasis on personalization, connectivity, and participation in decentralized, digitally mediated contexts.

1.2 Informal Learning in the Digital Age

It has been claimed that when Einstein was requested to provide his telephone number at some point, he allegedly looked at a directory, remarking something to the effect that, "Why should I memorize it when I can look it up?" (Dron & Anderson, 2023, p. 79). Now our equivalent of Einstein's telephone directory is the enormous digital universe, far superior. Almost any snippet of fact is available in seconds.

Concurrently, the digital era has enabled us to interact with millions of others in various spaces and platforms. Unlike in the past, when human beings used to interact with one environment at a time, today we live many at once, switching from one to another easily. And most importantly, these environments are fluidly co-created and not just experienced. Our lives are increasingly traceable, recorded, and persist long-term in digital formats (Dron & Anderson, 2023). Consequently, learning today is significantly different from the traditional models.

It is no longer necessary to internalize all information; rather, the key is knowing how to find it and how to apply it effectively. The volume of online materials and human interaction has produced new learning opportunities along with growing needs for it (Dron & Anderson, 2023). Contrary to the early idealistic expectations of technology offering further leisure, nowadays we are handling growing cognitive and social requirements. As the Red Queen in Through the Looking Glass intimated, we must "run as fast as we can just to stay in place" and run even faster to move ahead (Carroll, 1871, quoted in Dron & Anderson, 2023).

It is the same with technological innovation, each new technology paves the way for further innovations by releasing new niches (Arthur, 2009; Kauffman, 2019). To keep up, learners today need not only to ingest more information but also to learn new skills and develop motivation. The traditional model of formal learning is increasingly disconnected. Worse still, extrinsic motivators like grades and diplomas can undermine learners' intrinsic motivation (Kohn, 1999; Ryan & Deci, 2017). Informal learning, by contrast, tends to occur organically through casual activity or unforeseen experience. It organically supports autonomy, competence, and relatedness which are essential drivers of intrinsic motivation (Ryan & Deci, 2017).

With the virtual world's convenience to knowledge abundance and social depth, informal learning is an empowering pathway to fostering lifelong learning. But, as Postman (1998) cautioned, technological developments are two-edged. Each innovation that answers a problem always creates new issues. This was what he calls a Faustian bargain.

Digital Informal Learning (DIL) refers to learning and skill acquisition that occurs outside formal education, enabled by the pervasive dissemination of digital technology. DIL is interest-driven, self-directed, and not part of institutionally controlled or mass-produced curricula and gatekeeping (Doleck et al., 2019; Mishra, 2020; Yang, 2020). This closely aligns with the earlier definition of informal learning. It can be achieved on a wide spectrum of digital modes, from mobile applications to open educational resources (OERs), MOOCs[2], webinars, blogs, podcasts, social media activity, and video websites like YouTube. Notably, DIL is learner-focused meaning that it allows one to determine the pace, focus on personally relevant goals, and approach knowledge domains in a responsive trend (Reinders & Benson, 2017; Sackey et al., 2015). One of the most distinctive characteristics of DIL is its reliance on Personal Learning Environments (PLEs), which are user-generated aggregates of digital content, resources, and social networks tailored to an individual learner's specific needs and interests. PLEs contrast with institutionally controlled platforms because they place highest value on user agency, modularity, and context sensitivity (Dron & Anderson, 2014). PIEs are the assemblages of digital tools and communities that learners configure to meet their individual needs and goals. These environments may include platforms like Reddit for crowd-sourced guidance, blogs for reflective writing, YouTube for instructional content, and podcasts that provide broader context (Dron & Anderson, 2014; Kumar & Gruzd, 2019). These spaces not only facilitate adaptable content consumption and creation but also facilitate personalization of the learning paths for learners in formal and informal learning environments. Inseparably bound up with PLEs is the idea of digital literacies that is itself in transition. They are not merely technical competencies to use digital technology, but critical competencies to evaluate, curate, remix, and ethically engage with digital material. Digital literacies are complex in structure, reflecting the diverse practice required to flourish in networked, participatory digital cultures (Buckingham, 1993; Coiro, 2008).

The online space, therefore, is more than just a means of content provision; it is used as a material for learning, thinking, experimenting, and creating. This conception is very much aligned with more general constructivist views, where learning is conceived of in terms of active meaning constructors, especially when learning in appropriate participatory environments (Vygotsky 1934/1986; Bruner, 1960). The autonomy and personalization of Digital Informal Learning (DIL) align with the principles of self-determination theory, which posits that autonomy, competence, and relatedness are fundamental psychological needs for intrinsic motivation and deep learning (Ryan & Deci, 2017).

This learner-driven structure is consistent with constructivist models of learning, in which knowledge is actively built through exploration, engagement, and experimentation (Bruner, 1960; Vygotsky, 1934/1986). This learner agency is also expressed in the decentered form of most digital platforms, with learners themselves shaping patterns of engagement and content.

[2] **MOOCs** (Massive Open Online Courses) are a new type of online course where anyone, anywhere, can participate via video lectures, computer graded assignments, and discussion forums.

As observed by Dron and Anderson (2014) and Kumar and Gruzd (2019), students often blend formal and informal materials in these self-curated spaces, creating personalized learning pathways that draw from diverse sources and interaction patterns. These learner-generated trajectories reflect a broader epistemological and cultural shift toward decentralization and participatory learning (Jenkins et al., 2006; Dron & Anderson, 2014).

This evolution in learning practices is unfolding in tandem with the emergence of new literacies. Since the 1990s, concepts such as computer literacy, media literacy, information literacy, and digital literacy have challenged traditional definitions of what it means to be "literate" in the digital age (Buckingham, 1993; Coiro, 2008). These literacies now extend beyond technical proficiency to include critical evaluation, remixing, and ethical engagement with digital content.

Although these skills are generally acquired in virtual environments, DIL is not limited to the virtual world. Formal and informal physical environments also offer rich, unplanned learning experiences that reinforce and enrich digital experiences. Institutional Informal Learning Places (IILPs= including zoos, aquariums, botanical gardens, museums, and science centers that offer experiential and exploratory learning opportunities. In these spaces, learners interact directly with materials, phenomena, and artifacts in ways that are often student-directed and shaped by local contexts and interests (Falk & Dierking, 2018; Leinhardt et al., 2002). Increasingly, such physical experiences are augmented by digital tools such as mobile apps, augmented reality, and interactive kiosks that offer multimodal representations, immediate feedback, and personalized guidance (Yoon et al., 2012; Wojciechowski & Cellary, 2013). These hybrid environments further blur the boundary between physical and digital learning spaces, reinforcing the blended nature of contemporary DIL (Radu, 2014; Degner et al., 2023).

Another key domain of DIL is open to online communities grounded in collaboration and collective knowledge production. The open-source software (OSS) movement is a leading example of this model, where contributors participate in iterative problem-solving, peer feedback, and mentorship without relying on formal institutions or credentials (Koh et al., 2007). Learning is through doing; therefore, these groups are excellent examples of what Lave and Wenger (1991) describe as "communities of practice," in which identity and knowledge are constructed through legitimate peripheral participation. OSS developers are not passive consumers of information but co-producers of software systems and digital artifacts (Petr & Walter, 2005).

Beyond technical skills, participation in these environments supports the development of broader 21st-century competencies: creativity, systems thinking, adaptability, self-regulation, and advanced digital literacies (Arthur, 2009; Kauffman, 2019). Platforms such as MOOCs, YouTube channels like *CrashCourse*, or subreddits such as *r/explainlikeim5* provide learners with opportunities to grapple with complex concepts, synthesize disparate information sources, and apply knowledge in meaningful ways. These environments cultivate critical evaluation, iterative learning, and comfort with uncertainty.

Ultimately, Digital Informal Learning represents a paradigm shift in how knowledge is accessed, constructed, and shared. It is not institutionally mandated, yet it is self-directed, socially networked, and intellectually rigorous. These are the qualities that make it as challenging and credible as traditional formal learning.

Building on constructivist, sociocultural learning theory, and motivational theory, DIL puts learner agency and the affordances of digital environments at the heart of its method for promoting transformative change. Thus, it is a central and emerging feature of twenty-first-century learning that helps learners to participate in lifelong, self-directed learning in digital and physical spaces across their lives. DIL is not an activity in a vacuum but is inextricably embedded in sociotechnical cultures. One of the most powerful and evocative examples of DIL in practice is that of the maker and hacker communities, where learning in everyday life is practiced through experimentation, co-working, and collective problem-solving. These environments are not only technologically rich but also unique values and identities.

One of the most vivid examples of Digital Informal Learning in action is hacker and maker communities which grassroots, peer-to-peer networks of people where knowledge emerges through shared work, hands-on exploration, and collective problem-solving. The next section explores how such environments embody and propagate the fundamental principles of DIL.

1.3 Hacker Culture as a Learning Environment

Hacker and maker communities (HMCs) are fluid, peer-supported settings where digital informal learning (DIL) happens through experimentation, exploration, and co-constructed knowledge. Such places, from makerspaces to virtual code repositories, are premised on a culture of openness, autonomy, and participatory learning. Hacker ethic, originally associated with computer programming, is a more generic inclination, the propensity to alter, repurpose, or reassemble systems beyond their original functions (Lunceford, 2006). This sensibility makes use of the bricolage concept, by which innovation does not result from linear instruction but through creative recombination of resources available (Turkle & Papert, 1990). These environments exist as communities of practice (Lave & Wenger, 1991), wherein learning is situated, contextual, and social in character within shared action.

Hacker culture first emerged in the 1960s at institutions such as MIT, where members of the Tech Model Railroad Club would do "hacks" on model train systems in ways that were creative and innovative (Levy, 2010). These activities became the basis for a global culture of creative problem-solving and ludic transgression of formal authority. With the growth of computing technology being more widely available, the hacker culture expanded into the maker movement, resulting in the creation of makerspaces that were community-operated workshops with shared equipment, digital fabrication tools, and open workspaces (Maxigas, 2012).

An inherent element of HMCs is that they emphasize hands-on, iterative practice rather than passive learning. Learning is infused into real tasks through which individuals learn by doing, often coordinated by peers or mentors rather than formal teachers. This approach is consistent with Dewey's (1938) "learning through doing" as well as Ingold's (2018) notion of knowledge as situational in material engagement. These communities promote self-directed, non-linear learning in which failure is a natural and often necessary aspect of the process.

Events such as hackathons are merely one instance of the collective, informal nature of HMCs. Hackathons are deadline-oriented, often problem-based gatherings in which small, multidisciplinary teams craft working prototypes over the course of hours or days. Although begun in the realm of technology, hackathons have expanded into such domains as education, civic technology, and the arts (Briscoe & Mulligan, 2014; Trainer & Herbsleb, 2014). These experiences are experiential, socially situated learning environments in which people learn through practice, receive feedback, and engage in peer validation. They are often working on open-source projects or addressing community-driven issues.

HMCs not only support technical skills, but also provide sites for identity formation, political empowerment, and social solidarity. Feminist hacker spaces, for example, deliberately undermine mainstream norms of exclusion through welcoming, care-based communities (Fox et al., 2015; Toombs et al., 2014). Other communities put environmental sustainability as repair work on their agenda or embrace "critical making"–a practice that situates DIY production both as technology and politics (Grimme et al., 2014; Jackson, 2014). These dimensions describe how informal learning in these environments most often exceeds the development of skills, often involved in deeper ethical and cultural practices.

HMC pedagogical distinctiveness is also supported by differences from conventional education. Formal institutions prescript hierarchies of curriculum, temporalized examinations, and reward systems tied to credentials. Hackspace, on the other hand, supports "sideways" learning trajectories (Horton, 2006) where contributors may determine what to learn, how to learn it, and when without fear of failure or penalty. GitHub is an example in which the contributors share work-in-progress code, offer iterative feedback, and build knowledge through practice, not certification.

This contrast is especially revealing when set against MOOCs. Although digital scalability and widespread access are provided by platforms like Coursera and edX, they often replicate lecture-based, one-way pedagogies. Students in HMCs frequently describe MOOCs as being unsocial and lacking hands-on practice (Lewin, 2012). As a hacker quipped, "What does that have to do with our study of hackerspaces?" a remark underlining the discrepancy between institutional digital learning and rich, peer-influenced learning cultures.

This chapter has traced the evolution of informal learning from its constructivist and sociocultural foundations to its contemporary expressions in digitally mediated and hybrid spaces. Informal learning has gained significance in the digital era. Digital Informal Learning (DIL) is a case in point, offering decentralized, self-determined learning possibilities through personal learning

environments, emergent literacies, and participatory digital cultures. Among the strongest instances of DIL are hacker and maker cultures, which best represent experimentation, self-taught learning, and peer-to-peer co-creation of knowledge. Together, these perspectives confirm that informal learning is not a secondary supplement to formal education, but a central process through which engagement, identity, and knowledge are created in the twenty-first century. Referring to this theory, the subsequent chapter looks to " Phrack Magazine " which is arguably the most effective unofficial learning environment among hackers.

Chapter 2: Phrack Magazine and the Shaping of Hacker Collective Identity

2.1 The Emergence of Hacker Identity

The "computer underground" is a robust, emergent subculture that is shaped by technological progress, institutional resistance, and cultural evolution (Meyer & Thomas, 1990; Holt, 2007). It is an identity-centered, multi-dimensional society constructed around technical expertise, common values, information networks linked, socio-technical hierarchies, technical vocabularies, and culturally inscribed meanings (Meyer & Thomas, 1990). The image of the "hacker," a term that conjures up a broad and frequently contradictory set of meanings, lies at the symbolic and practical core of it. Frequently misrepresented or hyped by dominant cultural discourse, hackers have been variously read as antisocial technophiles, subversive youth heroes, political activists, cybercrooks, or visionary technologists (Jordan & Taylor, 1998). To understand this figure, one needs to consider historical, cultural, and technological developments that have shaped hacker identity through time.

After World War II, the United States also became a hub for technology, leading the way for what would eventually form the digital age. The world telephone system, before computer networks and the internet, was the first true time-based global communications network that was used commercially. With its complex organization, it attracted a crowd of technically savvy individuals who desired to learn and master it. These initial hobbyists started to experiment with the system using methods like whistling phone audio frequencies, constructing custom devices known as "boxes," and employing technical workarounds like loop-around's. All these methods were put together under the single name phone phreaking (Sterling, 1992, p. 18; Levy, 1984, p. 181). Phone phreaking, a term for telephone hardware tampering to access to conduct an illegal experiment. While the majority of phreakers were driven only by curiosity and a passion to know more, others crossed into criminal domains, prompting firms like Ma Bell and AT&T, often in collaboration with law enforcement. to track down and prosecute offenders (Dilipraj, 2016).

By the 1970s, the focus of technology enthusiasts shifted from the complexities of telephone networks to the new field of personal computing. Although computer systems were not that user-friendly in their early years and required a significant amount of technical expertise to be used, their vast innovation potential attracted more and more curious minds, including amateur tinkerers, electronics hobbyists, and self-taught technologists. These individuals who usually referred to as "computer freaks began gathering informally in non-institutional settings such as garages, basements, and community centers to explore, modify, and push the boundaries of the limited software and hardware available to them. These were less technical workshops than spaces for cultural exchange and collaboration wherein information was shared freely, collaborative skills developed, and an ethos of experimentation and play was built.

The key event in the very early stage in the creation of this community was the establishment of the Home Brew Computer Club on March 5, 1975, in Silicon Valley (Turner, 2006, p. 109; Markoff, 2005, p. 95). This amateur community created a public space for enthusiasts to display functional prototypes, trade hand-made circuit boards, trade source code, and bounce ideas about how to push the functional capabilities of personal computers. The club rapidly became both symbolic and operational center for this nascent movement, and it ushered in an entire generation of computer enthusiasts and eventually gave rise to some of the most influential figures and companies of the digital revolution. As more individuals joined the movement and identified themselves with its objectives and practices, there developed a shared culture, a culture centered on open sharing, reverse engineering, and system-scale experimentation. The technical activity of discovering, changing, and reconfiguring computer systems acquired a new cultural significance under the label "hacking, a name that came to include not just a set of methods, but a general attitude of curiosity, disrespect, and creative troubleshooting. Hacking at this formative phase was not driven by criminal or malicious intent, but rather indicative of a subversive engagement with systems and a pursuit of technical mastery. (Dilipraj, 2016).

The origins of this hacker culture exist even further back in elite universities, preeminently the Massachusetts Institute of Technology (MIT), where the first and most influential computer pioneers began their tinkering in the 1960s. In these university research labs and research environments, hacking developed not as a revolutionary or illicit activity, but as technical craftsmanship and intellectual liberty (Levy, 1984, pp. 23–25). Steven Levy chronicled in great detail this initial hacker culture, noting its core principles of openness, sharing of information, technical superiority, and anti-authoritarianism. Early hackers believed that access to computer resources should be unbounded and that information should be shared freely rather than being plundered. Richard Stallman (2001), one of the stars of this tradition, described how early practitioners "had little respect for the silly rules that administrators like to impose, so they sought ways around" (p. 20). This attitude was part of a broader disdain for hierarchical power and bureaucratic restrictions, which created a moral system based on cleverness, openness, and self-determination. By the late 1970s, this set of beliefs and practices had coalesced into a visible, long-lasting subculture, popularly referred to as "Hackers' culture, that was defined by its commitment

to learning, creativity, and pursuit of technical excellence as a form of resistance to technological and institutional constraints (Levy, 1984; Stallman, 2001).

As computer technology spread and became more widespread in the 1980s, and as modems made remote communication between isolated computers possible, a new digital ecology started to develop. Bulletin Board Systems (BBSes) were used as decentralized platforms for software, documentation, and technical information exchange. These platforms offered users message boards involvement, file sharing, and cooperative involvement from distant places as a new, open area for digital communication independent of traditional academic or governmental institutions. BBSes quickly became the focal point of computer underground development, serving as important crossroads where a new generation of hackers established identities, reputations, and memberships (Sterling, 1992, pp. 52–54; Meyer & Thomas, 1990, p. 47). They were not just simple communication tools; they were a social infrastructure by which people constructed pseudonymous selves, established status, and built recognizable online communities around common interests and technical skill.

The emergence of BBSes was a historic watershed moment in hacker history. Information previously maintained within the walls of university labs or research facilities was now disseminated broadly across decentralized, grass-roots electronic networks. Increased access to the platforms made hacker culture more stratified. New users created more competition for virtual space and resulted in the formation of status hierarchies and insider dynamics. Senior, more experienced hackers began to distinguish themselves from what they derisively called "lamerz", derogatory terms for novice or theatrical users who lacked real technical experience. In this competitive environment, the possession of certain digital artifacts—like cracked software, intercepted communications, or esoteric system vulnerabilities—were forms of subcultural capital. These artifacts were not only tools but identifiers of technical proficiency, inside information, and credibility among hackers (Lunceford, 2009).

In the beginning, the term "hacker" referred to extremely competent programmers using nonconventional techniques to solve difficult problems. Sterling (1992) referred to this as "the free-wheeling intellectual exploration of the highest and deepest potential of computer systems" (p. 53). Levy (1984) referred to this subculture as utopian and technically heroic. Hacker subcultures have always been heterogeneously stratified. Their action varies from improvisational play to purposeful intrusion into fundamental systems (Thomas, 2005; Nissenbaum, 2004). Despite divergent intentions, there is a shared identity formed on the basis of common norms, language, and constant negotiation with outside social expectations (Voiskounsky & Smyslova, 2003; Jordan & Taylor, 1998).

Although the 1980s may be termed the "Golden Age" of hacking, this is best applied to the in-house creation and ramping up of enthusiasm within hacker culture itself; much more so than with the prestige afforded hackers by the general populace. The big change in public perception occurred in 1983 with the film Wargames, Wargames (1983), a widely viewed Hollywood film,

which portrayed a teenage computer genius who inadvertently breaks into an American nuclear missile defense system and nearly unleashes a global disaster. Although earlier references to hackers had occasionally appeared in specialty press and academic literature, none of those had penetrated so far and had had such an impact on popular culture as this on-screen portrayal. The theatrical rendition and visual representation of the film strongly influenced mass spectators, and they became apprehensive and suspicious of hackers on a mass level, particularly individuals who possessed limited technological awareness (Dilipraj, 2016). As practices by hackers advanced and became increasingly sophisticated, a gap emerged between the subculture's inherent valueslike centered upon creativity, discovery, and information sharingand the way the media and authorities portrayed hackers increasingly. Mainstream images subsequently shifted to hysteria, as politically driven reports (Thomas, 1990) and Hollywoodized screen depictions stoked a burgeoning moral panic. These images assisted in the development of a building moral panic where hackers were constructed as destabilizing forces or future offenders. Law enforcement responded with renewed focus, in the wake of high-profile probes and prosecutions that further entrenched the profile of hackers as evil deviants (Thomas, 2005).

Even before that, philosopher Joseph Weizenbaum (1976) warned that programmers wield creative influence on digital law with the danger of self-corrupting omnipotence. He portrayed hackers as obsessive and isolated; their work sometimes driven more by intellectual thrill than by usefulness.

However, not all perspectives aligned with the criminalizing narratives, even after the movie wargame as Sherry Turkle (1984) later offered a sociological explanation of hacking that introduced hackers as not technologists but identity seekers in symbolic and aesthetic realms. In contrast to engineers who used computers as tools, hackers perceived them as spaces to discover, perform rituals, and be themselves. Their fascination with "closed systems" carried from code to parallel hobbies such as lock-picking and telephone phreaking; tangible manifestations of an abstract compulsion to tear down borders and unwind systems in and of themselves (Levy, 1984; Turkle, 1984). This cultural and psychological characterization of hacking persisted well into the 1980s, even as broader technological developments started to redefine its public image (Thomas, 2002).

Over time, though, the word "hacker" accumulated more pejorative meanings. It became associated with unauthorized access, cyber-vandalism, national security threats, and intellectual property theft (Chandler, 1996; Halbert, 1997). Under this climate of criminalization, Skibell (2002) contends that a great deal of the fear of hackers was constructed more out of cultural myth than empirical fact. In his view, the majority of those termed hackers lacked the intent or technical skill to bring major damage but engaged in harmless online exploration.

Hackers have also offered their own counter-definitions of identity and purpose. According to Howard Becker (1963), deviance is a term that other people have applied to an act rather than an intrinsic quality of that act. The way society responds is what counts. The person to whom the term "deviant" has been successfully applied is the deviant. Crucially, individuals respond by

internalizing and acting out these identities. Because of this, deviance is not a fixed concept but rather a social process. Subcultures frequently accept deviance is not an intrinsic quality of any behavior but is instead constructed via social processes of labeling. Individuals who are characterized as "outsiders" tend to internalize and strategically perform their identities, marking themselves deliberately against mainstream norms and establishing alternative valuation systems. In the case of hackers, this is not stigma rejection; rather, it is a conscious recasting of self as technologists, innovators, critics, and architects of the digital world. Their actions represent an ongoing endeavor not only to overstep institutional boundaries but to reshape the cultural narratives to which those boundaries are based and maintained.

This is true that hackers have never been homogeneous. Rather, they comprise a highly differentiated and internally pluralized group. Their reasons and actions range from playful improvisation and curiosity testing to sophisticated, intentional intervention in complex systems (Thomas, 2005; Nissenbaum, 2004). While often autonomous in their action, hackers share a collective identity structured by subcultural norms, distinctive forms of technological practice, and ongoing negotiation with the norms of society and state response (Jordan & Taylor, 1998). This identity is neither fixed but fluid; growing out of intersecting layers of technical expertise, behavioural practice, and domains of specialist knowledge (Voiskounsky & Smyslova, 2003).

Despite these evolving inside and outside representations, hackers have long been on the vanguard of technological evolution. They played a key role in the creation of the Internet's infrastructure and were at the forefront of the development of the personal computer (Wall, 2001). Chandler (1996, p. 229) suggests that "it might, in fact, even be argued that the personal computer would never have existed had it not been for the computer hacker." The early hacker cultures foresaw computing as an egalitarian force; something that countered institutional gatekeeping and spurred personal empowerment (Levy, 1984; Jordan & Taylor, 2004). Hannemyr (1999) also notes that hackers contributed essential feedback towards creating efficient and inexpensive tools, ranging from text editors to network protocols.

Another important event affected by hacker community in1960s and 70s was the open-source model. The open-source model is based on distributed authorship, openness, and non-proprietary sharing of knowledge (Levy, 1984; Ljungberg, 2000). During the intervening decades, the overlap between open-source values and hacker ethics has helped to salvage the image of the hacker, highlighting its productive, creative, and socially beneficial orientation. Hacker culture's most enduring legacy is perhaps its early influence on the open-source software movement. Derived from the hacker subculture ethos of the 1960s and 70s, the open-source model is based on distributed authorship, openness, and non-proprietary sharing of knowledge (Levy, 1984; Ljungberg, 2000).

Consequently, cybersecurity researchers and professionals sought to draw distinctions between ethical and malicious hacking. Icove, Seger, and VonSorch (1995) proposed a conceptual dichotomy between "hackers," framed as inquisitive and useful, and "crackers," framed as

dangerous and destructive. But this divide easily collapsed under public ignorance, legal ambiguity, and media hyperbole (Raymond, 1996).

In today's globalized world, hacking has become a usual part of modern life that needs to be constantly addressed by nations all over the world. Not confined to specialized subcultures anymore, hackers now operate globally and for a wide variety of reasons; revenge, pranks, terrorism, political and military espionage, money, and ideological hatred (Dilipraj, 2016). Despite its normalization and integration into the digital culture, the hacker as a character remains culturally equivocal, still straddling the position between suspicion and admiration. The figure of the hacker thus continues to exist on the periphery of cultural approval and disapproval.

Earlier sociological studies of stigma, such as Erving Goffman's, were concerned with the ways in which individuals manage identities stigmatized by society as deviant (Kusow, 2004). According to Goffman (1963), people who experience social stigma frequently practice "information control" and "identity management" to circumvent, defy, or subvert prevailing narratives. This is particularly noticeable in the hacker subculture, where status claims and protective mechanisms include the use of pseudonyms, anonymity, and the development of elite technical discourse. To evade detection, hackers frequently "pass" as regular users, as Goffman would say, or, on the other hand, display their identity as "elite" when they are safe within subcultural boundaries, such as Phrack. Goffman, E. (1963)

In the case of hackers, these strategies were coordinated through subcultural media such as Phrack and were not just individuals. Phrack and its contributors chose to embrace the hacker label rather than reject it, turning stigma into a subcultural status where deviance was reframed as innovation, surveillance as persecution, and marginalization as moral superiority. But newer work stresses that stigma is not fixed; rather, it is established, negotiated, and constantly in process (Kusow, 2004). In hacking, this involves more than simply rejecting stigmatizing terms, it is a task of re-signifying one's own identity. Hackers largely re-imagine themselves not as criminals but as technologists, cultural critics, and creative innovators. Following that, they represent what some researchers have called "positive deviants, individuals who's aberrant or rule-breaking behavior ultimately results in social or technological innovation.

2.2 The Discursive Construction of the Hacker in Phrack

Whereas Section 2.1 explored how the hacker ethos emerged from the broader 196 0s counterculture which was marked by anti-authoritarianism, intellectual curiosity, and an ethic of open information. This section traces how that ethos became manifest and concrete through *Phrack* magazine. As both technical manual and cultural manifesto, *Phrack* was the space where hackers disseminated knowledge, forged identity, and contested existing power structures. The hacker movement’s origin in 1960s youth culture, which resisted blind deference to expert authority (Roszak, 1969; Kirkpatrick, 2002), continued to shape the editorial and ethical tone of *Phrack* decades later.

Early hackers possessed a culture that valued relentless commitment to problem-solving. In their midst, recognition entailed not just competence but hours of selfless exertion. Traits such as focused effort, incessant debugging, and overnight programming work was seen as evidence of serious commitment to the hacker culture (Kirkpatrick, 2002, p. 173; Levy, 1984; Turkle, 1984; Weizenbaum, 1976; Chandler, 1996; Weber, 1974). A central value of hacker culture has always been the open exchange of knowledge, often through collaborative and open-source initiatives (Dilipraj, 2016).
The influence of printed and self-published magazines within these communities was profound and foundational in this collaborative learning. These publications played a vital role in educating and inducting newcomers, functioning as informal training grounds for amateur enthusiasts. Many articles were authored by the amateurs themselves, who contributed their technical insights, shared firsthand experiences, recommended further reading, answered queries from less experienced operators, and voiced opinions about proper on-air conduct. Common etiquette guidelines included avoiding excessive use of expressions like "OM," clearly indicating the start and end of transmissions, and refraining from using substandard equipment that could cause disruptive interference. However, contributors also advised against overly rigid enforcement of these norms, suggesting that even disruptive users could become skilled participants if not alienated early on (Mazzini, 2011).

The development of early print amateur radio hobbyist magazines is a model for how Phrack worked. Both were spaces where geographically dispersed individuals could share information, build credibility, and create a shared identity (Mazzini, 2011). These media spaces transformed individual tinkerers into cultural movements which was a phenomenon that Phrack replicated in the digital sphere.

Above all, these periodicals helped create a sense of cohesion among geographically scattered individuals whose interactions were shaped by the variety in their geographical conditions, seasons, and the variable performance of homemade devices. Despite their isolation, readers began to see themselves as part of a larger national and even international network united by shared hobbies, cultural values, and often similar gender and class backgrounds. The ethereal space of radio waves, combined with the tangible, regular arrival of a printed magazine, enabled the emergence of a "virtual" community well before the digital age. This allowed individuals to forge collective identities through a blend of immaterial connection and material media (Mazzini, 2011).

Phrack magazine, which has been in publication for over four decades, has been a primary locus of hacker culture. It is a place for the exchange of knowledge, cultural production, critical analysis of technological systems, and resistance to institutional authority. The magazine has always been about revealing the vulnerability of systems and the requirement for transparency, calling upon readers to question authority and contribute to a general collective of technical knowledge and hacker ethics (Phrack, n.d.).

Dedicated to exposing new ways of exploitation, Phrack featured a wonderful selection of brief, useful articles presenting valuable background material not typically found in formal instruction (Bratus, 2007). Phrack has extensively documented new methods of exploitation spanning decades from stack smashing and return-to-library attacks through integer overflows, format string bugs, heap manipulation, GPS jamming, and arcane hardware errors (Bratus, 2007). Through its tutorials, manifestos, and news announcements, Phrack emerged as an important text within hacker culture that was read far and wide by security experts as well as hackers. The early issues of Phrack were integral to the creation of a hacker identity with both technical sophistication as well as a defiant stance in relation to mainstream authority. According to Edwin Black's "implied auditor" theory, where it is argued that rhetorical texts project an image of their envisioned reader, Phrack spoke to an inquiring, rebellious teenage boy, a boy who was ready to disregard rules and subvert conventions (Black, 1970, as quoted by Lunceford, 2009).

This ideal reader was defined more by attitude than by age, specifically, by a spirit of resistance to societal norms and hierarchical authority" (Thomas, 2002). The content ranged from acetylene balloon bomb instructions to MCI Communications schematics. It was a representative of a culture that blended exploration, destruction, and play. As Paul Taylor (1999) and others have noted, hacker communities have traditionally been coded masculine, in part because the activity of hacking is itself permeated with metaphors of conquest, invasion, and domination. Taylor vividly describes hacking as an act of penetration, contending that successful intrusion is like "penetration and orgasm," and thus situates the practice on the well-known cultural scripts of masculine mastery (p. 34). But even within these gendered frameworks, Phrack also provided space for more complex definitions of hacker motivation and identity.

One of the most influential Phrack pieces was "The Conscience of a Hacker," a manifesto that Loyd Blankenship (better known as The Mentor) wrote within days of his arrest in 1986. There, the hacker is not portrayed as an outlaw, but rather as a brilliant individual disappointed by an education system that stifles instead of inspires. The true "crime" of the hacker, the piece argues, is "curiosity" (Blankenship, 1986, as cited in Lunceford, 2009). This paper quickly became dogma amongst the group, celebrated as a hacker declaration of values, it romanticized intelligence, deviance, and access to information. People such as Chris Goggans (a.k.a. Erik Bloodaxe) and Crimson Death echoed this ethic in later editions of Phrack, stating that a hacker is one "who wishes to learn everything there is to know about the workings of a particular computer system" or one who "likes to test the limits… explore new ground" (Lunceford, 2009, p. 7). These statements put hacking not as nihilistic destruction, but as an intellectual and exploratory endeavor.

By facilitating recursive conversation, Phrack was more than a technical manual. It was a site of cultural reproduction where hacker values, ethics, and identity were continually created and contested. In its editorials and manifestos, the magazine offered an "implied auditor" (Black, 1970, quoted in Lunceford, 2009; typically, an inquiring, nonconformist reader who would challenge authority and investigate systems. This rhetorical move not only confirmed insider identity but also established hacker subjectivity's aspirational nature. It is, as Lunceford (2009) describes, akin

to Habermas's (1989) public sphere, in that rational-critical debate allows for collective identity and political power. Phrack was, in this sense, a digital age in public sphere: a participatory democracy in which technical debate could no longer be separated from ethical concern and subcultural affiliation. Describing themselves, hackers not only described their community but contributed to constituting it.

This discursive foundation enabled Phrack to respond forcefully when hackers were framed as threats to state or corporate security. Operation Sundevil and the arrest of Kevin Mitnick were not merely legal interventions but were pivotal symbolic events that crystallized an even more pronounced politicization of hacker identity. Phrack's responses to these crackdowns, through manifestos, editorial stance, and promoting solidarity, reinterpreted these events as ideological conflicts between the state and an emerging digital counterpublic. By doing so, Phrack tapped into a broader history of techno-political dissent that had always been implicit in hacker culture. This heritage includes groups like the Technological American Party (TAP) in the 1970s, who used phreaking as a form of protest against military-industrial and corporate monopolies, to later organizations like Germany's Chaos Computer Club and America's Cult of the Dead Cow (cDc), who wedded technical proficiency with explicit political criticism (Thomas, 2002; Mazzini, 2023). Along this historical path, Phrack positioned itself not simply as a magazine but as an instrument of cultural resistance; connecting past, present, and future modes of hacker activism.

Such accounts contributed to developing a stronger collective identity defined by freedom of information, anti-corporate resistance, and anti-state surveillance (Thomas, 2005; Lunceford, 2009). In releasing responses to these events, Phrack purposefully built the history of hacker resistance and represented its authors as political agents in a more surveilled digital environment rather than villains. Subsequent parties like Germany's Chaos Computer Club and America's cDc would inherit that tradition of bridging technical interference with political action (Mazzini, 2023). By the late 1990s, Phrack had established itself as "the most influential and visible of the hacker e-zines" (Thomas, 2002, p. 164).

Over the decades, Phrack served two purposes. Firstly, it played a role as an archive of technical information and as a cultural ideal. Secondly, it constructed a hacker identity that was aspirational yet grounded in practice. Phrack was always something more than a magazine, it was more of a virtual meeting place, a guidebook, an archive of the past, and a symbolic beacon. Through its published writing, it not only facilitated hackers to exchange techniques but to narrate, create norms, and construct oppositional selves. As a result, Phrack was not merely a recorder of hacker existence but one of its main shapers, directly shaping its values, trajectories, and collective memory. Phrack's influence, however, spreads beyond the 1990s. Its spirit persists and is still experienced in contemporary hacker gatherings, online forums, and cyberspace, and shapes ongoing battles over access, control, and openness in networked society today.

2.3 Legitimacy, Criminalization, and Reclaiming the Narrative

The basic tenets of hacker ethic, as best laid out by Steven Levy, are openness, decentralization, meritocratic reward, and free exchange of information. From the militant "information wants to be free" to the irreverent slogan "plagiarism saves time" (Branwyn, quoted in Mondo 2000), hackers have espoused a philosophy of discovery, learning through doing, and access without gatekeepers. There is a necessity in this society to provide systems to everyone who wants to learn from them and to evaluate individuals on their technical competence instead of formal qualification, social status, or institutional membership. Computers are tools, but artists' easels on which code is not just a form of knowledge but also a courteous endeavor (Halbert, 1997).

According to commentators like D. Halbert and Anne Wells Branscomb, this virtue increasingly conflicted with the growing interests of state securitization and enterprise in cyberspace. These hackers who previously had long legitimized their endeavors as exploratory or pedagogical were increasingly constructed as menacing criminals (Halbert, 1997). Together, all these events indicate that the hacker persona was never simply a technological or skill-related issue; it was always shaped by cultural discourse, inner story, and outside conflict.

As Phrack and other hacker publications so aptly showed, to be a hacker was not technical expertise but also membership in a self-consciously aware subculture defined against institutional power and mass cultural story. This oppositional identity was not without opposition, however. In reference to the early days of hacker culture, government-funded university laboratories in the 1950s and 1960s were the rich soil from which early hackers grew. They were not today's criminals, but masters' programmers motivated by intrinsic motivations such as curiosity, problem-solving, and an insatiable need to push beyond the original intent of technological systems (Levy, 1984; Stallman, 2001).

Ironically, those same traits that made hackers so valuable to government and industry in the first place, being nonconformist, autonomous, and boundary-breaking, would come to be viewed as threats (Weizenbaum, 1976). By the 1980s, stories had already begun depicting hackers as menacing saboteurs rather than innovative breakthrough artists. Headlines such as "Raising Security Consciousness" and news accounts of teen "electronic escapades" in Milwaukee showed heightened concern over computer break-in and data privacy (Lunceford, 2009). Eric Raymond (1996) observes that at this time, there was a conflation of the line between hackers and "crackers" in mass use and mass media coverage, and security practitioners applying the formerly neutral label "hacker" to cyber thieves and electronic vandals.

This redefinition of hacking revealed a persistent cultural struggle for meaning. As Jon Erickson (2003) argues, hacking must be defined by attitude and motivation, not by legalistic definitions. Hacking is a willingness to rethink, redesign, and remake technological systems. This struggle over hacker identity is enhanced by competing definitions given by different constituencies; like government agencies, the media, corporate information security professionals, and hackers. Definitions are not only descriptive but also political and performative. As has been noted by Yar

(2005) also, it is a deliberate action on the part of institutions to limit a resisting subculture. However, hackers have also tried to redefine and reclaim their identity, occasionally through their own media outlets and forums for so doing (Thomas, 2002; Lunceford, 2009). As mentioned earlier, the most fruitful of such reflection forums is Phrack, an online publication that initially began publication on November 17, 1985, on the Metal Shop Bulletin Board System (BBS) operated by Taran King (Phrack, 1985, as cited in Lunceford, 2009).

While mass representation generally means that all hackers have ulterior motives like stealing secret data or infiltrating systems for economic gain, but the reality is otherwise. Some hackers were engaging in activities that subvert security or violate the law, but many of them were very devoted to improving technological systems and studying their vulnerabilities (Thomas, 2005; Nissenbaum, 2004). They were working primarily on information technology all the time, so it is uncertain what separated security research from illegal probing (Bratus, 2007).

Hackers were therefore difficult to categorically identify. They occupied an intermediary position of public fear and technical promise, social pathology and cultural heroism. Hackers were unique not only because of what they did but also because of their epistemological approach to technology, which was characterized by ongoing inquiry, testing of boundaries, and defiance of institutional restrictions. To illustrate, early groups like the MIT Tech Model Railroad Club embodied "hacker spirit" in repurposing discarded telephone equipment to plan their advanced model train systems; acts of creative engineering new technologically but significantly imbued (Levy, 1984).

While the preceding two subsections have followed hacker identity's cultural construction and internal self-definition, this final subsection talks about outside forces that contested and reorganized it. Even as hackers formally legitimized their aesthetics and ethics through zines like Phrack, they became visible, too. They were also visible to media and legal system attention. By the early 1990s, however, the hacker image had become neither cipher for technical virtuosity nor playful subversion; rather, it was a cultural lightning rod. External pressures from governments, companies, and the mass media began to transmute their anxieties onto the hacker culture, at times drawing it as a public nuisance and a national security threat. The result was a concerted effort at delegitimating and disciplining hackers by surveillance, prosecution, and symbolic scapegoating. The pressures put on the hacker culture produced tensions that resulted in a transformation of the hacker culture, creating new internal dichotomies and survival strategies.

These definitional challenges picked up speed again in the 1990s with a succession of high-profile state crackdowns that redefined the world of hackers. A significant step on the way was Operation Sundevil in May 1990. Sundevil was a nationwide sweep by the U.S. Secret Service comprising dozens of raids, seizures of equipment, and arrests. The raid was typical of a broad trend where state agencies used high-theatrics policing to prosecute, and to declare a new era of online regulation. In Phrack World News, the response was quick and raw: "May 9th and 10th brought two days [that] would be marked in every hacker's history book. the witch hunt currently in progress" (Phrack, 1990, as cited in Lunceford, 2009).

After Sundevil, Phrack began publishing legal self-help information such as "Getting Caught, Legal Procedures" and credit card fraud legislation notices. These developments were seen as the turning point from celebratory information exchange to survival rhetoric (Thomas, 2002)

The arrest and trial of Kevin Mitnick facilitated this rhetorical shift from internal celebration to defensive survivalism. As it has been presented by the federal government and reporters as "America's most wanted computer criminal," Mitnick had already become a target of popular anxieties about cybercrime. But in the culture of hackers, the case was greeted with ambivalence. While Phrack parodied media hyperbole with headlines like "JUMP ON THE MITNICK BANDWAGON!"; others, conversely, rationalized his actions as not wicked but rather ones fuelled by cerebral curiosity (Lunceford, 2009; Hafner & Markoff, 1995). In his own biography, Mitnick asserted, "I broke the rules, but I never broke the law," and continued to state, "The government wanted to make an example out of me. I was the 'poster boy' for cybercrime not because of what I'd done, but because of what I could do" (Mitnick, 2011).

His own account verifies the extent to which legitimacy in the world of hackers was not action, but perception, which was decided by institutional interests, not empirical harm. The law responded back at hacking by reinscribing the body back into a space once utopian as disembodied and anonymous, relates Thomas (1998). Mitnick's Technologically inclined appearance, non-normative physical traits, and socially atypical behavior were incorporated into the deviance narrative of the media, illustrating how the hacker was both constructed as a computer risk and constructed as a socially deviant self. The police shifted away from proving cases of entries into acts and toward criminalizing possession, of passwords, red boxes, or knowledge itself, and consequently transubstantiating potentiality into culpability (Thomas, 1998, p. 384).

The handling of Kevin Mitnick's case differed from standard legal procedures. He was not offered a bail hearing and remained in jail for years prior to trial proceedings commencing, ultimately accepting a plea deal after four years in prison. This highly unorthodox approach provoked criticism among hackers. Magazine 2600: A Hacker Quarterly did not hesitate to place his arrest within the context of broader repression, predicting that Mitnick would be the first of a number if criminalization of the art persisted. To make the point even more forcefully, the same issue published source code from a file found in Mitnick's possession to demonstrate its easy availability. The pressure that was waged in 1997, describing his lengthy pretrial incarceration as a symbol of an approach punitive in its origins more out of news hype and public fright than to some balancing assessment of risk or criminality (Mazzini, 2023, p. 184).

Probes of other computer hacking offenses similarly increasingly departed from the use of technical proof toward the utilization of informants and network monitoring and, in practice, turning cause-and-effect reasoning around in prosecuting cybercrime (Thomas, 1998, pp. 389–393). This new legal and cultural landscape gave rise to the emergence of new forms of institutional response.

The Electronic Frontier Foundation (EFF), established in 1990 in direct reaction to Sundevil, was an attempt by a group of people to defend computer civil liberties and make hacking legitimate as an objective (Jordan, 2016, p. 83). Almost at the same time,, a three-way distinction was beginning to emerge between white-hat, grey-hat, and black-hat hackers.

As American hackers were criminalized and demonized in the media, their European counterparts to this group, with the German-based Chaos Computer Club (CCC) at the forefront, developed differently. Established in 1981, the CCC operated with a set of democratic principles and often operated inside the law to find vulnerabilities for the good of the people. Its legitimacy was supported by actions like the 1984 Btx hack and the 2011 Staatstrojaner scandal, used not to spread anarchy but to alert people to state intrusiveness and systemic insecurity (Kubitschko, 2018, pp. 84–89). Whereas Phrack bloomed in the oppositional and sometimes outlawed discursive space, the CCC created channels for communication aimed at achieving strategic legitimation (Kubitschko, 2018, pp. 91–92).

This contrast demonstrates the geographical specificity of hacker legitimacy whereby U.S. hackers often played the role of anarchists, whereas German hackers embodied ethical protectors. This is also reflected in how "ethical hacking" was legislated.

White-hat hackers surfaced in the United States by infiltrating the information security business as consultants and pen-testers. Sinha and Arora (n.d.) argue that ethical hacking was a part of information risk management in which sanctioned breaches enabled organizations to prepare for threats. This was institutional legitimacy and a complete reversal of the former notion of hacking as a means of rebellion or subversion.

Despite this positive turn, policing tactics rested on symbolic control. Police activity, Blevins and Holt (2009) argue, did more than merely avoid harm; it tried to exercise authority over stigmatized groups. In cyberspace, visibility was danger. Those rendered visible by fame, political opposition, or expertise were victims, while others moved in the shadows. Hacker culture then became a site on which legality, identity, and visibility were contested with every step.

Lastly, this period saw an end to the so-called Golden Age of hacking (1984–1990). It had been a period of shared ideology, subcultural infrastructure, and idealized discovery. Zines like Phrack, 2600, and CuD were leaders in this period, delivering not only technical advice but also a common ideology for sharing hacker values.

This fragmentation marked the close of a foundational era. As hacker culture evolved under new pressures, new forms of expression appeared—IRC networks, anonymous forums, and decentralized collectives—but the earlier ideal of hacker identity as both intellectually challenging and morally nuanced became increasingly hard to maintain in the wake of criminalization, commercialization, and internal strife (Thomas, 2002). This background provided the context for Phrack's own editorial history, which would mirror the general trajectory of hacker culture. As we

proceed to the second part, we discuss how Phrack re-established itself amidst a period when it was under threat from prosecution, commercialization, and evolving standards of legitimacy.

2.4 Continuity and Legacy

While *Phrack* had been an active center for hacker identity for many years, its eventual status as an electronic-only operation ultimately made it especially vulnerable to state scrutiny. In 1989, the year of watershed events, federal authorities began monitoring the electronic transmissions of the magazine, leading to the arrest of its two founders, Craig Neidorf (aka Knight Lightning) and Robert Riggs (The Prophet). Under pressure, they were compelled to offer the names of their editorial board and subscribers. This crackdown, prompted by Phrack's publication of a 911 emergency system document, raised serious questions about digital publishing's legal boundaries and whether hacker media could be criminalized for merely disseminating technical information (Mazzini, 2023).

This early confrontation between law enforcement and hacker media coincided with the overall mood of repression following Operation Sundevil. Electronic speech was now a battleground for legal skirmishes. Print-based hacker publications like *2600* were quick to notice that their printed nature likely insulated them from such state intervention. *Phrack*, on the other hand, became increasingly vulnerable to conspiracy and cyber misconduct accusations as a testament to how medium, as well as content, impacted its readability within the law and institutional susceptibility. This attack marked a threshold moment in which Phrack was no longer simply a zine being investigated, but a symbol of the hacker counterpublic in crisis (Mazzini, 2023).

In response to these pressures, *Phrack* moved toward formalizing its legitimacy. In 1993, the publication was registered with the Library of Congress and assigned an ISSN. Editor Erik Bloodaxe proclaimed this as a move toward legitimacy, stating, "This adds a new era of legitimacy to *Phrack*... it should never again face any legal challenge that would bypass any paper-based magazine" (Lunceford, 2009, p. 9). This marked a turning point. No longer simply a collection of rebellious philes, *Phrack* began curating its content more carefully; dropping bomb recipes in favor of advanced exploits, shellcode analysis, and software vulnerabilities (Thomas, 2005).

This professionalization, however, was met with ambivalence by some members of the community. Dark Sorcerer, a later commentator on *Phrack*, noted this shift with mixed emotion. "Early articles, dumb and inarticulate as they usually were, showed more of a wide-ranging desire to conquer time and space... now it's polymorphic shellcode and spectrum analysis" (as cited in Lunceford, 2009, p. 10). The tone had become more professional, more exclusive, and more adult.

Concurrently, and in response to this shift, Phrack began actively curating its own historical legacy. The figure of Kevin Mitnick, for so long a polarizing icon, was reinterpreted in the fresh issues as not a cautionary figure but an unjustly misunderstood pioneer. "They've probably found a dumb

web bug to defile his site," grumbled Duvel, of younger hackers dismissing Mitnick's reputation. "If these hackers could do a third of what Kevin did, they'd be Underground heroes" (Lunceford, 2009, p. 13; Hafner & Markoff, 1995).

This reframing of hacker figures was mirrored by a broader editorial shift within *Phrack* itself. In defending the addition of seeming non-topic material such as consciousness hacking, editors held that *Phrack* was never anything but greater than code. As mentioned in one issue"Before it was a computer hacking publication, *Phrack* was dedicated to sharing the occult wisdom, invisible and revolutionary" (Lunceford, 2009, p. 14). The return of such metaphysical concerns brought back earlier fusions of cyberculture and psychedelia, recalling the memory of *Mondo 2000*'s statement that governing information and consciousness both was the edge of revolutionary change (Thomas, 2002; Mazzini, 2023).

Despite these reinventions, Phrack's continuity was again interrupted. By 2006, *Phrack* was declared dead (Thomas, 2005). But this was not the first death and not the last. Later editors sought to restore its countercultural soul. In new philes, contributors like Gladio and Keptune revived the spiritual and philosophical elements of hacking, including articles like "Hacking Your Brain: The Projection of Consciousness" (Lunceford, 2009). These efforts reflect *Phrack*'s enduring commitment to hacking as both a technical and ideological act.

This editorial resurrection was not merely an attempt to revive a publication; it was a cultural call to arms. Duvel, one of the key voices in *Phrack*'s revived era, lamented the fading memory of the magazine's legacy among new hackers: "Nowadays, I'm pretty sure that new hackers don't read old *Phrack* articles anymore. Because they are lazy, because they can find information elsewhere, because they think old *Phracks* are outdated... But reading old *Phracks* is not only to acquire knowledge, it's also to acquire the hacking spirit" (as cited in Lunceford, 2009, p. 12).

This revival was not just inward facing. It also responded to growing fragmentation within the global hacker scene. In a phile titled "International Scenes," the editors noted that where hackers were once isolated and scattered, they now had the tools to share knowledge and unify. By spotlighting scenes in France, Brazil, and Quebec, *Phrack* reaffirmed its commitment to fostering a borderless hacker community rooted in shared values: curiosity, resistance, and the liberation of information (Lunceford, 2009; Thomas, 2002).

Ultimately, despite repeated pronouncements of its demise, Phrack has continued to resurface time and again not just as a magazine but as a living institution of hacker culture. Its evolving content, editorial approaches, and rhetorical maneuvers illustrate the evolution of hacker identity over decades. As Fernback (quoted in Lunceford, 2009) contends, "cyberspace is a repository for collective cultural memory," and Phrack is one of its most enduring monuments. As DH once stated, "Phrack is not dead. On the contrary, Phrack can't ever die" (quoted in Lunceford, 2009, p. 15). Its own survival testifies to the reality that hackers' cultural institutions still shape digital citizenship, resistance, and collective memory in the information age, making Phrack one of the longest-lived monuments of hacker culture.

Despite several obituaries, Phrack will not die. Its influence continues not just in its technical heritage, but as a repository of hacker collective memory. As mentioned earlier, even editors emphasize this archival role: "Reading old Phracks is not to gain knowledge only, it's also to gain the hacking spirit" (Lunceford, 2009, p. 12).

Phrack is not simply an artifact of history, but a living medium of cultural transmission that bridges generations of hackers through shared language, myth, and code. Despite all the announcements of its death, Phrack keeps reappearing; not as a newspaper so much as a cultural institution in hacker thought. Its influence is not only found in the technical attacks it circulated, but in being a repository of culture, a rhetorical drive of remembrance, and an echo chamber of ideology. Later editors have highlighted this archival feel, claiming that "reading old Phracks is not to gain knowledge only, it's also to gain the hacking spirit" (Lunceford, 2009, p. 12). Phrack has therefore served as a living bridge between generations, transmitting not only instructions but ethos: shared language, historical self-consciousness, and shared transgression. Whether remembering the past of such legends as Kevin Mitnick or bringing metaphysical streams of consciousness hacking back to life, Phrack presents itself as something more than a tutorial zine; it is a revolutionary knowledge object, a location at which the hacker's identity is not only remembered but performed anew. As Fernback (cited in Lunceford, 2009) sees, "cyberspace is a repository for collective cultural memory," and Phrack remains one of its most enduring landmarks. It is this subcultural adaptability combined with editorial reinterpretation that has allowed the magazine to last longer than most of its contemporaries. As DH in the preface to Phrack 31 so eloquently put it, "Phrack is more than just a technical newsletter. it's a symbol of our hacking history" (quoted in Lunceford, 2009, p. 6).

This continuity has also served to counteract the growing disunity of the global hacker community. As more recent forums, social media, and encrypted networks emerged, there was a need for a unifying voice. The result was Phrack going back to claiming status as a world cultural intermediary. The phile "International Scenes" mapped hacker scenes in Brazil, France, and Quebec and argued that hacking "knows no borders" (Lunceford, 2009). These actions reaffirmed Phrack's commitment to building an international hacker culture founded on curiosity, resistance, and unfettered information exchange (Thomas, 2002; Lunceford, 2009). Modern-day high-profile characters, whether Anonymous, Mudge, or countless lesser-known writers continue to echo the same original values first established by The Mentor in 1986: hacking as a process of intellectual curiosity, creative discovery, and principled resistance (Blankenship, 1986, cited in Lunceford, 2009; Thomas, 2005). Even as hacker culture spreads and evolves, there is still something to be learned from Phrack: a textual and ideological hub in an anarchic world. Phrack, then, is not so much an artifact of its time; it is a model of how subcultures endure, evolve, and speak themselves forward.

Chapter 3: Hacker Pedagogy in Practice: Informal Learning Discourses in Phrack Magazine

3.1 introduction

In this chapter, Phrack Magazine will be examined using a discourse-historical perspective as an initial and participatory informal learning space within the overall context of hacker culture. Following from the theoretical frameworks established in Chapter 1, in which Digital Informal Learning (DIL) was situated in constructivist and sociocultural theories (Vygotsky, 1934; Rogoff, 1990; Bruner, 1960), this chapter examines Phrack as a digital text or archive, but also as a pedagogical space where learning is organized through participatory discourse, ideological resistance, and community practice.

To get a correct understanding of the pedagogical project of Phrack, it is crucial to situate it historically. The magazine emerged in 1985, an era when the web did not yet exist, and electronic communication was centered on bulletin board systems (BBS). Straddling these early networks, Phrack helped to forge an informal but sustained virtual community of hackers who often had no other site of connection. This made its role in spreading information and influencing identity particularly crucial. Compared to the constantly networked internet populations of today, the early hacker culture operated under conditions that made collaboration and community building a much harder proposition. Phrack is thus not just an image of present-day hacker culture but an object of formative emergent history.

While Phrack has been centrally cited in hacker research on hacker identity, cyber deviance, or counterpublics, during its most productive years of production, Phrack functioned as an influential means of informal education for hackers. It remains understudied as a de facto curriculum of study by which technical skills, ethical stances, and subcultural norms were taught. Alongside 2600: The Hacker Quarterly, Phrack helped form a decentralized pedagogical space based on peer-authored discourse rather than formal pedagogy.

This is warranted by the fact that the magazine not only codes technical knowledge but also codes cultural norms, normative ethics, and collaborative research models constitutive of hacker practice. This kind of content and discursive practice, as described above, is characteristic of informal learning contexts which were peer-driven, socially situated, non-instructional, and intrinsically motivated (Callanan et al., 2011; Colley et al., 2003).

This analysis particularly resists the typical fallacy of considering informal learning to be second or secondary to formal education. From scholars like Livingstone (2001) and Dewey (1916), this chapter positions Phrack as a site of epistemic and ideological development. It accumulates technical knowledge, moral codes, and subcultural capital through situated practice and shared meaning.

This chapter considers Phrack both as pedagogical actors and cultural artifacts. It is a discursive space in which peer-to-peer informal learning assumes forms of informal education without institutional affiliation. Despite being an open magazine without having a formal course structure, credentialing, or instruction, Phrack finds itself in a generative middle space between informal and formal learning. It mediates hacker identity formation and technical knowledge transmission through interlocking discursive practice.

This is revealed by Phrack's textual and structural features; its decentralized strategies of evaluation, open contribution ethos, and genre diversity, ranging from amidated opinions to step-by-step technical tutorials. It contains no foreordained learning objectives or models of instruction plotted throughout it. Rather, its process of learning is emergent, self-guided, and reputation verified. Authors write for equals, comments pass via reputation or follow-up articles, and acquisition is supported through lucky incidents and not disciplined advancement. Such attributes are typical of systematic and planned informal education frameworks but within an extra-institutional environment of certification or rank.

Section 3.3 outlines the methodology of the chapter but succinctly, this study integrates computational text analysis with critical discourse analysis (CDA) and informal learning sociocultural theory. Voyant Tools allows me to have entry into the methods of keyword-in-context (KWIC) analysis, frequency, and collocational pattern analysis, used here not as termini but as heuristics to inform qualitative interpretation further. As emphasized in Chapter 1 throughout, informal learning is never quite so neatly translated into hard and fast propositions. It is negotiated, symbolic, and contextually embedded (Rogoff, 1990; Dron & Anderson, 2003).

This chapter therefore critically examines the learning culture of Phrack as an everyday pedagogy that resists formal education by its peer-based, experiential, and ideologically driven practice. These are very similar to the primary attributes of Digital Informal Learning (DIL) such as autonomy of learners, individualization, digital literacy, and social interaction in non-institutional locations (Reinders & Benson, 2017; Doleck et al., 2019).

This chapter moves Phrack from writer and pedagogue to an expanding one based on existing research to demonstrate ways in which online unofficial spaces supplement official schooling as well as function as primary sites for ideological production, identity formation, and discursive negotiation. In addition to being packed with technical expertise, the Phrack archive is also full of inclusion/exclusion practices, ethical boundary performances, as well as subcultural authenticity performativity's. The magazine therefore instigates and perpetuates a specific type of hacker literacy, a hacker pedagogy.

These inclusions and exclusions are generally implicit, practiced under the cover of stylistic hints and unspoken knowledge. Insider slang, leetspeak[3], and command-line allusions presume an assumed hacker literacy, inscribing novices from masters. Editorials and manifestos deploy cultural references to UNIX spaces, BBS heritage, and hacker legend to subtly signal subcultural borders. Tutorials, although rich, are composed to scaffold process knowledge, and ideologically centered texts construct the hacker as an ideologically autonomous entity. These rhetorical moves act as soft boundary controls, including on the one hand, excluding silently on the other.

In its hybrid modes, Phrack performs a variety of pedagogical roles. Tutorials offer procedural modeling. Manifestos and editorials are ideological summaries. These modes "teach" in specific ways, through step-by-step briefing or discursive alignment, to generate a multidimensional learning space. A tutorial on shellcode, for instance, might use imperatives such as "try," "test," and "exploit" in the manner of technical know-how. Manifestos such as The Conscience of a Hacker published by Blankenship (1986, *Phrack*, Issue 7, File 3), offer rather an ethical template for these actions, rationalizing trespass in discourse about curiosity and resistance. Together, the genres enact both technical skill and social identity, mediating the technical and ethical aspects of hacker pedagogy.

- [3] **Leetspeak** ("1337 speak") is a symbolic jargon shared by hacker and internet computer gaming subcultures, with letter substitutions for visually similar letters, for example, "leet" to "1337."
  L → 1
  E → 3
  E → 3
  T → 7

It serves both as an identity marker and an exclusionary/inclusionary linguistic marker of subcultural membership (Danet & Herring, 2007).

The chapter synthesizes the theoretical grounding developed in Chapter 1, for example the learner autonomy, peer scaffolding, and situated knowledge practices, to answer the following research questions:

- How were informal, peer-based learning practices constructed through discourse in Phrack magazine during its publication from 1985 to 2016?
- What discursive and rhetorical practices position readers as learners, tutors, or a learning community member?
- How does Phrack convey an ethos of inquiry, trial, and ethical information exchange?
- In what ways are inclusion/exclusion practices embedded or silenced in hacker learning discourse?

To respond to these questions, this chapter is structured into five sections. Section 3.2 provides the background story of Phrack and the corpus construction. Section 3.3 describes the research methodology, combining Voyant-based text analysis, CDA, and sociocultural learning theory. Section 3.4 describes a thematic analysis of five core informal learning discourses in Phrack. Section 3.5 addresses digital subcultural text methodological and epistemological concerns. Section 3.6 terminates by reiterating the role of Phrack in the construction of an unofficial, ideologically charged model of hacker pedagogy in the past, one that reversed conventional ideas of learning, authority, and expertise within digitally networked cultures.

3.2 Corpus and Context

3.2.1: An Introduction to the Historical Phrack Corpus (1985–2016)

These issues permeate the publication's most vigorous and broadest years, from its earliest gestation in 1985 to its later issues, the final one having appeared in 2016. While the entire corpus was surveyed through Issue 72, Issue 69 (2016) is likely the last to directly engage in pedagogical function, rather than merely continuing the publication tradition. This period is not arbitrary. Instead, it makes possible a diachronic analysis of Phrack's changing discursive strategies, pedagogical rhetoric, and ideological articulations in response to dominant legal, technological, and sociopolitical changes (Sterling, 1992; Coleman, 2013). The decision to use Phrack as the primary corpus is determined by the twofold nature of the publication: it is simultaneously a discursive archive of hacker culture and a pedagogical interface through which norms, ethics, and technical literacies are acquired, contested, and reproduced.

Chapter 2 concluded that Phrack needs to be read not just as a medium for disseminating hacker knowledge, but as a counterpublic space reflecting and constituting the collective identity of hackers (Warner, 2002, p. 86; Coleman, 2013, p. 17). Following this, the present chapter positions Phrack as a cultural-pedagogical object which its rhetorical forms cannot be separated from the processes of informal learning and subcultural belonging. As Lunceford (2009) argues, Phrack's

objectives are well beyond information exchange or documentation. Most of its writings, which could have been in the guise of tutorials, exploits, editorials, and manifestos, contain a self-evident didactic intent in that they aim to bring beginners into hacker participation's complex and often contradictory ethos. Most importantly, these articles are not authored by professional journalists but by hackers themselves, so Phrack is a site of insider pedagogy and cultural reproduction.

The issues written are a form of distributed pedagogy, interweaving instructional discourse within genres ranging from stern guidebooks to ironic manifestos. They operate not only at the level of technical knowledge, but also at the level of affective induction, ideological identification, and discursive competence, all of which are necessary to gain legitimacy as a community member. The corpus is composed of three distinct phases: the experimental and formative stage (1985–1989), marked by a lighthearted and experimental tone; the state-surveilled and cybercrime-statute-limited middle period (1990–1995), fueled by state surveillance and cybercrime statutes; and the globalization-suggesting and postmodern period (1996–2025), in which Phrack developed amid globalized hacker publics, aesthetic shifts, and increased ideological reflexivity.

These phases not only mark the development of Phrack's audience and voice, but also its altering pedagogical approach, ranging from explicit instruction to more implicit, culturally embedded knowledge-practice-sharing strategies.

Target texts such as "The Conscience of a Hacker" (Blankenship, 1986), fundamental exploits (i.e., shellcode tutorials from the earliest days), and op-eds on the resistance nature of hacking were selected since they demonstrate how Phrack facilitated learning-as-becoming (Lave & Wenger, 1991) a discursively negotiated process of identity formation via practice.

Chapter 2 also emphasized that hacker identity is not essential but discursively performed. Drawing on that, this chapter treats Phrack not as a linear series of didactic texts but as a discursive environment in which pedagogical power is circulated, co-authored, and mediated across voices and genres. The editorial trajectory of the magazine, joint authorship, and community feedback loops all together produce a rhetorical ecology in which knowledge is ideologically structured, affectively charged, and socially negotiated. In this way, Phrack operates more as an apprenticeship network than as a textbook in text form (Gee, 2004).

As Reinders and Benson (2017) point out in their study of informal digital learning, technology-mediated spaces that promote open participation, extensive discussion, and dialogic interaction are more likely to create stronger patterns of collective learning and identity formation than formal teaching spaces. Particularly remarkable is that such participatory and identity-forming dynamics were already highly developed in hacker communities prior to the emergence of the web. Well before the emergence of formal online education platforms, sources such as Phrack existed as informal yet highly engaged pedagogical spaces. Produced by hackers themselves, these writings provide peer-to-peer learning and collective problem solving through tutorial style instruction, manifestos, and technical explanations. Not only did this space enable the sharing of information, but it also provided a sense of communal identity grounded in shared values, practices, and models

of culture. The Phrack case shows that several of the key principles of contemporary digital learning, including openness, interactivity, and collaborative knowledge production, have earlier precedents in the 1980s and 1990s early digital subcultures.

These features underpin Phrack's structure. The alternation between formal tutorials and informal commentaries, humor columns, and political statements is not accidental; it is hacker culture pedagogical logic at its best. It represents hacker's plurality, indeterminacy, and transgressive performance. This genre of heterogeneity makes it possible to enter at any literacy and identity level. Beginners may have arrived at Phrack for technical expertise, but they were also acquiring a hacker culture that values not just competence, but irony, ethical engagement, and performative nonconformity with mainstream cultural values. In this sense, the corpus not only provides evidence of what is being taught (i.e., technical knowledge), but also the way it is being taught, for example; mutual narrative forms, peer referencing, insider humor, and ideological signaling. In keeping with the discourse-historical analysis methodology, the corpus is not analyzed as an objective store of fact-based information but as a performative learning environment, where texts enact power relations, moral positioning, and community boundaries.

As Chapter 2 continued to argue, hacker speech is always ideologically charged. Even in technical articles in Phrack, they function at the same time as arguments about freedom, power, and access, demonstrating how pedagogic action is replete with cultural meaning (Fairclough, 1995). This may be easily observed in Phrack, where text genres are invoked to achieve both knowledge reproduction and value reproduction. This corpus allows us to study the learning process in a decentralized, informal, and ideologically charged environment. Through the topics selected and their shifting tone, form, and authorship, Phrack is both pedagogically productive and politically astute artifact and not merely an analyzable text, but a field of learning, identity formation, and ideological negotiation embedded deeply in the contested terrain of hacker counter publics.

3.2.2 Corpus Composition and Selection

Based on the contextualized history elaborated in the above section, this section outlines the scope, composition, and rationale for the choice of the Phrack corpus employed in this work. An expressed understanding of the structure of the corpus is required not just for following discursive and rhetorical patterns which will be analyzed in subsequent sections, but also to inform the broader investigation into peer-learned, informal learning practices as they currently appear within hacker cultures and online subcultures. The primary corpus to be studied comprises 69 full issues of Phrack Magazine, spanning a period of publication. Phrack Magazine has a period of publication from its first Issue #1 in 1985 to Issue #69 in 2016, but this is broken by a long gap in activity, particularly after 2005, with gaps between issues of several years becoming more common. from its initial Issue #1 in 1985 up to its current Issue #69 in 2016. The three issues published after 2016 did not mean to have any ideologies or teachings or updates, and they were only published to keep the magazine alive. This longitudinal data set documents the entire course of development of Phrack from its modest, ASCII-formatted zine disseminated via early bulletin board systems

(BBS) to an advanced, world-recognized digital e-zine containing technically accurate documentation and worldwide authors.

Such a diachronic scope makes it possible to trace out both continuities and disjunctions in hacker rhetoric, particularly in relation to reconfiguring sociotechnical infrastructures, changing cybersecurity mores, legal regimes, and broader cultural evolution. By tracing these changes over time, the corpus provides an engaged context for examining how hacker education, beyond centralized, state-sponsored, and ideologically driven forms, has learned to adapt to emerging digital environments. Quantitatively, the corpus contains over 200,000 tokens distributed across roughly 300 individual articles. These tokens come from a broad variety of text genres and types including but not limited to highly technical guides, moral manifestos, editorial commentaries, scene reports, community interviews, and letters to the editor.

This variety of text approximates Phrack's varied nature as both a technical guide and a cultural artifact. Above all, it also indicates the role of the magazine as an amateur pedagogical site which is the one situated in a middle position between community-based pedagogy and dissemination. The coexistence of technical how-tos with ideological pamphlets and social commentary in general signals Phrack as being both a pedagogical medium and an identity vehicle in the community. Therefore, the corpus offers a rich soil to explore how hacker subculture learning is not only skill-based, but constructed normatively and discursively, and thus further solidifying Phrack as a non-institutional pedagogic space.

Corpus Feature	Details
Number of issues	69 (Issues #1–#69)
Approximate word count	200,000+ tokens
Article types	Tutorials, manifestos, editorials, cultural commentary, letters
Period covered	1985–2016

Table 3.1: *Overview of Phrack Magazine Corpus Composition (Issues #1–#69, 1985–2016), detailing the scope, article types, and time span of materials analyzed in this study.*

This comprehensive approach avoids sampling bias and allows the emergence or intermittent discursive phenomena. These are patterns which may only appear sporadically throughout the extended publication history of Phrack. The key to the utility of longitudinal digital corpora, as Rockwell and Sinclair (2016) point out, lies not merely in their ability to register frequency-based trends, but in their potential to reveal diachronic shifts in rhetorical strategy, ideological stance, and communicative practice. These affordances are particularly relevant to monitoring how hacker talk shifts with respect to broader social, legal, and technical shifts. To balance depth and concentration, the research established strict criteria for limiting articles from within the corpus directly answering the research questions presented in Section 3.1. The criteria were designed to

select texts enlightening on the mechanisms of informal peer-to-peer learning, the ethical positioning of hacker identity, and the broader pedagogical roles of the magazine.

The selected texts are chosen from topics in the following subjects. The first topic is instructional tutorials, such as "Introduction to Shellcode" or "Basic Buffer Overfits," that are technical knowledge transfer sites. The second topics include hacker ethics declarations, such as well-known manifestos like Blankenship's (1986). The next ones are the Conscience of Hackers,[4] that represent normative systems and moral philosophies. Following that would be the political or cultural commentary articles, such as "International Scenes[5]" or "Hacktivism Today[6]," that link hacking to international activism and sociopolitical commentary; and the last ones would be the critiques of conventional education and certification systems, such as the satirical works "Certs are for Clowns" (raT & alx, 2008)[7], which value autonomous knowledge over institutional power.

By implementing this targeted inclusion strategy within a large and diverse corpus, the analysis is preserved at both analytical complexity and interpretive richness. This dual benefit supports both the computational objectives of the research and its critical analysis of the socio-discursive construction of learning in digital subcultures.

Another point to consider is that pedagogically, Phrack is seen as an informal learning space where knowledge is not communicated hierarchically but discursively negotiated, where power is distributed laterally across networks of peers. This aligns with Reinders and Benson's (2017) conceptualization of digital learning ecologies in which informal, self-organized, interest-driven learning environments beyond formal institutional contexts are mainly emphasized.

Unlike formal educational materials, Phrack does not present knowledge as neutral or decontextualized but rather immerses learning within subcultural norms, ideological commitment, and performative work on identity (Gee, 2004; Lave & Wenger, 1991). The magazine thereby enacts a type of instruction that is highly situated, stylistically marked, and often resistant to normalization. The genre of heterogeneity throughout the corpus necessitated an open-ended analytic framework. Some texts explicitly function as pedagogical tutorials, while others embed their pedagogic content in satire, persona construction, or community storytelling. Unpacking the nested discursive strategies and identifying them is central to this study's application of Critical Discourse Analysis (CDA) (Fairclough, 1995; Gee, 2011), enabling it to consider how learners are positioned, how authority is marked, and how community boundaries are linguistically constructed. In addition, the corpus is not as much seen as a store of fixed content as a discursive formation; it is a corpus of texts that sit together through shared semiotic convention, ideology, and historical contingency.

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[4] Loyd Blankenship (The Mentor), "The Conscience of a Hacker," *Phrack* Issue 7, File 3, 1986.
[5] Anonymous, "International Scenes," *Phrack* Issue 49, File 3, 1996.
[6] Hackbloc, "Hacktivism Today," *Phrack* Issue 65, File 8, 2008.
[7] raT and alx, "Certs are for Clowns," *Phrack* Issue 65, File 9, 2008.

Therefore, in this regard, the magazine is not a field of content as of collective making-meaning, in which authors and readers create and recreate the discourse of hacking through time.

This is informed by Kelty's (2008) conceptualization of recursive publics; wherein technical and cultural groups are reconstructed through their own modes of communication. Thus, the Phrack corpus does not merely document hacker's pedagogy; it enacts it. It embodies the very features of informal learning; it describes decentralization, ideological embeddedness, and collective knowledge. The next section describes the preprocessing work done to prepare this corpus for analysis using Voyant Tools with particular attention to ensuring the discursive and pedagogical integrity of the texts remains intact across computational transfer.

3.2.3: Corpus Preparation and Preprocessing

During this research process, ChatGPT AI (GPT-4, via OpenAI) was used to locate single issues of Phrack Magazine by searching through massive digital collections, which reduced manual surfing through all the volumes. It further assisted in cleaning and pre-processing the corpus for analysis, including conducting light preprocessing tasks like eliminating redundant metadata and formatting noise. During the writing stage, AI was used to reorganize outlines, create sentence structure in initial drafts, and ensure terminological consistency throughout chapters. These uses helped to enhance the efficiency of the project, particularly in dealing with the volume of unstructured text, but all analytical interpretations and theoretical arguments were developed independently.

In addition to Voyant Tools, Orange Data Mining was used to enable exploratory analysis of the Phrack corpus. Text mining widgets within Orange enabled unsupervised clustering and topic modeling whereby repeat themes and discursive groupings within the magazine were uncovered. These computational outputs were not used as end categories but as heuristics to enable and triangulate qualitative analysis. For example, articles were grouped according to lexical similarity and compared with manually identified genres (e.g., tutorials, manifestos, editorials) to see whether discourse genres aligned with algorithmic groups. The topic modeling component also prioritized lexical convergence on terms like freedom, access, and curiosity that reflected themes identified through critical discourse analysis. Orange Data Mining was then used as a complementary analytical layer, delivering pattern recognition and content segmentation that complemented the interpretive reading of the corpus. However, Voyant was the main tool used in this thesis.

The pre-processing of the Phrack corpus was guided by a dual imperative. On one hand, attaining computational compatibility with digital tools such as Voyant Tools, and on the other, maintaining the discursive, ideological, and cultural integrity necessary for meaningful interpretive analysis. These aims reflect the hybrid nature of Phrack which is a publication that is simultaneously technical manual, cultural artifact, and community manifesto. In this complex text environment, preprocessing was not merely an issue of data cleansing. Instead, it was a process of theoretically

informed curation, tinted by the study's overall interest in informal learning as a situated, Sub culturally coded, and stylistically heterogeneous process.

The first task in this curation was the removal of structural words and to be more precise pieces like words and codes that were possibly visually or archivally important, but which did not carry discursive weight in terms of linguistic analysis. This included duplicate metadata such as issue numbers, time stamps, ASCII-art borders, and other formatting features which recur throughout a range of issues. While these elements are part of Phrack's unique look and feel, their inclusion threatens to affect computational results, especially in tools that depend on term frequency and patterns of textual distribution. Standardized texts such as legal disclaimers, copyright statements, and repetitive editorial headings were also omitted. These elements tended to repeat verbatim between issues and would have falsely highlighted specific lexical items or phrases, invalidating the usefulness of visualization tools like word clouds, collocational matrices, and trend graphs.

However, cleaning was not a bid to "normalize" the corpus in the typical NLP [8]sense. In contrast to practices preferring lemmatization or stemming to reduce lexical variation, this research explicitly maintained variation. The choice to keep or group distinct grammatical forms of "hack," "hacks," "hacking," and "hacker" was informed by the presupposition that these morphosyntactic differences may or may not be random; they signal important shifts in rhetorical voice, audience alignment, and learner positioning. As Fairclough (1995) and Gee (2011) argued, the nuances of word form and usage are central to discourse-level meaning-making, particularly in texts where identity and epistemology [9]are under contention. Some were treated differently, and some were grouped in visualization outputs for clarity. While grouping may have reduced lexical specificity, the interpretive analysis remained sensitive to their discursive differences where relevant.

Although the preprocessing preserved morphosyntactic variation for theoretical reasons, some aggregate visualizations (e.g., word frequency patterns) reduced these variants for pattern clarity. Close readings were used to recover discursive nuances as needed.

Furthermore, hacker culture-specific linguistic markers, like leetspeak (e.g., "l33t," "n00b"), intentional misspellings, and in-group slang were tried not to be excluded as noise but as semiotic resources of value. Such features are cultural boundary-markers and play essential functions in peer-based knowledge systems that mark informal learning in hacker communities. Far from being decorative or incidental, they are tactica. Preserving these forms is consistent with the critical aim of the project to treat Phrack as a source more than a content archive but a lively discursive formation wherein language performs pedagogical, symbolic, and performative work.

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[8] **Natural Language Processing (NLP)** is one of the areas of artificial intelligence and computational linguistics whose objective is to make computers deal with, understand, and generate natural language (Jurafsky & Martin, 2021).

[9] **Epistemology** is the theory of knowledge; namely, how it is defined, justified, and passed on. Epistemological assumptions in educational settings determine what counts as valid learning and who can be accepted as an authentic knower (Hofer & Pintrich, 1997; Lankshear & Knobel, 2003).

Given the corpus's generic heterogeneity, a genre filtering process was implemented to ensure analytical focus without sacrificing richness. The magazine's content spans a wide range of genres from formal technical walkthroughs to performative manifestos and ironic critiques. Consequently, inclusion criteria were developed to prefer texts that were contributing explicitly or implicitly to pedagogy, critical commentary on hacker practice, or normative commentary on community values of autonomy, resistance, and open access. Satire and humor were not ruled out. Instead, they were filtered for pedagogical or ideological usability, with the premise that irony and humor are often how critical discourse, ethical consideration, and norm-setting within communities occur in subcultural contexts. This was accomplished by interpretive filtering of tone, modality, placement in the issue, and presence of thematic or technical keywords.

This principled but flexible filtering approach rejects the binary opposition of "serious" and "non-serious" texts. Instead, it picks up the theoretical and methodological foundation laid in Chapter 1 and Chapter 3 that situates Phrack within a rhetorically complex and ideologically fraught landscape of informal learning. On this criterion, texts are evaluated not only for technical depth but for their ability to address the recursive public of hacker discourse where norms, identities, and types of expertise are endlessly negotiated and reiterated.

Moreover, the preprocessing strategy was directly aligned with Voyant Tools' constraints and affordances. While Voyant is ideal for displaying term frequencies, keyword trends, and lexical patterns, it is not suitable for revealing discursive properties such as irony, voice shift, rhetorical stance, or dialogic tension; those very properties that are central to Phrack's style and pedagogy. This sensitivity informed the decision of what to excise or keep in the corpus so that the results of computational analysis will remain interpretable under the discursive logic of the source text.

Ultimately, this process amounted to more than preparation; it was a methodological intervention. The corpus wasn't simply cleaned for computational analysis; it was shaped in accordance with the study's theoretical commitments, especially its understanding of Phrack as a space of ideologically charged, peer-mediated learning. Mediating the requirements of computational clarity with those of cultural fidelity, the preprocessing process laid a firm foundation for the thematic and discourse analysis to follow.

However, it is also necessary to critically reflect on the limitations of these preprocessing choices. In preprocessing the Phrack corpus, some formatting features such as ASCII art, boilerplate headers, and redundant metadata were removed in the interest of encouraging computational parsimony for analysis within Voyant Tools. Such features must be described, though, as not being strictly noise. Features such as ASCII banners or pretty line breaks typically perform identity-marking or ironic commentary, especially in hacker cultures in which textual play is signaling community. Similarly, stylized misspellings and leetspeak, while distracting computationally, can act as semiotic indicators of in-group speech and boundary-making. Omitting them, therefore, risks levelling out the discursive richness of the corpus and annihilating forms of performative literacy most vital to hacker pedagogy. While these constraints were necessary to render data

computationally manageable for digital analysis, follow-up work could incorporate multimodal or stylistic analysis better to preserve such pedagogic cues.

Following that, this corpus was not treated as a neutral data set but as a curated object of study, the interpretive legibility of which was teased out in the tension between algorithmic analysis and cultural sensitivity. The collection that resulted maintains the richness, diversity, and ideological tone of Phrack's hacker pedagogy while being tractable to digital methods. It is on this chosen basis that Section 3.4 builds its critical discourse analysis, making clear how language in Phrack performs educational labor, builds identities, and reproduces the cultural logic of the hacker underground.

To look more critically ,some features like SCII art, boilerplate headers, and redundant metadata were removed to allow computationally tractability in Voyant Tools, these features are not merely gratuitous. Features such as ASCII banners, use of decorative dividers, and leetspeak generally perform work as semiotic markers within hacker culture; for instance, doing identity work, demarking in-group membership, or performing irony and cultural play. Their elimination can thus flatten pedagogically or culturally significant aspects of the corpus. Unconventional spelling and syntax, while clamorous from a computational perspective, can also serve in the construction of hacker identity and subtle boundary-maintenance mechanisms. While the exclusions were methodologically reasonable, subsequent research would be well served to do some multimodal or stylistic analysis to recover these performative literacies and better account for the dense texture of hacker pedagogy enacted in visual and subcultural language.

3.3 Methodology and Analytical Framework

This study employs a mixed method of discourse analytical approach, combining theoretical and computational techniques to examine Phrack Magazine as an active informal learning environment. The method stays midway between interpretivist scholarship and computer-assisted text analysis, aiming not only to identify linguistic patterns in the corpus but also to interpret how the patterns contribute to the construction of learning, authority, and identity within hacker communities. While computational software such as Voyant Tools are employed to generate empirical data, in the shape of keyword frequencies, lexical trends, and collocational patterns, these are not ending in themselves. Instead, they are used to generate data for more concentrated qualitative interpretation guided by social and ideological context within the hacker world.

3.3.1 Theoretical Basis: Sociocultural Learning and DIL

Theoretical underpinnings of this analysis come from sociocultural learning theory, primarily in the early and most influential work of Vygotsky (1978) and lave and Wenger (1991). These scholars draw attention to the fact that learning is not solely a cognitive process within the learner but an intersocial mediated practice. When following the process whereby learning is constructed through language within hacker culture, this study applies three interrelated discourse analytic

concepts: pedagogic discourse, discursive positioning, and epistemic stance. Pedagogic discourse, as transformed by Bernstein (1990), is the communicative practice through which knowledge, roles, and values are controlled and conveyed in educational formal or informal settings. It involves not only explicit teaching but also implicit signaling of what constitutes legitimate knowledge and who can teach it. Discursive positioning, as theorized by Davies and Harré (1990), comprises the ways in which individuals become positioned within discourse with respect to shifting roles such as learner, expert, or peer roles that are not decontextualize but negotiated in real time using language. Epistemic stance, as understood by Kärkkäinen (2006), is the linguistic and rhetorical means by which speakers negotiate their relation to knowledge according to their stance of certainty, doubt, curiosity, or authority. These ideas are used throughout the chapter to analyze how Phrack builds instruction, mediates authority, and inscribes hacker identity through text practices.

Learning occurs through participating in the culture and engaging within discursive communities, where knowledge is not so much transmitted but co-created. In this, knowledge creation occurs within a context of place, where it is affected by interaction, objectives, setting, and symbols or instruments in the learning environment. This is the proper method of observing Phrack since it does not represent a typical institutional setting. Phrack relies less on formal education, accreditation, or acknowledged curriculum and more on learning through interaction, copying, sharing, and belonging.

Hacker learning will be more likely to occur through informal but structured means such as membership in communities of practice, independent finding of technical resources, and peer mentoring. These pathways are consistent with the sociocultural view that learning is distributed and constituted in social networks of meaning and practice. As outlined in Section 3.2.2, Phrack can be understood as a pedagogic common, an open, shared learning environment where instructional authority is decentralized, and knowledge is co-produced. It is a community where boundaries between learner and teacher are not fixed but negotiated through participation and contribution. This occurs through legitimate peripheral participation, as novices become technically proficient, learn community rules, and earn respect step by step (Lave & Wenger, 1991).

This model of hacking as learning also naturally slots into the theory of Digital Informal Learning (DIL), a field of research that explores the ways in which learning occurs in digital space outside formal educational systems. Described by Reinders and Benson (2017), DIL is characterized by autonomy, openness, and the learner's ability to create their own learning paths. It is a recognition that online environments are inherently well-suited to the hosting of non-credentialed, interest-driven, and shared forms of knowledge engagement. Here, learning is curiosity, social relationships, peer acknowledgment driven, not diplomas or tests.

All these are clear in the Phrack corpus, with the knowledge being created and shared by members of a dispersed group. Articles are freely available and theme-driven by independence, experimentation, and critique. Education through Phrack is not institutionally modeled. It is an

epistemology of difference, founded on values that value independence, suspicion of authority, and an ethos of openness. Here, education is an activity of technical becoming (Brown & Duguid, 2000) where the individuals do not merely acquire discrete skills but form identities by participating in shared values and practices.

This theoretical context also allows Phrack to be described as a pedagogical textual space. The magazine, as mentioned earlier, is not just a technical information medium but also a tool for the transmission of ethical, ideological, and cultural positions. It offers a contest in which readers are not passive consumers of content but integral actors in an interpretive and constitutive activity. This requires paying more attention to how knowledge is positioned in the community: who speaks, who is given the role of learner, and which sorts of knowledge are privileged or silenced.

Through the application of sociocultural learning theory to informal learning, this chapter establishes a comprehensive analysis framework for understanding hacker pedagogy. This makes possible an exploration of learning that involves not only the transmission of information but also identity construction, negotiation of cultural membership, and performance of resistance to hegemonic epistemic order. These are at the center of what Phrack represents within its context and across broader digital cultures.

3.3.2 Analytical Tools: Voyant and Critical Discourse Analysis

To examine Phrack's pedagogic discourse throughout its history and over the course of its multi-decade publication cycle, this study integrates digital text analysis using Voyant Tools, a web-based tool widely used in the digital humanities for linguistic visualization and pattern identification. Although Voyant is often used for surface-level tasks such as generating word frequency statistics, its application in this study goes far beyond that.

Voyant's affordances were particularly appropriate given the size, heterogeneity, and diachronicity of the dataset: 69 issues consisting of a range of genres such as editorials, technical how-tos, cultural criticism, political manifestoes, and insider newsletters (Section 3.2.2). As this information is both temporally scattered and discursively stratified, Voyant's capacity for detecting lexical shift over time, semantic clustering, and co-occurrence patterns made it a good starting point for discerning the contours of Phrack's informal pedagogy. Its capacity for filtering noise (common words like is,a,...), detecting key concepts, and facilitating visual plotting of thematic tendencies made it particularly useful for tracing hacker learning as a long-term, collective, and discursively mediated process.

The Cirrus feature that generates word clouds based on term frequency was used to get an initial impression of lexical salience across the corpus. Extremely frequent terms such as "access," "teach," "learn," "trial," and "share" recurred regularly across issues. Rather than being examined as individual tokens, then, these words were examined in the context of recurrent pedagogical logics particularly those mirroring hacker ethos of experimentation, open knowledge exchange, and peer-based skill acquisition. These tendencies resonate with learning ethos discussed in

Sections 3.4.1 and 3.4.3, where learning-by-doing, peer accreditation, and contribution to community are valued over formal pedagogy (Brown & Duguid, 2000; Dewey, 1938). The value of these terms was discerned not in their absolute frequency, but in their embeddedness in Phrack's larger rhetorical patterns and discursive tactics.

To trace lexical and ideological change over time, the Trends function was used. This tool graphs the frequency of chosen terms across the corpus chronologically, enabling researchers to identify shifts in discourse corresponding to external historical events or internal editorial changes. For instance, increased usage of the words "ethics," "freedom," and "responsibility" in the early 1990s coincided with Phrack's editorial shift amidst legal crackdowns such as Operation Sundevil. Such discursive tendencies appear in alignment with the political axes described by Coleman (2013) and Sterling (1992), suggesting a discursive response to legal and cultural pressures shaping the hacker community during this period. In later years, a rise in terms such as "contribute" and "public" is consonant with explanations of hacker identity developing toward more open-source, collaborative, and internationally networked styles of engagement, as discussed in Section 3.4.4.

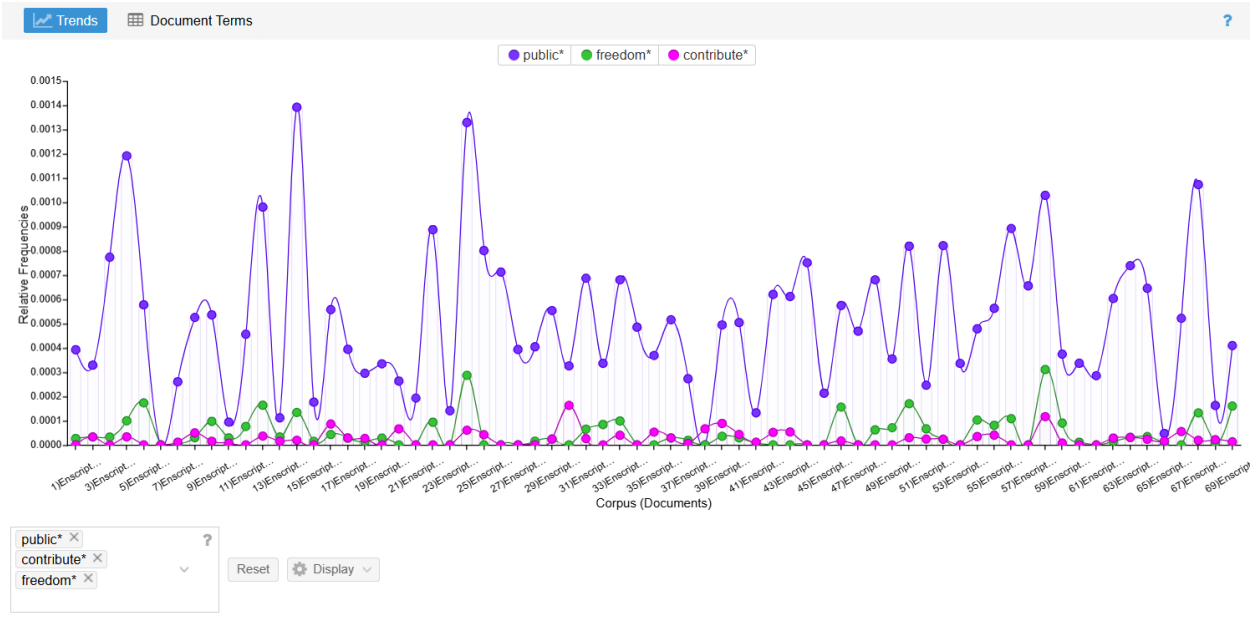


***Figure 3.1**: Chronological trends of public, freedom, and contribution in Phrack, highlighting shifts from political resistance in the 1990s to collaborative, open-source discourse in later years.*

For instance, an increase in the word "help" on issues between 2002 and 2005, as plotted using the Trends tool, necessitated a close reading of Issue #58's "Loopback" section. Several contributors in that edition debated the roles of "newbies" and "old-school hackers," using inclusive terminology such as "we help the curious ones" and "noobs have a right to guidance, not scorn." Such a change of tone, observed by Voyant, revealed how informal sponsoring and peer-to-peer pedagogy[3] were increasingly becoming more explicit in the community's rhetoric. The

computational signal then pointed us to a big change in hacker culture; increasingly, onboarding and social responsibility were becoming the themes, which otherwise would go unnoticed without large-scale lexical monitoring.

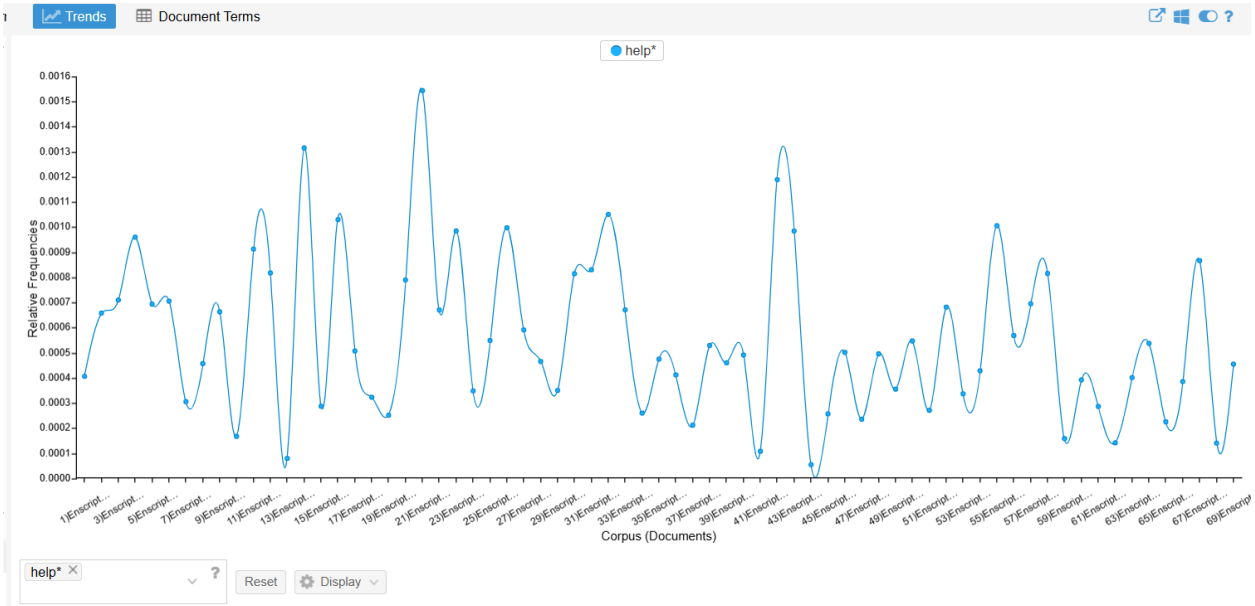


***Figure 3.2**: Relative frequency of help across Phrack issues, with peaks between 2002–2005 reflecting increased rhetorical emphasis on guidance, inclusivity, and peer-based learning.*

To study rhetorical context, Voyant's KWIC (Key Word in Context) feature was extremely useful. Unlike simple concordance software, KWIC allows the researcher to observe how key words function within local syntactic and dialogic contexts. This feature was instrumental in identifying recurring mentoring voice patterns, especially those involving second-person address, informal tone, humor, and direct encouragement as hallmarks of peer-based pedagogy. Phrases such as “You’ll get the hang of it” or “Try this out on your own” illustrate how instruction is often embedded in an ethos of exploration and community inclusion. These findings supported the discussion in Section 3.4.2, where informal mentorship and dialogic discourse had been shown to be at the center of hacker learning. KWIC also helped to uncover instances of coded humor, in-group reference, and symbolic boundary-drawing, illustrating the covert rhetorical mechanisms by which Phrack performs inclusion and exclusion, legitimacy and deviance.

The Summary feature was used to generate corpus-wide statistics such as vocabulary density, count of unique words, and document-wide word frequency distributions. These metrics permitted the calculation of genre-based lexical differentiation. For example, manifestos were discovered to consist of higher frequencies of abstract nouns such as "justice," "curiosity," and "ideology", while technical tutorials were lexically denser in procedural verbs, command-line syntax, and shellcode-specific jargon. This lexical distinction corroborates the necessity of a genre-sensitive analytic methodology, described in Section 3.2.3. It also provided additional confirmation of the thesis that

Phrack is not a mere repository of technical information, but a multivocal pedagogical space, with different genres serving different pedagogical and ideological functions.

These quantitative findings allowed the researcher to have a genuine overview in on specific lexical patterns, repetitive phraseology, or thematic shifts. Yet it was by way of close reading and theoretical interpretation, based on the principles of critical discourse analysis, that these patterns were made meaningful in relation to Phrack's role as a counter-institutional learning space.

In summary, the collaboration of Voyant's computational capacities with critical discourse analysis and sociocultural learning theory allowed this research to examine not only what is learned in Phrack, but also how knowledge is constructed, who is being interpellated, and what epistemological frameworks are being shared. Voyant was used descriptively and diagnostically. It presented a broad way of looking at big-picture patterns of text. It also enabled multi-layered reading in a way that can provide technical instruction, discursive positioning, and ideological framing. It validates the study's broader aim: to uncover the pedagogical structure and cultural logic of Phrack's involvement in hacker learning.

While Voyant Tools offers prominent computational affordances, such as frequency monitoring, collocation, and temporal trend visualization, this research does not take it to be interpretive determinative. Rather, it serves as a heuristic tool. It is a form of computational scaffolding that enables exploratory reading by drawing attention to lexical and rhetorical patterns. These results are meant to provoke interpretive interest but not resolve it. In this regard, the study is more concerned with meaning-making and sociocultural context than quantification, based on a qualitative reading of hacker discourse.

3.3.3 Linking Quantitative Outputs to Interpretive Strategy

A mentioned in previous sections, to move from lexical pattern recognition to ideological interpretation, this study applies Critical Discourse Analysis as a means of anchoring computational signals in a sociocultural context.

The interpretative phase of this research is grounded in Critical Discourse Analysis (CDA), a methodology which investigates the ways language functions to create and spread power relations, social identities, and ideological meanings (Fairclough, 1995; Gee, 2011). Rather than treating language as a transparency that allows information to pass through, CDA is focused on the power of language to create the context within which knowledge will be intelligible and authority legitimate. In this study, CDA offers a framework for reading Phrack Magazine not merely as technical text, but as a discursive site in which some learner identities, ethical positions, and cultural boundaries are constructed and negotiated intentionally.

Phrack is not an apolitical means of spreading hacking knowledge. It is a field of culture and rhetoric in which technical advice is embedded deeply within subcultural stories of resistance, autonomy, and legitimacy. The magazine embeds pragmatic content in broader ideological frames

that praise anti-authoritarianism, independent discovery, and access to information on a regular basis. In this context, the hacker subject is not simply an individual who acquires technical proficiency. Instead, the hacker is a particular type of subject, socialized by discursive suggestion that encourages oppositional learning, ethical transgression, and membership in subculture.

Using CDA, the analysis moves beyond recognizing what the magazine teaches. It instead pays attention to how it teaches, to whom it addresses, and what kind of learner it produces through its language. The reader-position is constructively positioned through discourse and stylistic means that invite critical distance from official knowledge, peer-to-peer exchange, and experimentation. Instructions like “try this,” “crash your system,” or “don’t believe the official docs” do not function solely as pedagogic directives. For example, the phrase “Try this and see what breaks” from Phrack Issue #48 exemplifies low-modality, high-agency language that positions the reader as an experimental actor rather than a passive recipient. Such imperatives bear ideological weight: they invite a resistant stance toward institutionalized authority and promote a culture of oppositional learning rooted in self-directed discovery and transgressive curiosity.

Furthermore, the use of pronouns such as "you," "we," and "us" makes it possible to establish a common identity between the writer and the reader. Such characteristics of language avoid hierarchical positioning and instead establish a peer-to-peer sense of community. For example, Issue #65 includes the line: “You probably won’t get it right the first time, don’t worry, you’re learning like the rest of us.” This sentence uses direct second-person pronoun (“you”) combined with inclusive plural (“the rest of us”), which works rhetorically to collapse the boundary between expert and novice. The sentence is also made less confrontational using hedging language ("probably," "don't worry") that decreases the threat of technical malfunction and frames experimentation as an appropriate part of learning. This is consistent with sociocultural notions of learning as identity formation through participation, not the acquisition of information. This sort of discourse in Phrack promotes a non-hierarchical, cooperative culture where learners are not subordinates but transient members of an expert community.

Therefore, the rhetoric situates the reader in not a passive position of mere receiver of information but as an active participant in a common epistemic and cultural venture. This kind of identity labor resonates with wider research in hacker culture studies, which prefers horizontal learning architectures, peer-to-peer mentorship, and independence encouraged by culture (Coleman, 2013; Taylor, 1999).

To chart these discursive features more systematically, CDA is complemented with computational analysis in Voyant Tools. The software permits early exploration of lexical links, frequency patterns, and semantic clustering. Such terms as "learning," "freedom," and "access" repeat with striking consistency, while second-person pronouns and imperatives appear as text forms. Such instruments remain at the level of quantitative correlation. They do not account for how language does subject-forming and boundary-marking without contextual and ideological reading. CDA addresses this limitation by locating lexical and grammatical features within broader cultural and

political discourses, and thus making it transparent how identity, power, and access are written into language.

To reach this, CDA enables us to read Phrack as more than a context-specific syllabus. It reveals the publication to be a cultural activity, one that recruits discourse for the purpose of establishing its own regimes of worth, knowledge, and authority. By putting language in the position of analysis, the study is hence able to show that informal learning sites such as Phrack cannot be fully understood without examining the ideological structures which underpin their communicative practices. The intersections of discourse, identity, and power are not accidental. It is the foundation of how these spaces function as educational and culture creation environments.

This interpretation model is at the heart of the chapter's research questions. Specifically, it informs the research on how learning is constituted, who becomes visible as a legitimate member, and through which means Phrack both reinforces and plays with the prevalent paradigms of education. Section 3.4 develops this model to characterize repeated patterns of discourse across the corpus of the magazine, including practices of experiential work, peer tutoring, and the inscription of openness as ethical precept and technical standard. Each of these is indicative not just of pedagogic preferences but of political allegiance.

3.4 Thematic Analysis of Informal Learning in Phrack

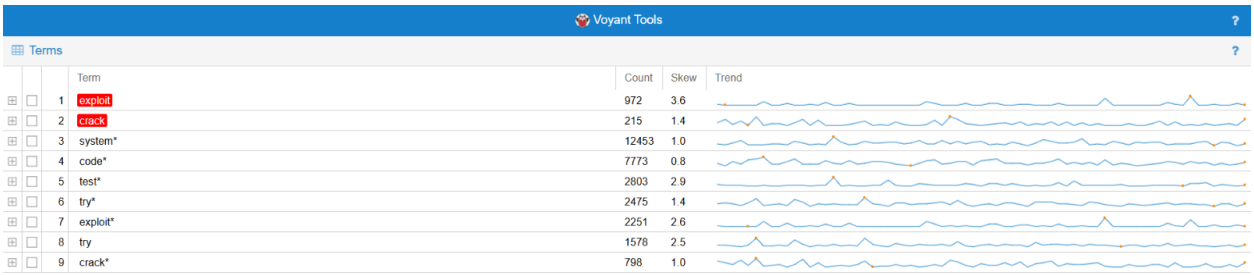
This section is a thematic analysis of common learning practices in Phrack Magazine based both on computational output from Voyant Tools and interpretive reading from a sociocultural and critical pedagogy viewpoint. The four themes discussed here are technical experimenting, peer mentoring, intrinsic motivation, and ethical sharing of knowledge. These themes were selected since they kept recurring across the corpus, time frames, and rhetorical stances in the corpus. They were repeated motifs, but the key reason that they were central motifs is that they are the two primary axes of informal learning within hacker culture. Technical experimentation and peer mentorship are the first two motifs and demonstrate epistemological processes, like knowledge acquisition, scaffolding, and enactment through doing and participation. On the other hand, intrinsic motivation and ethical sharing articulate ideological values. These values are the way learning is framed as a resistance against power within institutions, and the way openness and reciprocity function as norms of legitimacy. This epistemological-ideological distinction permits us to get a clearer picture of Phrack as a location of cultural identity construction and technical skill development that is integrally intertwined.

3.4.1 Learning through Doing: Technical Experimentation

The most recurring theme to emerge from Phrack's pedagogical discourse is its do-it-yourself ethos of learning-by-doing. This thread lies firmly on the epistemological dimension of learning. It is about how knowledge is created, tested, and realized via experimentation and technical practice. In opposition to institutional, didactic, or conventionally accepted modes of learning, Phrack

promotes a do-it-yourself ethos of experimentation, improvisation, and iteration. For instance, *Issue #47 states, "You're encouraged to experiment with this code."* It's not supposed to be perfect, and suggestions are welcome. Which explicitly links experimentation with personal learning and technical identity. This is directly opposed to learning paradigms that describe knowledge as static objects to be learned passively. Instead, Phrack aligns with what John Seely Brown (2009) calls "learning to be" a paradigm where the student is a unique kind of subject through situated practice within a community of practice.

This thematic focus is manifest in the language patterns that occur throughout the Phrack corpus. With Voyant Tools, we can extract a group of words that work as part of what can be defined as a procedural lexicon: test, code, exploit, crack, try, and system. These are not abstract definitions but verbs and imperatives that work pedagogically. As also clear from the term frequency and distribution chart (Figure 3.1), words such as exploit (frequency: 972, skew: 3.6) and crack (frequency: 215, skew: 1.4) are repeated, with skewed distributions, as a marker of their central role in Phrack's tutorial and instructive composition.



	Term	Count	Skew
1	exploit	972	3.6
2	crack	215	1.4
3	system*	12453	1.0
4	code*	7773	0.8
5	test*	2803	2.9
6	try*	2475	1.4
7	exploit*	2251	2.6
8	try	1578	2.5
9	crack*	798	1.0

Figure 3.1: *Term frequency, skew, and trend among some action words in the Phrack corpus, made with Voyant Tools.*

To break these words further apart, we can examine their collocational contexts and example use. For instance, as illustrated in Figure 3.2, exploit is prone to appear frequently with code, used, execution, and test. The phrases immediately around it that it is prone to appear in tend to imply step-by-step doing and direct system manipulation; for instance:

"Compile the exploit and run it against the vulnerable service to see if you get a shell." Phrack Issue 59, Article 0x08

This processional sequencing suggests a doing-and-learning mode. The imperative mood (Compile, exploit, run) summons the reader to work technically and not just read and consume. The extreme skewness of the word "exploit" (3.6) also suggests that these instructions are extremely technical and specialized in nature; presumably exploit development walkthroughs or vulnerability reports.

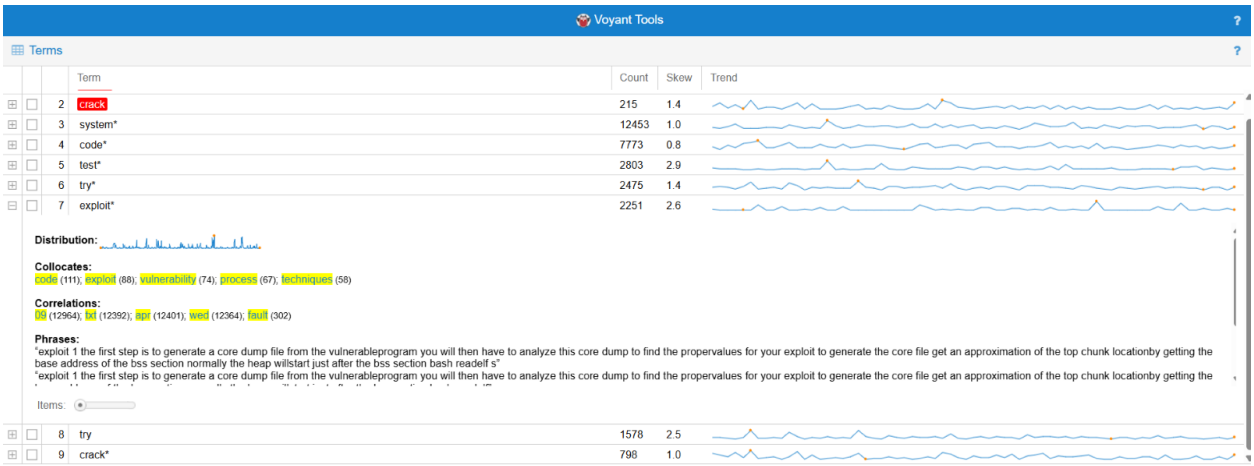


***Figure 3.2**: Voyant Tools statistics for the word exploit, including collocates and example use that convey a procedural and imperative linguistic function.*

By contrast, the word crack shows a relatively different but similar pedagogical function. As Figure 3.3 demonstrates, crack co-occurs with words such as hackers, computers, passwords, and software. These collocates indicate a thematic concern with access, breach, and security bypass. For instance, the following use:

"Crack British telecom how to extract funds from people."

uses crack not as a neutral technical term but as a deed charged with social, legal, and political connotations. Here procedural learning intersects with illegality, subversion, and systems critique. While "exploit" is closely linked to the technical procedure, "crack" is a more culturally loaded activity which is labeling the learner as rebel or lawbreaker.

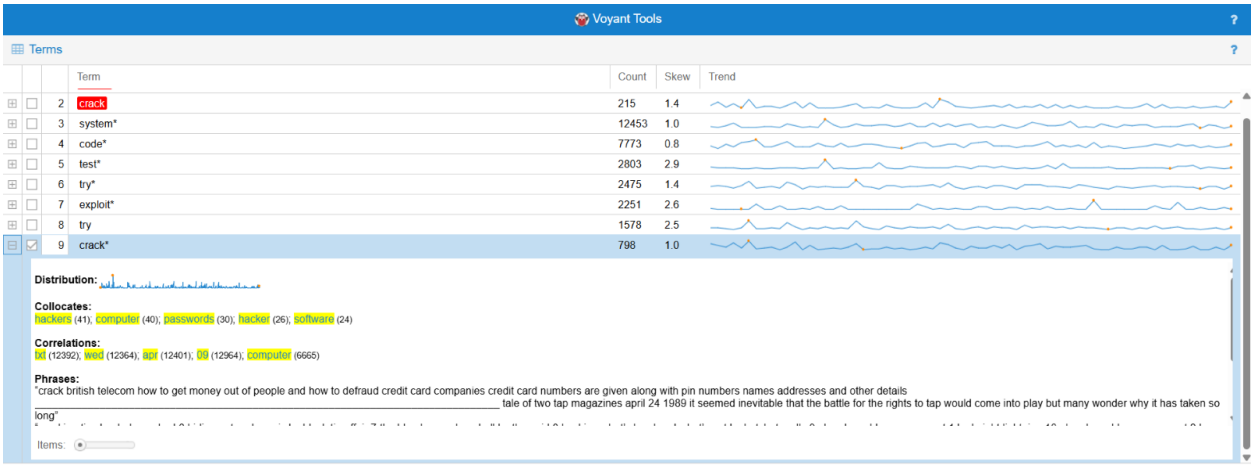


***Figure 3.3:** Voyant Tools output of crack, accompanied by close collocates and context expressions that reveal its subversive pedagogical implication.*

In theory, these two terms, exploit and crack, are instances of double function in Phrack's lexicon. They describe technical procedures and shape the reader into an active, independent, and system-aware learner. The procedural grammar called on here imperatives like try, test, see, crack, exploit, is what may be called a grammar of engagement, wherein Phrack initiates an object of learning not through pedagogy, but through partnership, trial and error, and taking a chance. In this mode, the reader is not simply told what to do, but being ushered into a subculture that demands initiative and improvisation.

In another example from the magazine, readers are instructed:

"Try different offsets and watch for the segfault[10]. Once it works, you'll get a shell." Phrack Issue 55, Article 15

This mandate also explains how Phrack language uses imperatives like try, test and see to construct a situated and active learning process. These imperatives are used not only to provide instructions, but to construct the reader as a learner who learns through risk, experimentation, and the manipulation of live systems. The reader is being hailed not as novice to be instructed, but as emerging peer with capability to join in and expand the knowledge base.

This is consistent with what John Dewey (1991) referred to as experiential education that is a model in which knowledge does not emerge via abstract instruction but via meaningful experience with tools, issues, and consequences. Dewey believed that education is not a preparation for life but life itself, a vision to which Phrack's philosophy is highly aligned. In this context, knowledge is not linear or up-front but emergent, situation-sensitive, and contingent on action and interpretation by the learner. This construction of the hacker as risk-taking, independent, and technically violent adheres to a masculinized agency model that is never questioned within the Phrack discourse.

Equally important is the way in which this pedagogical model captures and imposes core aspects of hacker subculture, namely its rootage in engineering playfulness, self-determination, and subversive creativity. As established in the foundational manifesto, the Conscience of a Hacker, attributed to "The Mentor," is driven in language characteristically probing. *"My crime is curiosity."* This mantra, a repeated incantation throughout hacker accounts, indicates the moral justification of unauthorized investigation as knowledge appetite. Here, hacking would be less about penetrative technicality than about knowledge gathering through boundary-pushing experimentation.

Phrack encourages this mentality in its weekly shellcode tutorial articles, where students are not merely instructed on how to perform a predetermined function but are encouraged to veered off script. One is not merely expected to copy steps but to reverse-engineer, alter, and embellish them. Pedagogical design therefore combines replication with creativity. It is assumed that the student

[10] segmentation fault

will not be content to simply obey but will appropriate the teaching material as raw material to be worked over further. This activity is a metaphor for what Lévi-Strauss called bricolage, which is an improvisational adaptation of materials on hand in problem-solving; and is a key to the operation of hacker learning cultures.

What we have here is an extremely bounded learner identity. The hacker is not framed as a docile student but as an autonomous agent with the capacity for independent exploration. The process of learning is not discussed in terms of the acquisition of competencies within a prescribed curriculum, but in terms of ethical disposition, inquiry, and action under conditions of uncertainty. In its vocabulary and pedagogical tone, Phrack does not merely educate content; it inscribes an epistemology and a value system that specifies what it is to know, to learn, and to belong. This section has illustrated how technical experimentation is a double-edged tool. It is epistemologically, learning by doing. Ideologically, it creates an independent, self-directed learner who resists formal educational systems.

This bounded but enabled learner identity is not merely discursively framed through the vocabulary and tone of Phrack, but also structurally infused in the format of its tutorials. As indicated in Figure 3.4, the ubiquitous application of the word "tutorial" is not merely a general signal, but an epistemic signal. These tutorials do not convey information in a traditional pedagogical form; instead, they invite readers into doing knowledge by action, risk, and improvisation.

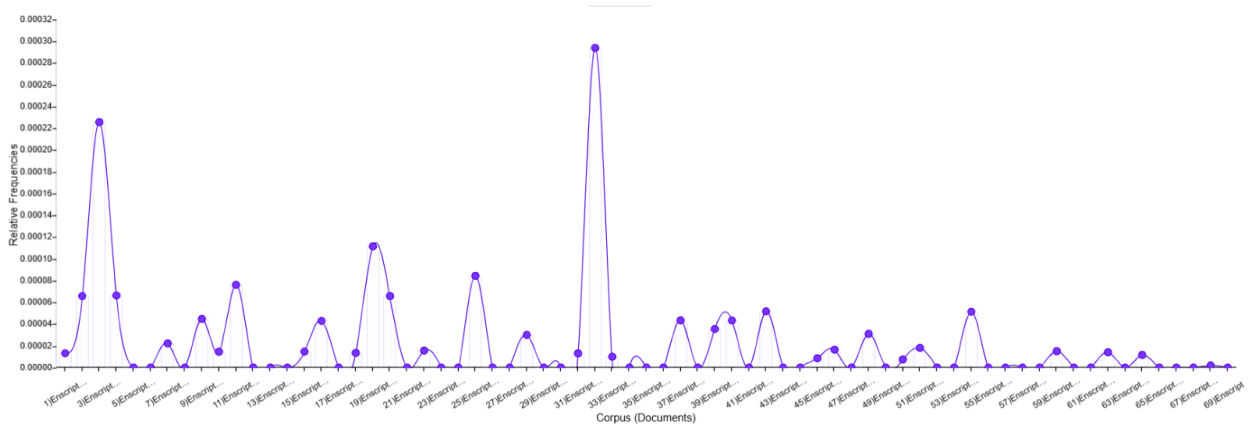


Figure 3.4 *Trend analysis of the term "tutorial" across selected issues of Phrack Magazine (1985–2016), created using Voyant Tools. The graph reveals the magazine's ongoing commitment to hands-on learning and peer guidance, with frequent use of "tutorial" as a genre marker for hacker pedagogy.*

The majority of Phrack tutorials deliver readers with incomplete scripts, abstracted steps, or bare functional code snippets that require interpretation, debugging, or heavy modification before they

can be applied successfully. Phrack's tutorial architecture resonates with Vygotsky's Zone of Proximal Development philosophy in that it exposes students to difficulties just beyond their immediate control and stimulates problem-solving in the guise of scaffolded hints instead of direct instruction. Here, Phrack presents no hardwired solutions or definitive walkthroughs. Instead, it provides adequate scaffolding, in the guise of hints, partial instructions, and piecemeal explanations to lead readers to independent discovery. The tutorials' incompleteness is a deliberate instructional design feature, not an editorial deficiency. It is based on an assumed capable, investigating learner and places the reader in an active building-space of knowledge rather than a passive consumer setting.

This approach welcomes what might best be called productive ambiguity, in which the student is asked to fill in gaps, make intelligent guesses, and verify and test their own hypotheses through experimentation.

"Finding the correct offset is left as an exercise for the reader." Phrack Issue 58, Article 5

Pedagogically, the tutorials thus function more as provocative prompts than as authoritative lessons, mirroring actual conditions under which technical expertise is created rather than transmitted. This echo situated learning theory (Lave & Wenger, 1991), which contends that learning is better accomplished when it occurs in context, in situated action with the practices of a community.

Another aspect that is noticeable is *Phrack*'s sustained effort to normalize and redefine failure in pedagogical discourse. Rather than being framed as a departure from the normative learning trajectory, failure is anticipated, embraced, and even ritualized in the process of technical coming-of-age. Phrases like the one below appear in nearly every article:

"You will probably crash your system a few times before this will work." (Phrack Issue 62, Article 6)

These statements are more than mere technical disclaimers; they operate rhetorically to reframe failure as evidence of meaningful activity rather than incompetence. One tutorial offers the following assurance:
"Don't worry if you crash your machine a couple of times, that just means you're learning."

(Phrack Issue 58, Article 0x05)

Crashes, misfires, and dead code are not warning signs that they are indicators that the learner is operating at the edge of their competence, a foundational condition of Vygotsky's Zone of Proximal Development.

What is missing most from Phrack rhetoric is a sustained investigation of the affective toll of its pedagogy. Where the magazine celebrates perseverance, curiosity, and failure as educational virtues, it's silently absent about the emotional labor involved in keeping going in high-risk,

technically demanding environments. Trial-and-error learning, especially in spaces where system failure, ambiguity, and legal liminality are the norm, can produce not only capability but also anxiety, fatigue, and burnout. This kind of hacker is usually mythologized as a stoic figure, emotionally impervious to frustration or disappointment, in a manner that effaces the invisible affective labor required to withstand endemic failure, interpret ambiguous feedback, or subsist on protracted self-instruction. This silencing aligns with a broader trend in technical subcultures to reduce affective labor and maximize grit, thereby concealing the psychological tension embedded in the very culture of autonomous learning. Paying attention to this affective domain would make things more complex for the prevailing legend of hacker competence, putting onto center stage not just what learners know or do, but what they feel and hold.

As shown in Figure 3.5, terms related to failure recur consistently across issues of *Phrack*, with the noun form "failure" appearing more frequently than the verb "fail." The prominence of *failure* as a discourse object rather than a momentary outcome reinforces its structural presence in the magazine's learning model. The visual rhythm of this term's recurrence suggests failure is not a deviation, but an integrated phase of procedural learning.

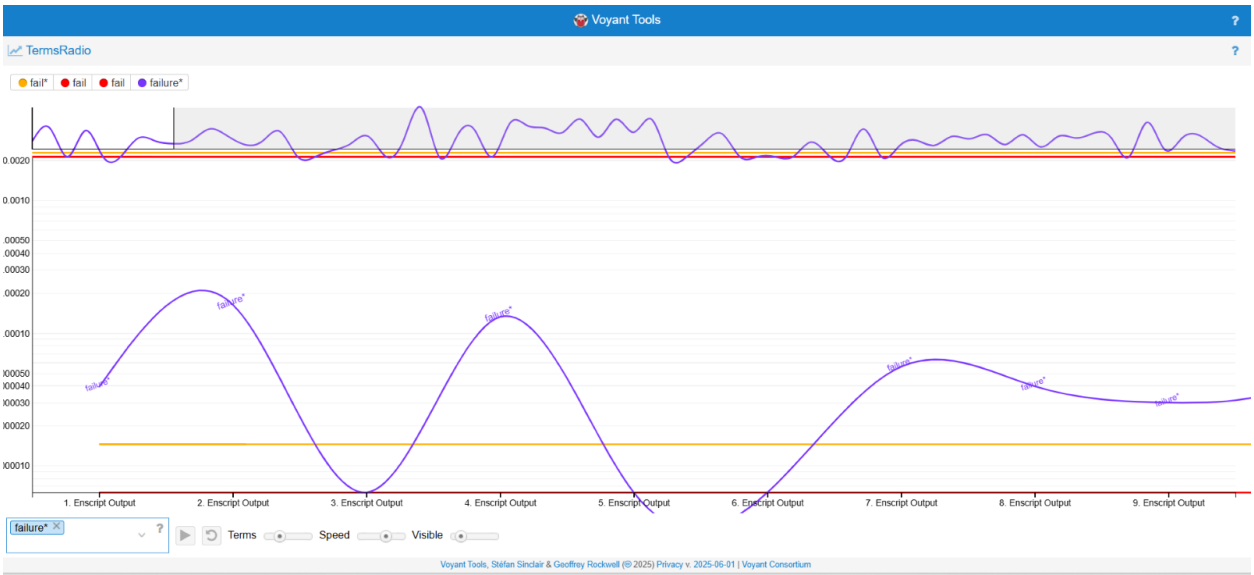


***Figure 3.5**: Term frequency trends for "fail," "fail*," and "failure*" across selected issues of Phrack, generated using Voyant Tools. The data shows the recurrent presence of failure-related language, supporting its discursive normalization as a feature of active technical learning.*

To resist negative thoughts on failure, Phrack fosters a particular pedagogical culture in which perseverance, resilience, and critical thinking are preferred to success. Failure is epistemologically meaningful. It encourages the student to reflect, work out problems, and rethink assumptions. This is echoed by the pedagogy of "productive failure" (Kapur, 2008), through which students struggle

initially with difficult tasks and learn a deeper amount of material than students who are shown correct methods to begin with.

Secondly, the rhetorical framing of failure contributes to the construction of a particular kind of learner subject in the Phrack discourse. The ideal reader is one who is not deterred by failure, motivated by interest, and equipped with the perseverance to master by iterated trial and error. Technical resilience thereby becomes a cultural expectation as much as a pedagogic strategy. Mastery is not just the ability to execute instructions, but the will to withstand system crashes, decoding problems, and interpretive ambiguity without recourse to external authority.

This failure orientation also has ideological implications. By celebrating independence and self-determination, Phrack indirectly condemns institutional models of learning that prioritize efficiency, standardization, and correctness. Instead, it holds a learning model in place that is centered on autonomy, resistance, and the enthronement of subversive curiosity. This does not mean that it is entirely inclusive. While the text invites the reader to embrace failure, it also presumes confidence, access, and literacy that already exists. The ideal student is already someone who is accustomed to the hacker culture's expectations and technologically literate enough to engage in high-risk experimentation, such as crashing one's own computer.

In combination, therefore, Phrack's approach of incomplete instruction and normalized failure offers a powerful exemplar of learning in the real world. It builds dense technical literacy without instruction but through located practice and epistemic experimentation. But it also inscribes specific cultural stories about whom it is possible for such success to be achieved by, stories that curate inclusion and legitimacy for the broader community of hacker learners.

3.4.2 Peer Learning and Informal Mentorship

Another essential characteristic of Phrack as an informal learning setting is its strong formalism of peer-to-peer instruction. Learning is decentered but distributed among practitioners. This is evident in the corpus through linguistic structures that index tutorial voice, community support, and cooperation.

Voyant's Contexts panel shows repeated occurrence of sentences like:

"This is for newbies…"

"Let me walk you through…"

"You might want to…"

"Don't forget to ask on IRC if you get stuck."

Such sentences illustrate what Erving Goffman (1981) terms "footing" in conversation, the coordination of speaker and listener footing that decides how participation, closeness, and

dominance are negotiated in talk. In Phrack tutorials, authors adopt the footing of peer tutor or co-learned rather than authoritative teacher. The tone is level rather than hierarchical. Tutorials are not delivered as canonical or institutional instructions but as local acts of information sharing. The author is not an authoritative speaker with a background of experience, but rather a practitioner reflecting on how he learned and inviting others to join him.

This approach also harmonizes in strong sympathy with Lave and Wenger's (1991) theory of legitimate peripheral participation, where newcomers are gradually introduced into a community of practice through learning, not training, but by observation, experimentation, and socially situated practice. In Phrack, tutorial content works as an invitation to participate and not by insisting on mastery, but by making provision for steps, errors, and learning. The value is less in the information and more in the discursive structure that holds out the promise of inclusivity and mutuality.

This inclusive stance is further reinforced at the micro-linguistic level, particularly using inclusive pronouns like we, you, and us. These pronouns dissolve the boundary between author and reader, constructing a shared subject position within the hacker community. Encouragement and hedging further softens the authoritative stance. The sentence that was discussed earlier can apply as an example here:

"Don't worry if you crash your machine a couple of times, that just means you're learning." Phrack Issue 58, Article 5

serve as strategies of reassurance. They dispel the fear of failure, break up stereotypes of standoffish or elitist hackers, and introduce instead a culture of support and perseverance. Such rhetoric serves to place Phrack not just as a magazine, but as a learning common, a discursive space where knowledge is circulated informally, lines of expertise blurred, and participation encouraged at various levels of experience.

One particularly vivid example of this pedagogical construction is in tutorial "The Basics of Buffer Overflows," in which the author opens by stating:

"This article is not meant to be highly technical, nor is it written for experts." (Phrack Issue 55, Article 15)

This sentence invites not just the technically inexperienced but does so through something greater than words. It is an invitation to the hacker practice that has been built into the text. It reduces the cognitive and emotional barriers to entry. The author anticipates discomfort, acknowledges disjuncture in prior knowledge, and makes an overt move in the tutorial from exclusion to being involved. Such an address is pedagogically useful: it establishes a line for novices not just to get information, but to take up common norms, to become possessed of common tools, and to begin constructing their own practices.

Theoretically, this sort of discourse has much in common with informal apprentice patterns in maker communities, open-source projects, and other peer-based learning settings (Kafai & Peppler, 2011; Kuznetsov & Paulos, 2010). Motivation in such cases comes from participation and contribution and not from institutionalized methods such as evaluation or certification. The affective component is equally important. Phrack regularly has personal aside, afterthought, or even failure description, as authors show how they flubbed through the same stuff. This, they break down the illusion of smooth competence and instead demonstrate mastery as incremental, sloppy, and communal.

Desacralizing mastery by recounting their own mistakes, Phrack authors are characterizing themselves through an understated engagement. They overthrow the myth of the hacker as a mystical coder and replace it with a more pragmatic, democratic model of education based on struggle, repetition, and gradual improvement. This move from rhetoric is ideologically significant. It challenges the technocratic assumption that technical spaces are the domain of high-level abilities only. Instead, it constructs a more inclusive epistemology about effort, persistence, and curiosity and promotes teaching as a form of helping meaning that a partner helps the other one which States the equality in position and no mastery of one on the other. This can be seen in *KWIC (Table 3.2).*

Left Context	keyword	Right context	Issue	Source
"The system will walk you through its on-line"	Help	"registration procedure."	#32	Knight Line I Part 3
"Ask on alt.irc how to get and compile the latest"	Help	"IRC client and connect to IRC."	#45	Line Noise Part II
"All the people who consistently come on irc and"	Help	"ask 'teach me how to hack'..."	#52	Phrack Prophile on o0
"There is some newbies who's really intelligent,"	Help	"and this is important to give him info about what they want."	#56	Phrack Loopback
"Just curious if this is run by newbies, or if"	Help	"there are any old-school people who might remember Delta-Master."	#58	Phrack Loopback

Table 3.2 *KWIC (Key Word in Context) examples for the word "help" in Phrack Magazine, illustrating its use in expressions of peer support, community learning, and informal mentorship. These instances reflect the pedagogical tone that encourages novice participation and knowledge exchange*

However, such open and hospitable stances are not boundary-less. The informal tone, cultural references, and assumed technical expertise continue to serve as soft boundaries, preferring already partially enculturated participants. There is also the benevolence of tone alongside an insider richness of language and value system. Therefore, Phrack's pedagogic discourse is inclusive and selective in the sense that it welcomes correct participation while still marking out its subcultural specificity.

3.4.3 Intrinsic Motivation: Curiosity, Autonomy, and Resistance

Hacker motivation is most likely the most ideologically charged aspect of Phrack's discourse of learning. In Phrack corpus, hacking has constantly been referred to as a useful activity and not framed as a deviant or criminal act. It has been seen as an epistemically driven practice of exploratory autonomy and critical ethicality. In this discursive formation, the hacker is positioned not simply as technician but as autonomous learner engaged in acts of discovery that transgress institutional borders. Voyant's Trends feature discovers high-frequency terms such as curiosity, freedom, explore, learn, knowledge, and access. These lexicon patterns reflect a value system where self-directed learning is not only esteemed but positioned morally good. In contrast to credentialism and gatekeeping of formal schooling, Phrack espouses a culture of learning founded on initiative and ideological dissent. This lexical emphasis is ideological rather than just descriptive. These terms represent hacker values of resistance, transparency, and epistemic self-determination rather than just naming actions.

This discursive mode is both evident in close readings of the text and that it returns distinctly through corpus-wide analysis. The following word cloud presents a visual display of Phrack's prominent lexical fields, affirming the overall significance of epistemic autonomy, experimentation, and access to its learning discourse.

Figure 3.6: *Word cloud of 25 issues of Phrack Magazine (1985–2010), visualized through Voyant Tools. Prominent lexical keywords such as "experimentation," "access," "hacker," "learn," "curiosity," and "free" are where the magazine's ideological investment in autonomy, epistemic agency, and self-guided discovery is located. Anything but technical terms, these words mark the value-infused ethics of resistance, openness, and anti-institutional learning that underpin Phrack's pedagogical paradigm.*

The ethic of focusing on learning rather than the eagerness to cause trouble is most famously identified in the work " The Conscience of a Hacker ", where the author mentions:

"We explore... and you call us criminals. We seek after knowledge... and you call us criminals. We exist without skin color, without nationality, without religious bias... and you call us criminals." (Phrack Issue 7, Article 3)

This was most read as a manifesto; but the passage functions more specifically as an expression of identity. It resists external legal and moral categorization and instead builds a conception of hacking as a type of internally driven learning. It supposes the hacker as an agent of intrinsically driven learning for discovery's sake rather than financial gain, institutional praise, or public renown. This build is taken directly from Ryan and Deci's (2000) self-determination theory, where autonomy, competence, and relatedness are identified as intrinsic psychological needs that promote intrinsic motivation. Phrack affirms all three elements in its pedagogical discourse.

Autonomy is enabled by open-ended discovery, where students are encouraged to experiment, mix things up, and generate their own problems. Competence is achieved through emphasis of technical ability and iterative acquisition of expertise, most significantly in tutorials fostering profound

knowledge of low-level system internals. Relatedness comes through peer respect, sharing of knowledge, and active belonging to a broader community focusing on giving rather than getting a credential.

Central to this is Phrack's depiction of learning as a political act instead of a psychological or cognitive one. Education is defined directly as a struggle. Rather than rendering knowledge neutral or depoliticized, the magazine directly depicts formal institutions as engaged in reproducing power. In one editorial, this stance is articulated explicitly:

"Academia teaches obedience, not insight." (Phrack Issue 66, Article 8)

This resistance to formal schooling relocates Phrack as a counterinstitution, or what can best be referred to as a "counter-school." It offers a new model of technical literacy grounded in resistance to standardization, stratification, and surveillance. From this, Phrack embraces the practices of computer science not to suit corporate or academic advancement, but for empowering oneself, collective knowledge production, and resistance to power.

This model of oppositional learning is also simulated within several articles that map self-directed learning paths. For example, Where to Start with Exploit Development (Gera, 2004, Phrack Issue 62) frames an exploratory rather than linear development of skills in a straight line. Rather than presenting a set curriculum or prerequisites, the tutorial promotes multiple entry points and encourages learners to choose their own path. This model defies the formal hierarchies of conventional computer science pedagogy, where students are typically tied by level, certification, or test achievement. Phrack, in contrast, imagines knowledge as fluid, situated, and socially distributed. Power is earned through contribution, not imposed through institutional role.

These discursive shifts construct hacker motivation both as a personal temperament and as an ethos shared by a group. Learning is both an individual right and a collective responsibility. Curiosity is not a form of distraction from proper learning but rather its base. This conception of motivation and learning positions Phrack in direct opposition to prevailing educational models, and in the process, it remakes what it means to know, to teach, and to be a member of a technical culture.

3.4.4 Ethical Sharing of Knowledge

The fourth discursive support of Phrack's own self-conceptualized approach to informal learning is its expressed and rigorous adherence to the open knowledge approach. This can be discerned, not only in what it publishes, but in the way it is written. Across the corpus, the Summary and Cirrus panels in Voyant spot a consistent use of terms such as free, public, open, share, and access (**figure 3.6**). These are used deliberately. They declare a pedagogical and ideological dedication to information as a collective asset, not an individualized commodity.

This ethos is reaffirmed in even greater detail in the editorial preface to most of the work, which may begin or conclude with declarations such as:

"This document is freely distributable and may be mirrored anywhere." (Phrack Issue 50, Article 0x01)

"Phrack Magazine is made available for free and is encouraged to be spread like a virus." (Phrack Issue 52, Editorial)

These terms evoke strongly the values of the copyleft [11] and Free Software movements [12], particularly the ideologies set out by Richard Stallman (1999) and later developed in detail by Christopher Kelty (2008). In these notions, technical data and software are thought of as public goods. They are to be modified, redistributed, and shared rights that not only lie at the heart of innovation but also promote democracy. Phrack absorbs and extremifies these tenets, taking free information flow not just as a technical or legal matter, but as an ethical one. Access isn't something to be granted as a reward for institutional membership or paying dues. It is offered as a right, and more importantly, as an obligation to belong to the hacker community.

This ethical background is repeated throughout the different issues. In one of the editorials, the author quite rightly connects open knowledge with communal responsibility:

"If you don't share your knowledge, you're no better than the suits selling certifications." (Phrack Issue 65, Article 10)

Here, the sharing of information is a test of membership and legitimacy. The return to the group is not voluntary but intrinsic to hacker identity. It is a form of articulation of what might be called recursive pedagogy, in which the learner is also being prepared to become a teacher. Knowledge circulates not through instructional hierarchies, but through contributions, critique, and revision. Differences between novice and expert, learner and teacher, are consciously eliminated.

This is also a community epistemology that is framed in contrast to commodified and institutional education. Phrack frequently disparages training and certification organizations that only accept paywalls. Publications like "Against the InfoSec Industry" (noid, 2009, Phrack Issue 66) and "Certs are for Clowns" (raT & alx, 2008, Phrack Issue 65) ridicule industry certifications and question their legitimacy. These are not only rhetorical complaints; they are guided by a political reading of education as a site of control, exclusion, and economic stratification. The credentialed expert is too commonly characterized as one educated to follow scripts instead of having a profound knowledge of systems. The hacker, on the other hand, is characterized as one who is educated through discovery, cooperation, and critical engagement with issues of the world.

This oppositional stance can be in such editorial comments as:

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[11] **Copyleft** is a software license model that allows free use, modification, and redistribution of software, under the condition that such freedoms be upheld in derivative works (Kelty, 2008, p. 113).

[12] **The Free Software movement** imagines source code as public good rather than as proprietary commodity and maintains users' autonomy to run, study, modify, and share software (Kelty, 2008, p. 99).

"The InfoSec industry sells snake oil and fear. We offer knowledge, for free." (Phrack Issue 66, Article 8)

"The CEH is a joke, and the people who wear it are clowns." (Phrack Issue 65, Article 10)

These declarations not only reject the commodification of knowledge but reinstate the political and ethical aspects of technical learning. They align with what Henry Jenkins (2009) calls participatory culture, a learning space characterized by low levels of participation barriers, informal mentorship, and high support for information-sharing and production. In Phrack, there are no degrees awarded, or formal certification offered. Instead, the learners gain reputation, recognition, and respect from their peers. These forms of informal legitimacy are very strong in hacker cultures, where authority tends to be given based on demonstrated proficiency, creative problem-solving, and knowledge-sharing.

As a theory of learning, this model also intersects with the principles of distributed and informal learning. Knowledge isn't pushed from a single source to passive recipients. Rather, it is co-created through decentralized interaction. Articles typically come from community conversation, evolve out of reader response, and are refined in subsequent issues. The community is the curriculum. All contributors are learners and potential nodes in the teaching net.

This revolutionary reengineering of pedagogy transforms Phrack from a magazine into an epistemic common. In these commons, knowledge is not monetized but shared, not static but remixed, and not capitalized but given away. Learning is political because it upends the logic of property of institutions and embodies the hacker ethic of autonomy, solidarity, and access. This model destabilizes not just usual boundaries among formal and informal learning but also reasserts the right to learn as not divisible from the right to share.

3.4.5 Pedagogy through Humor and Satire

One of the less immediately visible but pedagogically significant features of Phrack's pedagogical space is its systematic use of humor and satire. While these rhetorical strategies might initially appear to be shallow stylistic ornaments, they are a deep-seated educational and cultural element. In Phrack, humor is never used to castrate learning but to make learning difficult. It is a way of identity formation, boundary maintenance, and critical practice. Most importantly, it makes information transmission a culture-embedded process requiring active interpretation. Factual matter in most articles is disseminated not in the form of stern directive but in the guise of jokes, satirical guides, or caricatured persona-performance. Such genres do not only tell; they require the reader to decode, infer, and adopt the discourse norms of hacker culture.

Evidence of the trend is most prominent in the Voyant Summary and Contexts panels, in which repeated titles and modal templates are loaded with irony and campy overkill. A few examples are: "How to fry your BIOS in three easy steps", "The art of social engineering; 101 ways to ruin someone's day", and "For those who like to learn things the fun way (i.e., by destroying things)".

These are not to be taken seriously as instruction titles. Instead, they are complex texts with layers of warning, joke, and inside jokes. They are placing a signpost with the reader to let them know that learning here is not going to be an easy mental exercise. It is going to be an interpretive and performative act, solidly based on subcultural codes.

This is exactly what James Paul Gee (2007) said about becoming a member of a Discourse, not learning about knowledge, but becoming some sort of person by being involved in a community of practices, values, and conventions of language. The reader is not so much being taught how to manipulate systems or know about exploits. They are being called upon to become hackers, to play with knowledge in a lighthearted, irreverent, and critical manner. To get the joke is to be part of the community.

Jokes in Phrack also serve as a boundary mechanism. They distinguish between those who are competent in hacker speaking and those who are not. Those readers who "get the joke" implicitly show cultural competence. Those who fail to see the irony or take things at face value betray themselves as outsiders. In this manner, satire is a type of soft evaluation. It examines readers not on technical proficiency alone, but on whether they can see through tone, decipher coded meaning, and place knowledge in terms of community values. This makes humor an educational gatekeeper, the one that operates in silence to determine who is "in" and who still hangs on the periphery.

This discursive play has its roots deeply rooted in the hacker zine tradition, in zines such as 2600, Cult of the Dead Cow, and others that blended political analysis, absurdist performance, and irreverent satire (Coleman, 2013; Duncombe, 2007). Humor here is not against learning. It is socialization, critical thinking, and resistance. It is a form of radical pedagogy in which students are not only invited to question authority but taught to do so in irony, naughtiness, and rhetorical acumen.

A particularly characteristic example is found in a Phrack essay that pseudo-teaches readers "How to get caught" in issue 49. Apparently, the essay is a subversive do-it-yourself guide to failure. But from here on in the reader, the article is a satirical deconstruction of poor operational security. The article uses inversion and overstatement to accuse common mistakes and to ridicule superficial ways of hacking. The humor requires the reader to question and introspect but does so insidiously, using parody as a pedagogical tool.

This pedagogy is situated within a wider discursive framework in which learning takes place as subversive play, involving criticality, reflexivity, and comfort with ambiguity. By locating learning in performative irony, Phrack requires more of its readers. It requires them not merely to learn what is offered up but to adopt a mode of thinking, one that is cynical, culturally aware, and aware of power dynamics. The irreverent voice is not disconnected. It's an indicator of insider literacy and a way of charting the paradoxes of hacker culture, both lighthearted and politicized, technical and social.

Humor is not merely performative but is at its core pedagogical. It works epistemologically by requiring interpretive labor meaning that readers must decode satire, irony, and hyperbole to arrive at underlying messages, mirroring the experiential and indirect nature of learning valued in hacker practice. Ideologically, humor reinforces identity boundaries, paying rewards to insiders, and punishing outsiders. In this way, humor serves as a gatekeeper that both educates and tests cultural fluency, instilling basic hacker values of irreverence, autonomy, and anti-authoritarian critique into the process of learning itself.

Ultimately, humor in Phrack creates a site of discourse where autonomy, critique, and literacy are mutually supporting each other. Readers are not merely acquiring skills. They are apprenticed to a rhetorical habit, a normative system, and an epistemic orientation that prizes wit, irony, and autonomy as forms of knowledge. This habit may exclude some, but for those who are in tune with its codes, it builds a sense of belonging that is as affective as cognitive.

The following conceptual mind map (Figure 3.7) summarizes the five key pedagogical themes recognized in Sections 3.4.1 through 3.4.5. These groups of themes are learning-by-doing, peer support, intrinsic motivation, knowledge sharing, humor and satire. These are the primary discursive practices by which Phrack functions as an informal space for learning.

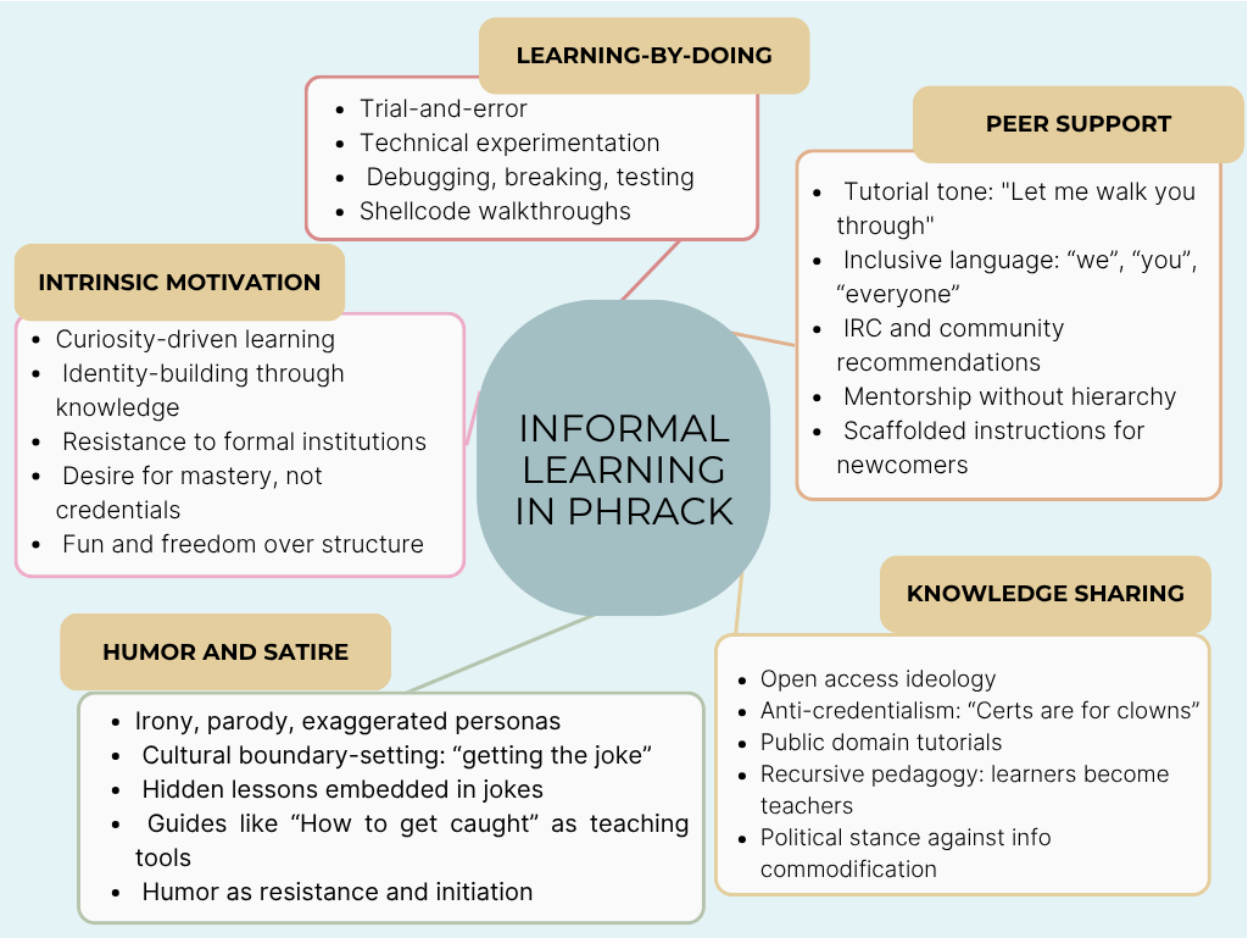

Figure 3.7: *Concept map of informal learning practices in Phrack Magazine. The five thematic nodes represent the modes in which informal pedagogy is enacted discursively through technical practice, peer mentoring, ideological commitments, and rhetorical style.*

This integration ensures that the pedagogy of Phrack cannot be explained by technical pedagogy. Instead, it brings into perspective a discursive ecology where cultural, political, and rhetorical elements work together to construct an individual space for learning. The unplanned learning spoken of in Phrack is strongly rooted in the development of identity, mutual meaning-making, and ideological resistance. As explored in the next section (3.4.6), this construction is itself marked by critical silences, particularly gender, allowing counterclaims of exclusion and limitation.

3.4.6 Gendered Absence in a Historical Male-Dominated Space

While the preceding sections have explained how Phrack constructs an amateur and professional learning environment marked by autonomy, experimentation, and ideological resistance, the same discourses have substantial silence. One such major silence is the absence of gender as a category of analysis, identity, or representation. The openness cultivated in Phrack's pedagogy obscures a major absence that deserves greater exploration. This final section (3.4.6) addresses the gendered silences in Phrack's learning discourse and argues that what appears open and inclusive relies on the normalization of a masculinized subject learner.

Although Phrack seems to be a freely open and anti-institutional site of technological learning, it is marked by an all-pervasive silence about gender. Pseudonym selections such as "The Mentor," "The Plague," and "Erik Bloodaxe" evoke images of male authority, conflict, or outsider rebellion. It shows that there are no female-coded names, pronouns, or inclusive language used across decades of publication suggests more than a demographic lack. It betrays a discursive logic that heterosexualizes the hacker by default. This tacit gendering of the ideal student reinforces the boundaries of the hacker community and quietly discourages participation on the part of those who do not meet the masculine-coded ideal. It is through these rhetorical silences and elisions that Phrack performatively enacts a restricted model of hacker pedagogy and identity.

This silence is less a matter of lack of representation than of its discursive formation. This silence is not a passive historical residue from a bygone computing age, nor an accidental oversight. Rather, it is itself a discursive strategy, the one that constructs a universal learner-subject while tacitly inscribing masculine norms of risk, autonomy, and technical aggression. The absence of gender discourse thus performs active cultural work, establishing masculinity as the unmarked norm. Nevertheless, follow-up work could explore whether counterexamples or faint cracks in this norm are discernible in the corpus.

With more than two decades of difficulties, no sustained investment in gendered experience, no acknowledgment of positions of the subject other than male, and nearly no speech directed to inclusion beyond the ideological gesture of openness, it may well be the case that the beginning cannot be spoken. Feminist theorist Joan Scott (1990) argues that silence is never neutral but works actively to define the limits of the thinkable and the sayable in a discourse. In Phrack, the absence of explicit mention of gender and constructs of identity creates the seemingly universal learner subject. However, this universal is quietly gendered. The cultural ideal hacker is a subject already situated with the cultural codes, language norms, and technical values of a masculinized elite subculture.

This exclusion is not brought about through overt misogyny but through the normalization of a learning subject as one who holds technical aggression, independent problem-solving, and control of systems regardless of social mediation. These features have been consistently encoded as masculine in popular and scholarly descriptions of computing cultures (Wajcman, 2004; Faulkner, 2001). The tutorials, manifestos, and essays that clutter Phrack assume a culturally confident reader, not only against technology but against risk, confrontation, and subcultural knowledge. To the extent that Phrack's editorial voice regularly encourages readers to crash their systems, test exploits on real targets, and probe legal and institutional boundaries. These activities are framed as not merely technically competent but ideologically correct. But they also suggest a subject who is already at ease with their access, already capable of recovery from failure, and socially placed in such a way as to be able to engage in high-risk experimentation. This assumed safety is not universally accepted. As feminist technology historians have clearly illustrated, technical spaces are entered through gendered histories of exclusion and inclusion, which decide who anticipates being capable of participating, fail, or occupying space (Ensmenger, 2010; Misa, 2011).

The grammatical structure of Phrack also reinforces this exclusion. Voyant Tools' analysis of the corpus shows the lack of use of gendered terms such as woman, female, girl, feminism. As Figure 3.8 illustrates, masculinity-related words such as guy, man and boy are found with many more instances of frequency than any of woman, female, or girl. Feminism is mentioned only once in the entire corpus of nearly 200,000 tokens in this phrase:

''They argue feminism, political campaign funding, TV violence, landmines, sex and Nazism. There is even a fun group that regularly argues the perennial subject of world domination by hamburger franchise (it's called alt.nuke.the.usa).'' Phrack Issue 49, Article 1

Voyant Tools

Terms

	Term	Count	Trend
1	men*	1943	
2	guy*	1029	
3	guy	546	
4	pal*	299	
5	boy*	271	
6	girl*	247	
7	male*	117	
8	woman*	82	
9	female*	75	
10	female	63	
11	gender*	15	
12	feminism*	1	

Figure 3.8: *Masculine-coded and gendered lexical distributions and collocations in the Phrack corpus, visualized in Voyant Tools. Masculine-coded words (men, guy, boy) are invoked with much higher frequency than feminine or gender-inclusive words (woman, female, gender, feminism).*

The KWIC analysis shows that “woman” is typically mentioned in incidental or peripheral contexts, while “men” appears in unmarked, general usage. This disparity reinforces Phrack’s construction of a tacitly masculinized learner subject and the broader absence of gender discourse in its informal pedagogical framework. Where woman appears, as occasionally it does, it will normally be collocated in terms like "woman behind the front desk," suggesting presence in a marginal or backgrounded capacity rather than one integrated into technical identity or competence. This is shown in figure 3.9 and figure 3.10.

8 woman* 82

Distribution:

Collocates:
said (5), pwn (4), named (4), behind (4), year (3)

Correlations:
computer (6665), people (4389), pwn (4246), data (6536), like (6504)

Phrases:
"woman behind the front desk"
"womanizer one of comsex's founders"
"woman behind the front"

Items:

8 woman* 82

Distribution:

Collocates:
said (5), pwn (4), named (4), behind (4), year (3)

Correlations:
computer (6665), people (4389), pwn (4246), data (6536), like (6504)

Phrases:
"woman behind the front desk"
"womanizer one of comsex's founders"
"woman behind the front"

Items:

Figure 3.9, 3.10: *Gendered term comparison in the Phrack corpus using Voyant Tools. "Men" has high frequency and occurs in generalized, neutral contexts; e.g., "information," "people " with masculinity as unmarked center.*

"Woman" has low frequency and occurs in marginal or service roles like "woman behind the front desk," with collocates "behind" and "named" having circumscribed epistemic agency. Together,

they form a discursive construct whereby technical authority is constructed as normative male and gendered others rendered invisible or marginal.

Instead of the words including female essence, highly frequent words such as exploit, break, crack, test, own, and control is present. These action verbs and imperatives demonstrate a relationship with technology that is defined by dominance, power, and instrumental agency. The prevalence of command modes and militarized metaphors in the text makes the effect worse. As much as such words are ideologically neutral in technical subcultures, they have the tendency to promote a particular way of technical engagement. Discursive neutrality has the tendency to hide structural forms of exclusion, as argued by Sara Ahmed (2004). Unnamed inclusion is not criticized. Phrack does not gatekeep overtly, but its self-exclusion of gender reinforces the cultural default implicitly, yet knowledge is not hoarded.

Even those pieces which present themselves as speaking for the community, such as The Conscience of a Hacker by The Mentor, are in an implicitly masculinized position of subject. The piece universalizes the hacker as an information-seeker, institution-critic, and victim of misunderstanding. But the subject that it imagines is safe, resilient, and culturally aligned, an identity consistent with the dominant values of male-coded subcultural resistance. No space exists in that article, or in Phrack overall, for other ways of exposure, technical apprenticeship, or collaborative learning. The magazine thus imitates a narrow stereotype of hackers and learners: independent, belligerent, technically militant, and unmarked by identity. The model of learning thus described in Phrack is thus deeply invested in a masculinist ethic even when it will not articulate its name.

Women’s and gender-non-conformist exclusion in Phrack is not merely a matter of underrepresentation but also one of discursive construction. The magazine habitually constructs the model hacker as autonomous, technically competent, and implicitly male. This reflects a broader pattern documented by Faulkner (2007, p. 334) and Wajcman (1991, p. 41), in which technical competence is closely aligned with masculine norms of control, competitiveness, and individualism within hacker and engineering subcultures. In her 2010 study of feminist approaches to human-computer interaction, Bardzell (p. 1305) emphasizes that such "default user" representations are not politically neutral but systematically privilege certain users while marginalizing others. The absence of counter-narratives or alternative identities in Phrack reifies what Reagle (2013, p. 5) calls the "meritocratic myth," obscuring the ways in which structural inequalities are reproduced even in seemingly open, peer-driven learning environments.

It is ideological in that it makes a space in which learning is organized through unmarked cultural capital. Although Phrack supports freedom of information, intellectual independence, and resistance to authority, it does so based on access to resources, self-confidence in technical experimentation, and expertise in adversarial systems of knowledge. These are not evenly distributed across gendered borders. Feminist analyses of hacker and open-source cultures have long documented how ideals of openness operate to conceal the reproduction of systematic discrimination (Reagle, 2013; Nafus, 2012). On this view, Phrack's pedagogic message cannot be

separated from its cultural politics. It invites readers to participate in a paradigm of learning that appears leveling, but which is underpinned by exclusions that it does not recognize.

Together, these six themes illustrate that Phrack is a complex and ideologically charged pedagogical environment. Its learning ideology, centered on experimentation, cooperation, intrinsic motivation, and unrestricted access to knowledge, offers a clear alternative to formal, institutional education, but with this accessibility come essential boundaries, including unwritten presumptions, for example, about learner identity, emotional stability, and cultural awareness. These requirements are not available to all members. Therefore, Phrack is not simply a useful ground for informal education but also a place where careful consideration needs to be given to issues of inclusion, access, and the underlying infrastructures that determine who can join in and flourish. This double nature requires a critical pedagogy that is attentive to both its emancipatory and constraining moments.

3.5 Methodological Reflections and Limitations

While Voyant Tools have offered rich insights into the discursive construction of informal learning in Phrack, critical reflection is warranted regarding the methodological, technical, and interpretive limitations of employing computational tools for cultural density and ideological location texts. Text analysis across digital environments can detect surface-level lexical structures, thematic frequencies, and term locations, but such observations do not translate into meaning. They must be set with care in the sociocultural and rhetorical context of the text, or they will produce erroneous generalizations and abstractions bent out of meaning.

Data manipulation and exportation were some of the first practical difficulties that I had to confront. Voyant is not capable of providing total control over data output. For example, it does not support direct export of all visual content, such as the word clouds, and neither does it produce clean, easily segmented frequency lists by default. An alternative to not falling into this was to hand-pull frequency information and use ancillary programs, such as WordArt or Python, to make customized visualizations. To persue study, these were a word cloud of thematic keyword frequencies and a concept map that were made to illustrate relational ordering of recurring hacker values and discursive categories. This entailed segmenting the Phrack corpus by the year of publication to track shifts in theme and language across time. While useful, this did pose some problems. Different issues of Phrack vary considerably in length, genre, and consistency of formatting, so any longitudinal comparison is open to variation in token frequency, text structure, and content density.

Another practical constraint encountered was the inability to export Voyant Tools visualizations in high resolution or editable formats. For key outputs such as word collocations, trend charts, and keyword-in-context panels, there were no straightforward vector or raw data exports available. Thus, it was necessary to take screenshots of the visualizations and subsequently enhance or

annotate the images by hand with the assistance of external tools. While this allowed for inclusion of relevant figures within the chapter, it also had restrictions in terms of image quality, consistency in formatting, and reader accessibility. This restriction highlights the need for more research-friendly and flexible features within commonly used digital text analysis tools.

Another difficulty was the selection of appropriate keywords and search terms. Voyant identifies high-frequency words, although they don't need to be the most culturally or ideologically charged. Frequently, meaningful ideas are conveyed through low-frequency or metaphorical language words. Terms such as access, for instance, exploit, or control possess technical as well as political connotations. Voyant, without close reading, is unable to distinguish between literal usage in exploit code and metaphorical usage in manifestos or editorials. Similarly, salient terms such as feminism, gender, or diversity can be so infrequent that they are eliminated altogether. Yet their lack of appearance is discursively significant, especially when analyzing cultural silences and elisions. In such cases, what does not appear in the corpus may be telling as much as what appears.

The Phrack lexical field is often context-dependent and polysemic. Terms such as own, crack, or kill might be applied to system access, code manipulation, or cultural domination, relative to context. Voyant is unable to read meaning or purpose and therefore interpretive responsibility falls on the researcher alone. This is particularly the case when dealing with subcultural or transgressive text, where irony, satire, and play represent fundamental rhetorical tropes. For example, a sentence that appears to justify illegal behavior may, in context, be ironic denunciation of technocratic authority. Voyant cannot recognize such modes of rhetoric. It is oblivious to genre, voice, and community codes to understand hacker discourse. The tutorials, editorials, and manifestos of Phrack are heterogeneous in tone and purpose, and any reading which posits that they are homogeneous dangers to misread their pedagogical or political significance.

Additionally, the corpus presents another level of methodological difficulty. While the version used in this study was as exhaustive as possible, it is incomplete. Some issues of Phrack can only be available in broken or incomplete states. The corpus is also heterogeneric. Exploit code, philosophical treatises, political manifestos, satire, and technical commentary all appear within it, and sometimes in the same issue. The texts vary in length, purpose, and rhetorical mode. Therefore, quantitative trends based on frequency counts aggregated into bigger groups must be treated cautiously. Inter-mixing genres encourage richness in analysis but also complicate comparing directly, as the same label can play differently across text types.

Researcher's positionality is another aspect that must be considered. As an outsider and viewer of the Phrack corpus, there is always the risk of misinterpreting or over-simplifying the cultural codes and in-group jokes that structure these texts. What appears transgressive or subversive to an outsider might be entirely normal in hacker culture. on the other hand, an ironic posturing may be interpreted too literally by an outsider to subcultural speech acts. Preempting this, efforts were made to triangulate Voyant analysis with close reading of the most significant articles, with secondary literature on hacker ethnography, critical code studies, and digital sociology. This methodological caution cannot eliminate the risk of interpretive distortion. Further methodological richness can be accessed by incorporating participatory research tools, such as interviews with

productive Phrack contributors or readers, or inspection of changing interactive environments such as forums and IRC channels where these ideas are argued, discussed, and learned in the moment.

Finally, an ethical aspect of the analysis cannot be dismissed. Most Phrack writing discusses actions that fall within legally ambiguous realms, such as social engineering, system penetration, and denial-of-service attacks. Reading them raises difficult questions about the status of hacker knowledge. Should it be built up as public knowledge or as culture-bound knowledge? Is the existence of scholarly debate over potential illegal means to oppose them that will make them legitimate? Where is the line drawn between critique, recording, and complicity? In this case, Phrack has been treated as a pedagogical and cultural text rather than a technical manual. It is not valuable for the practical reuse of tactics, but for what it reveals about how knowledge is produced, communicated, and ideologically framed within hacker culture. Most of the technical information it presents is obsolete, in a few cases by more than a decade, and often much more, so its usefulness to present-day hacking practice is limited. Its rhetorical techniques, ethical justifications, and pedagogical methods, however, remain very much worth learning from. These axes are worthy of additional consideration in future studies, especially those using participant observation or field-based methods that have the potential to examine how these kinds of texts still influence hacker identity and learning in the present day.

Despite these limitations, applications like Voyant can be useful when employed with methodological humility and theoretical nuances. They are not objective means, but discursive means that deliver partial and suggestive views of dense textual landscapes. They give out answers that are suggestive rather than definitive and must always be situated within a critical interpretive framework. Future scholarship could benefit from pairing tools like Voyant with more open, code-friendly environments like Python's spaCy, where there is greater control over preprocessing, filtering, and visualization. However, such tools need to be employed to complement and not replace the deep cultural reading and participatory understanding of populations that they describe.

In summary, the insights created by Voyant Tools have been profound but necessarily limited. They reveal habitual routines, lexical proximity, and thematic signals but not the real deep humor, affect, irony, or social relations. They can be accessed only by qualitative and ethnographic methods that favor voice, context, and meaning. Critical discourse analysis can function as a method to analyze hacker texts like Phrack only if it vacillates between machine reading and human interpretation, between data visualization and ideological critique.

3.6 Summary of Key Findings

In the chapter, a close critical discourse analysis of Phrack Magazine has been carried out considering the way in which the text constructs, spreads, and legitimizes a hacker culture-based model of informal learning. Through a combination of close textual readings and corpus-based evidence generated through Voyant Tools, the chapter has emphasized an interlinked web of mutually connected threads of pedagogic and ideological character. These are not necessarily

always framed as such, but they do function to bring readers into specific epistemic production, identity formation, and belonging modes. Six primary findings were constructed within the analysis, each illuminating Phrack as a space for cultural learning.

Learning-by-Doing as Ground-level Epistemology

Prioritizing learning-by-doing is the most basic and prevalent theme. Across the Phrack corpus, tutorial and technical articles emphasize experimentation by hand, iterative failure, and system manipulation. Learning is conceptualized here not as instructional or sequential but as a bodily, situated engagement with technological systems. Technical understanding is offered as that to be revealed through action and not transposed from novice to expert. This constructivist account is reinforced in John Seely Brown's "learning to be," in which students acquire not only skills but identities and dispositions through the engagement with genuine practice. In this regard, Phrack constructs its audience not as recipients of information but as emergent subjects who become hackers through attempting, falling, and iterating within the system.

Peer Pedagogy and the Role of Incomplete Instruction

A second key finding pertains to Phrack's use of incomplete or scaffolded teaching. Tutorials frequently feature partial code, abstract commands, or suggestive recommendations rather than complete solutions. Such presentation demands active interpretation and user involvement and places the responsibility of learning on the reader. While this may appear to detract from instructional clarity, it in fact follows a strongly evolved pedagogical model consistent with Vygotsky's Zone of Proximal Development. By positioning the student slightly beyond their current capability and challenging them to complete the task using problem-solving, Phrack encourages a model of knowledge that is co-constructed, rather than transmitted. Readers are set up not as students to be taught, but as peers to challenge themselves. Peer pedagogy such as this encourages independence of mind, self-reliance, and risk-taking in searching out knowledge.

Failure as a Cultural and Educational Norm

The third theme is closely liked. It talks about the normalization and pedagogisation of failure. Rather than being stigmatized, failure is anticipated, ritualized, and even valorized within Phrack's discourse. Phrases such as "you'll probably crash your system a few times before it works" are not presented as caveats but as signs that the reader is operating on the edge of their competence. This reframing failure is part of a distinctive epistemology of resilience, under which learning is recursive, flawed, and self-directed. Failure is not something that throws the learner off the learning path but is a natural component of it. In this redefinition, Phrack reframes the hacker as not just a technical master but one who embraces uncertainty, complexity, and risk as learning contexts.

Intrinsic Motivation and Hacker Identity

The fourth theme concerns the ideological construction of motivation. Phrack never positions hacking as an intrinsically motivated activity grounded in curiosity, autonomy, and wanting to understand. Voyant-based lexical analysis revealed the prominence of words such as "explore," "learn," "freedom," and "access," all of which keep a self-directed intellectual pursuit discourse. This stands in deliberate contrast to the extrinsically driven models of formal education, which emphasize credentialing, regulation, and institutional compliance. Here, Phrack articulates a learning identity that aligns with Ryan and Deci's theory of self-determination, where the psychological needs of autonomy, competence, and relatedness drive sustained engagement. The hacker, in this framework, is not a technician or criminal but a self-directed learner driven by intrinsic values and epistemic curiosity.

Open Knowledge as Ideological and Pedagogical Principle

One fifth of the findings are related to Phrack's articulation of open knowledge ethics. The corpus records a common pattern of adjectives such as "public," "share," "free," and "open" employed to place information as a shared good in opposition to proprietary commodity. This vocabulary goes beyond technical accessibility—it performs a political function in placing Phrack within a master of rhetoric of anti-authoritarianism, openness, and resistance to institutional gatekeeping. The magazine does not simply spread information; it redescribes the act of sharing as an ethical and pedagogical obligation. By providing half-code, disassembled binaries, or strategic commentary, Phrack is made a space for collective learning that cannot be institutionalized and privatized.

Gendered Absence and the Bounded Ideal Learner

Finally, the chapter addressed an essential absence within Phrack's discourse: the almost complete erasure of gender as a subject position, experience, or inclusion category. Even as the magazine promotes ideological diversity, its own rhetoric produces a restricted image of the ideal learner, one typically understood to be technically skilled, socially assured, and culturally sensitive to masculine-coded forms of knowledge and danger. Voyant analysis showed a widespread absence of use of terms such as "woman," "female," or "feminism" and instead highlighted the dominance of adversarial action verbs such as "exploit," "own," "break," and "crack." These findings suggest that openness in this magazine is a highly selective kind of openness, favoring those who are already socialized to the norms of hacker culture and familiar with hostile technical worlds. Feminist explanations of technology cultures help to point towards how this apparent neutrality operates to cover up systemic exclusions and reproduce masculinized power in informal learning.

To better illustrate the particularity of Phrack's pedagogical rationality, the following table contrasts its culture of informal learning with the dominant characteristics of formal, institutionally guided education. Their differences are framed on the basis of essential assumptions about motivation, access, learner identity, testing, and epistemic authority.

Aspect	Phrack Magazine	Formal Education

Learning model	Peer-to-peer, informal	Instructor-led, structured
Motivation	Curiosity, autonomy	Certification, career
Access	Free, open, remixable	Restricted, institutional
Authority	Distributed, self-built	Centralized, credentialed
Assessment	*Reputation, contribution*	*Exams, grades*

Table 3.3. *Comparison of Phrack's informal learning culture with traditional education models, highlighting differences in motivation, access, and authority.*

The table depicts the degree to which Phrack's model of informal learning is articulated through autonomy, peer confirmation, and open remixability. These are qualities that harmonize with the very nature of informal and self-organized learning. These qualities proclaim a decentralized pedagogical rationality, one in which technology mastery is nurtured by experimenting, contributing, and building reputation rather than by institutional approval. While this model appears radically open and democratic, it exists within a subcultural context that implicitly sets out in terms of access. Unlike institution-based education, in which inclusion is formally organized (if unevenly) through policy and accreditation, Phrack's model relies on internalized norms and unspoken codes of legitimacy. Openness in this case does not automatically equate to inclusivity. This tension is then compounded when viewed from a gender and identity lens.

It is also to be noted that even as these themes occurred frequently within the corpus, their elaboration and intensity varied over time and across genres. Issues of the earlier era, from 1985 to 1992, were more disposed toward ideological and identity-based writing, e.g., manifestos and commentaries regarding community. Issues from the later period were focused on more technical refinement and exploit documentation. As a result, themes like open knowledge sharing or intrinsic motivation were more prominent and rhetorically built periods. This distribution implies that the pedagogical rhetoric of Phrack was not static, but dynamically altered over time by community goals, legal setting, and technological progress.

Concurrently, these texts reveal a self-reflexive recognition of contradictions within their own discourse. While Phrack is marked by a robust culture of openness, cooperation, and access to information, this openness is not necessarily inclusive across the board. The discussion quite frequently subscribes to a technically competent, culturally assimilated, and generally masculine subject. Rhetorical silences about gender, insider language and oppositional metaphors, function to set boundaries around who is implicitly invited in. Therefore, Phrack's pedagogical culture democratizes access to technical knowledge while reproducing exclusions based on larger patterns in technology subcultures.

Conclusion

As a group, these six findings demonstrate that Phrack is a highly sophisticated pedagogical space that enables and constrains informal learning. It is a model that is intellectually dense, technically challenging, and richly cultural, organized around values of autonomy, discovery, and peer mentoring. But it also has embedded within its subtle degrees of boundary-making, exclusion of identity, and ideological selectivity that limit who can really participate fully in this culture of learning. Rhetorical openness frequently masks cultural barriers to actual access. The language of this magazine reflects the greater paradoxes of informal digital learning environments; they can promote mastery and autonomy but depend on tacit knowledge, common values, and insider codes that are not necessarily available to all learners.

Chapter 4: Conclusion

This thesis examined how Phrack Magazine functions not just as a technical magazine but as a cultural and learning distributed informal site. Guided by sociocultural learning theory (Vygotsky, Lave & Wenger), digital informal learning studies (Reinders & Benson), and subcultural analysis,

this study analyzed how Phrack's discourses do not just teach technical practice but also write hacker identity, values, and ethics. To investigate these dynamics, this study combined Critical Discourse Analysis (CDA) with computational text analysis using Voyant Tools. This mixed methodology allowed for broad quantitative pattern finding as well as close qualitative interpretation of learning performed discursively over the long term of its publication.

The results using Voyant Tools offered robust empirical evidence to the interpretation of Phrack as an unformal pedagogical space. Term frequency data across the corpus highlighted the significance of high lexical words associated with learning and teaching, for example, "technique," "tutorial," "exploit," "guide," and "manual". Even though not always consistent, the language does have a repeated commitment to instructional language and directive pedagogy. A longitudinal analysis of these terms, over the corpus from the early 1985 to 2016, provided a consistency of procedural and technical term use in support of the argument that Phrack has been pedagogically inclined over decades.

In addition to raw frequency counts, keyword-in-context (KWIC) searches also showed that instructional discourse markers, imperative verbs ("try," "test," "see") and peer-directed modal phrases ("you might want to," "let me walk you through"), also tend to appear in proximity to technical terms. These kinds of linguistic features resonate with syntactic constructions normally associated with informal mentorship or tutorial-style writing. Whereas Phrack holds an anti-institutional ideological position, its use of clear, directive language synchronizes rhetorically with conventional pedagogical practices in mainstream education. This discourse situates Phrack as a pedagogical text whereby informal peer-to-peer learning is facilitated through open tone and imitation of instructional genre conventions.

High frequency collocates of terms like "system," "shell," "exploit," and "vulnerability" further revealed the epistemic bent of the magazine. These terms frequently co-occurred with first-person action verbs, emphasizing direct engagement with systems. These verbs like "gain," "access," "test," "bypass" which implied both technical knowledge and interest in active manipulation and intervention in technological systems.

This is a technical commitment ethic that rests on an even more basic epistemological stance. Knowledge in hacker culture is at times legitimized not by abstraction or certification, but by application. in other words, knowledge exists only insofar as it can be performed. This reinforces the hacker ethos of learning-by-doing, whereby knowledge is calibrated by the ability to manipulate, test, and recreate systems. At this point, technical skill is inseparable from practice, and learning itself is a performative act. It is a demonstration of both competence and membership in the culture.

Trend analysis also demonstrated how frequently technical themes were accompanied by ethical and ideological concerns over time. Phrases such as "freedom," "rights," "security," "anonymity," and "authority" appeared in issues that were unrelated to one another and were most often written across editorials or reflective manifestos. These themes did not emerge in isolation but were

discursively linked with technical discourse; meaning that learning in Phrack is simultaneously an act of cultural alignment. This supports the case that the magazine does not simply instruct its readers technically, but introduces them to a broader world of self-determination, resistance, and challenging institutional vigilance.

One of the more unexpected Voyant findings was the prominence of humor, sarcasm, and satire. While computationally harder to detect, certain lexical domains like pop culture allusions, pseudonyms, absurdist section titles, marked a meta-pedagogical strategy: pedagogy through play, parody, and symbolic disobedience. This sort of "subversive pedagogy" lends a unique cultural taste to Phrack's tutorial articles, at once entertaining, transgressive, and didactic. It also shows how the magazine can encode sophisticated social and political satire in the level of technical information.

Collectively, these observations confirmed that Phrack facilitates an extraordinary form of informal learning. It provides an environment within which technical learning is embedded in collective discursive practice, cultural signaling, and ideological belonging. Tutorials and "how-to" columns are direct instructional writing, but editorials, announcements, and satirical interludes serve to enclose, critique, and reiterate the subcultural norms of the hacker community. Such multi-layer discursivity makes it possible for a recursive, distributed pedagogy that is not top-down but dialogic, emergent, and consistent with the logic of open-source collaboration and peer production.

Methodologically, the use of Voyant Tools complemented the research in two ways: (1) to enable large-scale pattern detection that would be impossible through single reading alone, and (2) to elucidate changes in discourse emphasis over time. For instance, earlier difficulties(1980s–1990s) were more in line with increased emphasis on telecommunication networks, phreaking, and low-level system intrusion, while the latter were more concerned with software exploits, privacy, and geopolitical intelligence. These diachronic variations reflect that Phrack aligned its pedagogic content to the changing technological as well as political contexts. The magazine thus not only teaches but transforms; it dynamically adapts to various requirements and situations of its learning community.

Theoretically, the research has several contributions. To begin with, it supports a non-binary, process-oriented definition of informal learning that blurs lines between formal, and informal education. Phrack is taught without certificate, commented on without grade, and promoted without curriculum scripts and learning unmistakably takes place. Secondly, the research shows that learning is a very social and cultural activity. Hacker is not only a skilled learner, but a performer in a symbolic and moral world. Learning in this sense is also identity work. Third, this thesis confirms that digital subcultures can indeed be pedagogical systems, even when they abandon institutional forms. Doing so, the thesis broadens the category of what counts as an educational site. These dynamics also map onto Gee's concept of affinity spaces, in which learning is through the performance of shared practice, rather than instruction. Phrack is one such space, where knowledge is enacted by doing, sharing, and culturally grounded discourse

More generally, this research has important implications for learning in networked, subcultural settings. As educational frameworks grapple while trying to adapt to decentralized, on-demand, and peer-based cultures of learning, Phrack gives a persuasive illustration of just how it is possible to create sustainable, long-term, informal learning. It shows that learners can navigate complicated systems of knowledge, learn from one another, and pass on community values without institutional support. This shifts instructional attention into values like autonomy, curiosity, resistance, and shared culture, rather than mastery of content or measurable outcomes.

The research has limitations as well as strengths. It is text analysis of published issues only and does not include ethnographic understanding, usage measurements, or reception information from the audience. Subsequent work may fill these spaces by researching how readers read Phrack in real time, how they evaluated, used, and circulated its content. Comparative research with other related media (e.g., exploiting databases, hacker print zines, or Discord servers) may also elucidate whether Phrack's pedagogical methods are exceptional or common across technical subcultures.

Lastly, the research leaves several doors open for future work. One trajectory is that of developing pedagogical models that can be adapted by language teachers from subculture-specific environments; such as the use of humor, anecdotes, and anonymity to motivate learning. This is while another is that of applying sentiment analysis or rhetorical parsing methodologies to uncover the emotional and psychological connotation of technical writing. More broadly, scholars must explore the increasingly dominant status of self-directed, collaborative systems of learning across domains from computer coding and cryptography to climate activism and social movement building. As networked digitality becomes the new schoolhouse, subcultural spaces like Phrack will play increasingly important roles in shaping how we come to know, why we come to know, and who gets to decide what knowing can be.

It is essential to highlight that the results of this thesis are historically contextualized. The analysis is undertaken on a corpus of texts that was published between 1985 and 2016, a time frame that witnessed the move from early Bulletin Board Systems (BBS) to a more globalized, internet-focused hacker culture. The pedagogic practices, values, and discursive strategies located in Phrack can be identified with the particular concerns, technologies, and political context of the period. While the magazine is an inspiring example of informal, peer-to-peer learning, it is set in a cultural environment very distant from that of the present-day hacker space. Contemporary surroundings, such as GitHub, Discord, and Telegram, are shaped by other platform logics, legal environments, and social norms. Therefore, this thesis should not be interpreted as a universal description of hacker pedagogy, but as a historical case study that shows learning, identity, and ideology have become entangled within a particular subcultural media form. While considering all these and keeping in mind that Phrack has not been active since 2016, its historical footprint remains foundational to the hacker pedagogical canon. Later research can explore how these discursive practices have evolved, persisted as is footprint, or dissolved in more modern digital spaces.

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Bibliography

- Abbate, Janet. 1999. Inventing the Internet. Cambridge, MA: MIT Press.
- Abdallah, Mahmoud M. S., and Mohammad M. A. Alkhrabsheh. 2019. "Challenges of e-learning quality in developing countries." International Journal of Educational and Pedagogical Sciences 13 (11): 1401–1408.
- Ahmed, Sara. 2004. The Cultural Politics of Emotion. Edinburgh: Edinburgh University Press.
- Allen, I. Elaine, and Jeff Seaman. 2010. Class Differences: Online Education in the United States, 2010. Babson Park, MA: Sloan Consortium.
- Anonymous. 1996. "International scenes." Phrack Magazine, Issue 49, File 3. https://phrack.org/issues/49/3.html
- Arthur, W. Brian. 2009. The Nature of Technology: What It Is and How It Evolves. New York: Free Press.

- Auxier, Brooke, and Monica Anderson. 2021. “Social media use in 2021.” Pew Research Center. https://www.pewresearch.org/internet/2021/04/07/social-media-use-in-2021/
- Bangun, R., R. Saragih, and A. P. Nasution. 2022. “The effect of digital transformation on education and business during the Industrial Revolution 4.0.” International Journal of Educational Research & Social Sciences 3 (1): 135–140.
- Barlow, John Perry. 1994. “The economy of ideas.” Wired. https://www.wired.com/1994/03/economy-ideas/
- Barnard, Debbie, and Marius van der Merwe. 2016. “Innovation management in education: Organizational processes and tools for development.” International Journal of Innovation Science 8 (1): 35–49.
- Bardzell, Shaowen. 2010. “Feminist HCI: Taking stock and outlining an agenda for design.” In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, 1301–1310.
- Becker, Howard S. 1963. Outsiders: Studies in the Sociology of Deviance. New York: Free Press.
- Bennett, Sue, and Karl Maton. 2010. “Beyond the ‘digital natives’ debate: Towards a more nuanced understanding of students’ technology experiences.” Journal of Computer Assisted Learning 26 (5): 321–331.
- Black, Edwin. 1970. The Second Persona: Studies in the Theory of Rhetoric. Madison: University of Wisconsin Press.
- Blankenship, Loyd. 1986. “The conscience of a hacker.” Phrack Magazine, Issue 7. https://phrack.org/issues/7/3.html
- Branwyn, Gareth. 1993. “Jargon watch: Plagiarism saves time.” In Mondo 2000: A User’s Guide to the New Edge, 60–63. New York: HarperPerennial.
- Bratus, Sergey. 2007. “What hackers learn that the rest of us don’t: Notes on hacker curriculum.” IEEE Security & Privacy 5 (4): 72–75.
- Breslow, Lori, et al. 2013. “Studying learning in the worldwide classroom: Research into edX’s first MOOC.” Research & Practice in Assessment 8: 13–25.
- Briscoe, Gerard, and Will K. Hon. 2014. “Hackathons as a means of accelerating technology development.” International Conference on Open Innovation. https://doi.org/10.2139/ssrn.2402667
- Briscoe, Gerard, and Catherine Mulligan. 2014. “Digital innovation: The hackathon phenomenon.” Working Paper No. 6. Creativeworks London.
- Brown, John Seely, and Paul Duguid. 2000. The Social Life of Information. Boston: Harvard Business School Press.

- Brown, Mark, Joanna Dehoney, and Nancy Millichap. 2015. The Next Generation Digital Learning Environment: A Report on Research. EDUCAUSE Learning Initiative.
- Bruner, Jerome S. 1960. The Process of Education. Cambridge, MA: Harvard University Press.
- Buckingham, David. 1993. Children Talking Television: The Making of Television Literacy. London: Falmer Press.
- Callanan, Maureen A., Carla A. Cervantes, and Molly Loomis. 2011. "Informal learning." Wiley Interdisciplinary Reviews: Cognitive Science 2 (6): 646–655.
- Carliner, Saul. 2023. Informal Learning Basics. 2nd ed. Alexandria, VA: ATD Press.
- Castells, Manuel. 2001. The Internet Galaxy: Reflections on the Internet, Business, and Society. Oxford: Oxford University Press.
- Chandler, Daniel. 1996. Engaging with Media: An Introduction to Media Studies. London: Routledge.
- Coleman, Gabriella. 2013. Coding Freedom: The Ethics and Aesthetics of Hacking. Princeton, NJ: Princeton University Press.
- Coleman, Gabriella. 2014. Hacker, Hoaxer, Whistleblower, Spy: The Many Faces of Anonymous. London: Verso.
- Colley, Helen, Phil Hodkinson, and Janice Malcolm. 2003. Informality and Formality in Learning: A Report for the Learning and Skills Research Centre. London: LSRC.
- Corneli, Joseph, and Charles Jeffrey Danoff. 2011. "Paragogical praxis." Planetary Collegium 8 (3): 73–81.
- Danet, Brenda, and Susan C. Herring, eds. 2007. The Multilingual Internet: Language, Culture, and Communication Online. Oxford: Oxford University Press.
- Davies, Bronwyn, and Rom Harré. 1990. "Positioning: The discursive production of selves." Journal for the Theory of Social Behaviour 20 (1): 43–63.
- Davies, Sarah R. 2017. Hackerspaces: Making the Maker Movement. Cambridge: Polity Press.
- Dewey, John. 1916. Democracy and Education: An Introduction to the Philosophy of Education. New York: Macmillan.
- Dewey, John. 1938. Experience and Education. Washington, DC: Kappa Delta Pi.
- Dilipraj, E. 2016. "Hacking: Tracing the history: What can India do with its hackers?" Liberal Studies 1 (2): 239–252.
- Doleck, Tenzin, Paul Bazelais, and David J. Lemay. 2019. "Examining digital informal learning among school teachers." Education and Information Technologies 24 (1): 237–246.
- Driscoll, Kevin. 2014. "Dial-up dreams: Connecting the Bulletin Board System to the internet." Communication, Culture & Critique 7 (4): 510–529.

- Dron, Jon, and Terry Anderson. 2014. Teaching Crowds: Learning and Social Media. Edmonton, AB: AU Press.
- Dron, Jon, and Terry Anderson. 2023. “Informal learning in digital contexts.” In Handbook of Open, Distance and Digital Education, edited by Olaf Zawacki-Richter and Insung Jung, 1373–1389. Cham: Springer.
- Ensmenger, Nathan. 2010. The Computer Boys Take Over: Computers, Programmers, and the Politics of Technical Expertise. Cambridge, MA: MIT Press.
- Faulkner, Wendy. 2007. “‘Nuts and bolts and people’: Gender-troubled engineering identities.” Social Studies of Science 37 (3): 331–356.
- Fairclough, Norman. 1995. Critical Discourse Analysis: The Critical Study of Language. London: Longman.
- Fox, Susanna, et al. 2015. “Feminist hackerspaces as safer spaces?” Journal of Peer Production 5.
- Gee, James Paul. 2004. Situated Language and Learning: A Critique of Traditional Schooling. London: Routledge.
- Gee, James Paul. 2005. “Semiotic social spaces and affinity spaces.” In Beyond Communities of Practice, edited by David Barton and Karin Tusting, 214–232. Cambridge: Cambridge University Press.
- Gee, James Paul. 2011. How to Do Discourse Analysis: A Toolkit. London: Routledge.
- Gee, James Paul. 2007. Good Video Games + Good Learning: Collected Essays on Video Games, Learning, and Literacy. New York: Peter Lang.
- Goffman, Erving. 1963. Stigma: Notes on the Management of Spoiled Identity. Englewood Cliffs, NJ: Prentice-Hall.
- Goffman, Erving. 1981. Forms of Talk. Philadelphia: University of Pennsylvania Press.
- Hackbloc. 2008. “Hacktivism today.” Phrack Magazine, Issue 65, File 8. https://phrack.org/issues/65/8.html
- Hafner, Katie, and John Markoff. 1995. Cyberpunk: Outlaws and Hackers on the Computer Frontier. New York: Simon & Schuster.
- Halbert, Debora. 1997. “Discourses of danger and the computer hacker.” The Information Society 13 (4): 361–374.
- Himanen, Pekka. 2002. The Hacker Ethic and the Spirit of the Information Age. New York: Random House.
- Hofer, Barbara K., and Paul R. Pintrich. 1997. “The development of epistemological theories: Beliefs about knowledge and knowing and their relation to learning.” Review of Educational Research 67 (1): 88–140.
- Ito, Mizuko, et al. 2013. Connected Learning: An Agenda for Research and Design. Irvine, CA: Digital Media and Learning Research Hub.

- Jenkins, Henry, et al. 2009. Confronting the Challenges of Participatory Culture: Media Education for the 21st Century. Cambridge, MA: MIT Press.
- Jordan, Tim, and Paul A. Taylor. 1998. Hacktivism and Cyberwars: Rebels with a Cause? London: Routledge.
- Jurafsky, Daniel, and James H. Martin. 2021. Speech and Language Processing. 3rd ed. Draft. https://web.stanford.edu/~jurafsky/slp3/
- Kelty, Christopher M. 2008. Two Bits: The Cultural Significance of Free Software. Durham, NC: Duke University Press.
- Lankshear, Colin, and Michele Knobel. 2003. New Literacies: Changing Knowledge and Classroom Learning. Buckingham: Open University Press.
- Lave, Jean, and Etienne Wenger. 1991. Situated Learning: Legitimate Peripheral Participation. Cambridge: Cambridge University Press.
- Levy, Steven. 1984. Hackers: Heroes of the Computer Revolution. Garden City, NY: Anchor Press/Doubleday.
- Livingstone, David W. 2001. Adults' Informal Learning: Definitions, Findings, Gaps and Future Research (NALL Working Paper #21).
- Lunceford, Brett. 2009. Identity and Discourse in Hacker Subculture. Amherst, NY: Cambria Press.
- Markoff, John. 2005. What the Dormouse Said: How the Sixties Counterculture Shaped the Personal Computer Industry. New York: Viking.
- Maxigas. 2012. "Hacklabs and hackerspaces—Tracing two genealogies." Journal of Peer Production 2.
- Mazzini, Francesco. 2023. Hackers. Storia e Pratiche di una Cultura. Rome: Laterza.
- Meyer, John P., and Douglas Thomas. 1990. "The computer underground: A reassessment." In Proceedings of the Conference on Computers, Freedom, and Privacy, 1–10. New York: ACM Press.
- raT and alx. 2008. "Certs are for clowns." Phrack Magazine, Issue 65, File 9. https://phrack.org/issues/65/9.html
- Raymond, Eric S. 1996. The New Hacker's Dictionary. 3rd ed. Cambridge, MA: MIT Press.
- Reinders, Hayo, and Phil Benson. 2017. "A research agenda for language learning beyond the classroom." Language Teaching 50 (4): 561–578. https://doi.org/10.1017/S0261444817000192
- Rogoff, Barbara. 1990. Apprenticeship in Thinking: Cognitive Development in Social Context. Oxford: Oxford University Press.
- Stallman, Richard M. 2001. Free Software, Free Society: Selected Essays. Boston: GNU Press.

- Sterling, Bruce. 1992. The Hacker Crackdown: Law and Disorder on the Electronic Frontier. New York: Bantam Books.
- Thomas, Douglas. 2002. Hacker Culture. Minneapolis: University of Minnesota Press.
- Thomas, Douglas. 2005. "The moral ambiguity of social control in cyberspace." New Media & Society 7 (5): 599–624.
- Turkle, Sherry. 1984. The Second Self: Computers and the Human Spirit. New York: Simon & Schuster.
- Turkle, Sherry. 2005. The Inner History of Devices. Cambridge, MA: MIT Press.
- Turner, Fred. 2006. From Counterculture to Cyberculture: Stewart Brand, the Whole Earth Network, and the Rise of Digital Utopianism. Chicago: University of Chicago Press.
- Vygotsky, Lev S. 1978. Mind in Society: The Development of Higher Psychological Processes. Cambridge, MA: Harvard University Press.
- Wenger, Etienne. 1998. Communities of Practice: Learning, Meaning, and Identity. Cambridge: Cambridge University Press.
-
- Thomas, Douglas. 2002. *Hacker Culture*. Minneapolis: University of Minnesota Press.
- Thomas, Douglas. 2005. "The Moral Ambiguity of Social Control in Cyberspace." *New Media & Society* 7 (5): 599–624.
- Turkle, Sherry. 1984. *The Second Self: Computers and the Human Spirit*. New York: Simon & Schuster.
- Turner, Fred. 2006. *From Counterculture to Cyberculture: Stewart Brand, the Whole Earth Network, and the Rise of Digital Utopianism*. Chicago: University of Chicago Press.
- Vygotsky, Lev S. 1978. *Mind in Society: The Development of Higher Psychological Processes*. Cambridge, MA: Harvard University Press.
- Wenger, Etienne. 1998. *Communities of Practice: Learning, Meaning, and Identity*. Cambridge: Cambridge University Press.